

Topological Energy Condensate Theory (TECT): 27

Jusang Lee^{1,*}

¹*Independent Researcher, Incheon, Korea*

(Dated: March, 2026)

I. ONE-DEFECT ACCEPTANCE REDUCTION AND THE μ_∞ -FIRST AUDIT

A. General one-defect criterion

Assume the one-defect/one-mode reduction has already produced the effective radial-core coefficients

$$a_0 = \frac{\xi_{EGE} + \xi_{cg_c}}{z_{0,0}}, \quad b_0 = \frac{\xi_{EGE}}{z_{0,0}}, \quad c_0 = \frac{\xi_{OGO}}{z_{0,0}}, \quad (1)$$

together with the bulk mass parameter

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}. \quad (2)$$

For the projected lowest channel κ_* , a practical sufficient one-mode criterion is

$$\mu_\infty > 0, \quad a_0 > 0, \quad b_0 > 0, \quad \frac{|\kappa_*| b_0}{a_0} > \frac{1}{2}, \quad (3)$$

supplemented by competitor exclusion and remote-gap control.

B. Lowest-channel reduction

We now specialize to the practically most relevant lowest-channel case

$$|\kappa_*| = 1. \quad (4)$$

Then (3) becomes

$$\mu_\infty > 0, \quad a_0 > 0, \quad b_0 > 0, \quad 2b_0 > a_0. \quad (5)$$

Introduce the dimensionless longitudinal burden ratio

$$R_c := \frac{\xi_{cg_c}}{\xi_{EGE}}, \quad (6)$$

whenever $\xi_{EGE} \neq 0$. Then

$$a_0 = \frac{\xi_{EGE}}{z_{0,0}}(1 + R_c) = b_0(1 + R_c). \quad (7)$$

Hence the conditions in (5) can be rewritten algebraically.

[Lowest-channel acceptance window] Assume

$$z_{0,0} > 0, \quad \xi_E > 0, \quad |\kappa_*| = 1. \quad (8)$$

* jusanglee@kakao.com

Then the inequalities

$$a_0 > 0, \quad b_0 > 0, \quad 2b_0 > a_0 \quad (9)$$

are equivalent to

$$\xi_{EgE} > 0, \quad -1 < R_c < 1. \quad (10)$$

Therefore the one-defect acceptance criterion reduces to

$$\mu_\infty > 0, \quad \xi_{EgE} > 0, \quad -1 < R_c < 1, \quad (11)$$

plus competitor exclusion and remote-gap control.

Since $z_{0,0} > 0$, one has

$$b_0 > 0 \iff \xi_{EgE} > 0.$$

Using (7), the inequality $a_0 > 0$ becomes

$$b_0(1 + R_c) > 0.$$

Because $b_0 > 0$, this is equivalent to

$$1 + R_c > 0.$$

Likewise,

$$2b_0 > a_0 = b_0(1 + R_c)$$

is equivalent, after dividing by $b_0 > 0$, to

$$2 > 1 + R_c \iff R_c < 1.$$

Combining with $1 + R_c > 0$ gives

$$-1 < R_c < 1.$$

This proves the equivalence.

C. Equivalent sharp form in the transport variables

The inequality $2b_0 > a_0$ is exactly

$$2\xi_{EgE} > \xi_{EgE} + \xi_{cg_c} \iff \xi_{EgE} > \xi_{cg_c}. \quad (12)$$

Hence in the lowest-channel case one may equivalently write

$$\boxed{\mu_\infty > 0, \quad \xi_{EgE} > 0, \quad \xi_{EgE} > \xi_{cg_c}.} \quad (13)$$

D. Specialization to the explicit canonical branch

Now specialize further to the explicit canonical branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0. \quad (14)$$

On this branch one has

$$g_O = 0, \quad (15)$$

so the odd channel drops out of the acceptance count.

Define R_c by

$$R_c = \frac{\xi_c g_c}{\xi_E g_E}. \quad (16)$$

If the explicit branch formulas yield

$$R_c = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2, \quad (17)$$

then the window $-1 < R_c < 1$ becomes

$$\left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2}. \quad (18)$$

Similarly, if the branch-level even coefficient satisfies

$$g_E = \alpha \propto \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad (19)$$

then the positivity condition $\xi_E g_E > 0$ reduces, for $\xi_E > 0$, to

$$\frac{\beta_X}{M_X^2} \theta_0 \kappa > 0. \quad (20)$$

Therefore the explicit branch acceptance conditions become

$$\boxed{\left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2}, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad \mu_\infty > 0,} \quad (21)$$

plus remote-gap control.

E. The μ_∞ sign audit

The bulk mass sign is controlled by

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}. \quad (22)$$

Assume

$$z_0(\sigma_\infty) > 0. \quad (23)$$

Then

$$\text{sgn } \mu_\infty = \text{sgn} \left(m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) \right). \quad (24)$$

This yields the following practical cases.

a. Case A: $u_X(\sigma_\infty) \geq 0$, $\chi_\infty \geq 0$. Then the Class-II asymptotic correction lowers the gap, so one needs

$$m(\sigma_\infty) > 3\chi_\infty u_X(\sigma_\infty) \geq 0. \quad (25)$$

b. Case B: $u_X(\sigma_\infty) \leq 0$, $\chi_\infty \geq 0$. Then the Class-II asymptotic correction helps the gap, so it is enough that

$$m(\sigma_\infty) > 3\chi_\infty u_X(\sigma_\infty), \quad (26)$$

which is weaker than demanding $m(\sigma_\infty) > 0$.

c. Case C: sign-indefinite asymptotic sector. Then one must perform the exact finite sign audit

$$\text{sgn } \mu_\infty = \text{sgn} \left(m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) \right). \quad (27)$$

[Earliest kill condition] If

$$m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) \leq 0 \quad \text{and} \quad z_0(\sigma_\infty) > 0, \quad (28)$$

then

$$\mu_\infty \leq 0, \quad (29)$$

so the branch is rejected before any g_E , g_c , or channel-comparison insertion.

Immediate from (22).

F. Final practical theorem for the explicit branch

[Practical explicit-branch one-defect acceptance] Assume:

1. $z_0(\sigma_\infty) > 0$;
2. the lowest projected channel satisfies $|\kappa_*| = 1$;
3. the branch is the explicit canonical branch (92);
4. the remote-gap condition holds:

$$\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}. \quad (30)$$

Then a sufficient practical one-defect acceptance criterion is

$$\boxed{\left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0\kappa)^2}, \quad \frac{\beta_X}{M_X^2} \theta_0\kappa > 0, \quad m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.} \quad (31)$$

Equivalently, in the more invariant transport notation,

$$\boxed{-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.} \quad (32)$$

Combine:

- the lowest-channel equivalence theorem (11),
- the explicit branch specialization (18) and (20),
- the bulk sign reduction (24),
- and the independent support condition (30).

This yields (31), hence also (32).

G. Honest status

The symbolic reduction is now complete: the explicit branch has been reduced to four finite scalar checks. What is not yet inserted is the actual asymptotic branch data

$$m(\sigma_\infty), \quad u_X(\sigma_\infty), \quad \chi_\infty, \quad z_0(\sigma_\infty),$$

together with the actual branch values of

$$\xi_c, \quad \xi_E, \quad \Delta_{\text{bench},\rho}^{\text{full}} - \eta_{\text{rem},\rho}.$$

So the next frontier is explicit evaluation, not further structural reduction.

II. FINAL SUMMARY

The C4 one-defect acceptance problem is no longer a broad conceptual question. In the lowest-channel case $|\kappa_*| = 1$, it reduces to

$$\mu_\infty > 0, \quad \xi_E g_E > 0, \quad \xi_E g_E > \xi_c g_c,$$

equivalently

$$-1 < R_c < 1, \quad R_c := \frac{\xi_c g_c}{\xi_E g_E},$$

together with remote-gap control.

For the explicit canonical branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

this becomes the four-scalar practical test

$$\left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0\kappa)^2}, \quad \frac{\beta_X}{M_X^2} \theta_0\kappa > 0, \quad m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

Thus the remaining frontier is not structural. It is the explicit branch evaluation of

$$m(\sigma_\infty), \quad u_X(\sigma_\infty), \quad \chi_\infty, \quad z_0(\sigma_\infty), \quad \xi_c, \quad \xi_E, \quad \Delta_{\text{bench},\rho}^{\text{full}} - \eta_{\text{rem},\rho}.$$

III. THE μ_∞ -FIRST AUDIT AS A COMPLETE FINITE SCALAR REDUCTION

IV. GOAL

At the current C4 / one-defect acceptance stage, the correct first task is to decide the sign of

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}. \quad (33)$$

This is the earliest kill condition in the full one-defect audit.

The purpose of this section is to show that the bulk sign problem is no longer conceptual: it is completely reduced to a finite asymptotic scalar audit.

V. 1. BULK MASS PARAMETER FROM THE LOCKED CLASS II DEFECT BACKGROUND

In the locked Class II background, the mediator-corrected defect mass is

$$m_{\text{eff}}(x) = m(\bar{\sigma}(x)) - 3\chi_X(x) u_X(\bar{\sigma}(x)). \quad (34)$$

The asymptotic one-defect bulk mass parameter is therefore

$$\mu_\infty = \frac{m_{\text{eff}}(\infty)}{z_0(\sigma_\infty)} = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}. \quad (35)$$

Thus the bulk acceptance problem depends only on the four asymptotic scalars

$$m(\sigma_\infty), \quad u_X(\sigma_\infty), \quad \chi_\infty, \quad z_0(\sigma_\infty). \quad (36)$$

VI. 2. SIGN REDUCTION THEOREM

[Exact sign reduction for μ_∞] Assume

$$\boxed{z_0(\sigma_\infty) > 0.} \quad (37)$$

Then

$$\boxed{\text{sgn } \mu_\infty = \text{sgn}(m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)).} \quad (38)$$

Equivalently,

$$\boxed{\mu_\infty > 0 \iff m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) > 0.} \quad (39)$$

This is immediate from (35). Under (37), the denominator does not alter the sign, hence

$$\text{sgn } \mu_\infty = \text{sgn}(m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)).$$

[Earliest reject criterion] Under (37), if

$$\boxed{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) \leq 0,} \quad (40)$$

then

$$\boxed{\mu_\infty \leq 0,} \quad (41)$$

so the branch is rejected before any insertion of g_E , g_c , κ_* , or competitor exclusion.

VII. 3. CASE DECOMPOSITION OF THE ASYMPTOTIC SIGN AUDIT

The sign reduction theorem is exact but abstract. To make it executable, we separate the asymptotic sign patterns.

Case A: $u_X(\sigma_\infty) \geq 0$ and $\chi_\infty \geq 0$

Then the Class II correction lowers the bulk gap:

$$-3\chi_\infty u_X(\sigma_\infty) \leq 0. \quad (42)$$

Therefore one needs the sharp lower bound

$$\boxed{m(\sigma_\infty) > 3\chi_\infty u_X(\sigma_\infty) \geq 0.} \quad (43)$$

Case B: $u_X(\sigma_\infty) \leq 0$ and $\chi_\infty \geq 0$

Then the correction helps the gap:

$$-3\chi_\infty u_X(\sigma_\infty) \geq 0. \quad (44)$$

Hence the positivity condition becomes

$$\boxed{m(\sigma_\infty) > 3\chi_\infty u_X(\sigma_\infty),} \quad (45)$$

which is weaker than the naive sufficient requirement $m(\sigma_\infty) > 0$.

Case C: sign-indefinite asymptotic sector

If the signs of χ_∞ and $u_X(\sigma_\infty)$ are not known a priori, then there is no further universal simplification. However, the sign audit is still finite:

$$\boxed{\text{sgn } \mu_\infty = \text{sgn}(m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)), \quad \text{assuming } z_0(\sigma_\infty) > 0.} \quad (46)$$

VIII. 4. BENCHMARK-NORMALIZED ACCEPT/REJECT FORM

The bulk decision can now be written in a completely executable form.

[Executable bulk accept/reject criterion] Assume

$$z_0(\sigma_\infty) > 0. \quad (47)$$

Define

$$\boxed{\Delta_\mu := m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty).} \quad (48)$$

Then exactly one of the following holds:

1. if $\Delta_\mu > 0$, then

$$\mu_\infty > 0$$

and the branch passes the bulk positivity gate;

2. if $\Delta_\mu = 0$, then

$$\mu_\infty = 0$$

and the branch is gapless at infinity, so the bulk positivity gate fails;

3. if $\Delta_\mu < 0$, then

$$\mu_\infty < 0$$

and the branch is rejected immediately.

Immediate from

$$\mu_\infty = \frac{\Delta_\mu}{z_0(\sigma_\infty)}$$

with $z_0(\sigma_\infty) > 0$.

IX. 5. RELATION TO THE FULL ONE-DEFECT ACCEPTANCE CHAIN

The full one-defect acceptance data are

$$\boxed{(g_c, g_E, g_O, \mu_\infty, \kappa_*)}. \quad (49)$$

But the correct finite audit order is

$$\boxed{\mu_\infty \longrightarrow g_E \longrightarrow g_c \longrightarrow \kappa_* \longrightarrow \text{competitor exclusion}.} \quad (50)$$

Therefore the present bulk reduction is logically prior to all transport insertions. If the bulk sign fails, then the rest of the one-defect audit is irrelevant for acceptance.

X. 6. HONEST CLOSURE STATEMENT

What is now fully closed is the *structural* reduction of the bulk sign problem:

$$\boxed{\mu_\infty \text{ is completely controlled by the finite scalar set } \{m(\sigma_\infty), u_X(\sigma_\infty), \chi_\infty, z_0(\sigma_\infty)\}.}$$
 (51)

What is not yet inserted numerically or branch-specifically is the actual asymptotic evaluation of those four scalars on the chosen vacuum-asymptotic branch.

Hence the honest status is:

$$\boxed{\text{bulk positivity is no longer conceptually open,}}$$
 (52)

but

$$\boxed{\text{it is not yet numerically closed until the chosen branch data are inserted.}}$$
 (53)

XI. FINAL SUMMARY

At the current C4 / one-defect acceptance stage, the bulk gate is the first and cheapest decisive test. The asymptotic bulk mass parameter is

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}.$$

Therefore, assuming

$$z_0(\sigma_\infty) > 0,$$

the sign of μ_∞ is exactly the sign of

$$\Delta_\mu := m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty).$$

Hence:

$$\Delta_\mu > 0 \iff \mu_\infty > 0, \quad \Delta_\mu = 0 \iff \mu_\infty = 0, \quad \Delta_\mu < 0 \iff \mu_\infty < 0.$$

So the μ_∞ problem is completely reduced to a finite four-scalar audit:

$$m(\sigma_\infty), \quad u_X(\sigma_\infty), \quad \chi_\infty, \quad z_0(\sigma_\infty).$$

This closes the structural bulk-sign problem.

What remains is not conceptual. It is the explicit insertion of the chosen vacuum-asymptotic branch data into those four scalars.

XII. BULK GATE CLOSURE ON THE SIMPLEST VACUUM BRANCH AND TRANSITION TO THE g_E AUDIT

XIII. 1. PRIORITY ORDER AND BULK DATA

The one-defect acceptance problem has already been reduced to a finite audit of

$$\boxed{(g_c, g_E, g_O, \mu_\infty, \kappa_*)},$$
 (54)

with the correct priority order

$$\boxed{\mu_\infty \longrightarrow g_E \longrightarrow g_c \longrightarrow \kappa_* \longrightarrow \text{competitor exclusion.}}$$
 (55)

The bulk package is

$$\boxed{m(\sigma_\infty), \quad u_X(\sigma_\infty), \quad \chi_\infty, \quad z_0(\sigma_\infty).}$$
 (56)

The exact constitutive bulk coefficient is

$$\boxed{\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}}.$$
 (57)

XIV. 2. EXACT SIGN REDUCTION

Assume the standard normalization

$$\boxed{z_0(\sigma_\infty) > 0.} \quad (58)$$

Then

$$\boxed{\mu_\infty > 0 \iff m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) > 0.} \quad (59)$$

Thus the general non-vacuum asymptotic branch requires the finite scalar comparison

$$\boxed{m(\sigma_\infty) > 3\chi_\infty u_X(\sigma_\infty).} \quad (60)$$

XV. 3. SIMPLEST VACUUM BRANCH

Now specialize to the simplest vacuum branch. By definition,

$$\boxed{X_{\perp,\infty} = X_{\parallel,\infty} = 0.} \quad (61)$$

Hence

$$\boxed{\chi_\infty = \frac{1}{3}(2X_{\perp,\infty} + X_{\parallel,\infty}) = 0.} \quad (62)$$

Substituting (62) into (57) gives

$$\boxed{\mu_\infty = \frac{m(\sigma_\infty)}{z_0(\sigma_\infty)}.} \quad (63)$$

Therefore, under (58),

$$\boxed{\mu_\infty > 0 \iff m(\sigma_\infty) > 0.} \quad (64)$$

[Bulk-gate closure on the simplest vacuum branch] Assume the asymptotic branch satisfies

$$X_{\perp,\infty} = X_{\parallel,\infty} = 0, \quad z_0(\sigma_\infty) > 0.$$

Then the one-defect bulk positivity gate is completely determined by the single scalar sign

$$m(\sigma_\infty).$$

More precisely,

$$m(\sigma_\infty) > 0 \iff \mu_\infty > 0,$$

$$m(\sigma_\infty) = 0 \iff \mu_\infty = 0,$$

$$m(\sigma_\infty) < 0 \iff \mu_\infty < 0.$$

Immediate from (63).

XVI. 4. HONEST LOGICAL CONSEQUENCE

This is stronger than the generic non-vacuum statement.

On a general asymptotic Class-II branch, the bulk gate requires the balance

$$m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) > 0.$$

But on the simplest vacuum branch, the Class-II asymptotic correction disappears identically:

$$\chi_\infty = 0.$$

Therefore the bulk gate is no longer a balance problem. It is a one-sign problem.

Hence, once the chosen working branch is the simplest vacuum branch, the bulk frontier is effectively closed up to the sign of $m(\sigma_\infty)$.

XVII. 5. TRANSITION TO THE FIRST REAL TRANSPORT FRONTIER

After the bulk gate, the next audit is g_E .

For the shell-projected transport coefficients,

$$\boxed{\alpha = C_E g_E, \quad \beta = C_O g_O.} \quad (65)$$

In the explicit pure-even canonical realization,

$$\boxed{\lambda_{\parallel} = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3, \quad \alpha = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad \beta = 0.} \quad (66)$$

If, in addition, one adopts the canonical normalization

$$\boxed{C_{\parallel} = C_E = C_O = 1,} \quad (67)$$

then

$$\boxed{g_E = \alpha = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad g_O = 0.} \quad (68)$$

Therefore the explicit pure-even canonical branch satisfies

$$\boxed{g_E > 0 \iff \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0.} \quad (69)$$

XVIII. 6. COMBINED PRACTICAL CRITERION AFTER BULK CLOSURE

In the practical lowest-channel case, the explicit branch criterion is already reduced to

$$\boxed{\left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2}, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho}.} \quad (70)$$

On the simplest vacuum branch, the third inequality becomes

$$\boxed{m(\sigma_\infty) > 0 \quad (\text{assuming } z_0(\sigma_\infty) > 0).} \quad (71)$$

So the practical explicit one-defect audit on this branch becomes

$$\boxed{\left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2}, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad m(\sigma_\infty) > 0, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho}.} \quad (72)$$

XIX. 7. HONEST STATUS

What is now closed:

1. the bulk sign problem is completely reduced;
2. on the simplest vacuum branch, it collapses to the single scalar sign $m(\sigma_\infty)$;
3. the next genuine frontier is g_E , not μ_∞ .

What remains open explicitly:

1. the actual sign of $m(\sigma_\infty)$ in the chosen TECT working branch;
2. the actual sign of $\beta_X \theta_0 \kappa$;
3. the longitudinal burden ratio through ξ_c/ξ_E ;
4. the final remote-gap numerical insertion.

XX. FINAL SUMMARY

The bulk coefficient of the one-defect acceptance theorem is

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}.$$

Under the standard normalization $z_0(\sigma_\infty) > 0$, this is equivalent to

$$\mu_\infty > 0 \iff m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) > 0.$$

However, on the simplest vacuum branch,

$$X_{\perp,\infty} = X_{\parallel,\infty} = 0 \implies \chi_\infty = 0,$$

so the bulk gate simplifies to

$$\boxed{\mu_\infty = \frac{m(\sigma_\infty)}{z_0(\sigma_\infty)}, \quad \mu_\infty > 0 \iff m(\sigma_\infty) > 0.}$$

Thus the asymptotic Class-II correction disappears entirely at infinity on this branch.

Therefore the bulk frontier is no longer a balance problem. It is a one-sign problem.

After this reduction, the first genuine remaining transport frontier is g_E . In the explicit pure-even canonical branch,

$$\alpha = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad \beta = 0,$$

so, under canonical normalization,

$$\boxed{g_E > 0 \iff \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0.}$$

Hence the practical explicit one-defect audit on the simplest vacuum branch is

$$\boxed{\left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2}, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad m(\sigma_\infty) > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

XXI. EXPLICIT g_E SIGN AUDIT AND THE FINAL PRACTICAL ACCEPTANCE FORMULA

XXII. GOAL

We now complete the g_E -sign audit on the explicit pure-even canonical branch and combine it with the already reduced R_c -window and bulk sign condition.

The point is to pass from the abstract operational criterion to a concrete branch-level constitutive criterion.

XXIII. 1. OPERATIONAL ONE-DEFECT CRITERION

The practical lowest-channel one-defect/one-mode criterion is already reduced to

$$\boxed{-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho},} \quad (73)$$

provided the projected lowest signed channel is unique and the competitors are spectrally excluded.

Equivalently, before introducing R_c ,

$$\boxed{\mu_\infty > 0, \quad \xi_E g_E + \xi_c g_c > 0, \quad \xi_E g_E > 0, \quad 2|\kappa_*| \xi_E g_E > \xi_E g_E + \xi_c g_c.} \quad (74)$$

XXIV. 2. EXPLICIT BRANCH DATA

Fix the explicit canonical pure-even branch

$$\boxed{T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0, \quad C_\parallel = C_E = C_O = 1.} \quad (75)$$

Then the extracted transport/Bloch coefficients are

$$\boxed{g_c = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3,} \quad (76)$$

$$\boxed{g_E = \alpha = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad g_O = 0.} \quad (77)$$

Hence the odd transport sector drops out of the protected-channel counting problem.

XXV. 3. EXACT g_E SIGN AUDIT

Since the prefactor

$$\frac{\sqrt{3}\pi}{16} > 0, \quad M_X^2 > 0,$$

the sign of g_E is exactly the sign of

$$\boxed{\frac{\beta_X}{M_X^2} \theta_0 \kappa.} \quad (78)$$

Therefore:

$$\boxed{g_E > 0 \iff \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0.} \quad (79)$$

Under the standard matching convention $\xi_E > 0$, this is also equivalent to

$$\boxed{\xi_E g_E > 0.} \quad (80)$$

So the transverse support test is no longer abstract. It is a one-sign constitutive test.

XXVI. 4. EXACT LONGITUDINAL BURDEN RATIO

Define

$$R_c := \frac{\xi_c g_c}{\xi_E g_E}. \quad (81)$$

Substituting (76) and (77) gives

$$\begin{aligned} R_c &= \frac{\xi_c}{\xi_E} \frac{\frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3}{\frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa} \\ &= \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2. \end{aligned} \quad (82)$$

Therefore

$$R_c = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2. \quad (83)$$

This already shows an important simplification: the sign and size of the longitudinal burden are controlled by

$$\frac{\xi_c}{\xi_E} \alpha_X$$

and the defect amplitude only through the square $(\theta_0 \kappa)^2$.

XXVII. 5. SIMPLEST VACUUM BULK BRANCH

Now specialize the asymptotic constitutive branch further to the simplest vacuum branch:

$$\chi_\infty = 0. \quad (84)$$

Then

$$\mu_\infty = \frac{m(\sigma_\infty)}{z_0(\sigma_\infty)}. \quad (85)$$

Assuming the standard normalization

$$z_0(\sigma_\infty) > 0, \quad (86)$$

one gets the exact equivalence

$$\mu_\infty > 0 \iff m(\sigma_\infty) > 0. \quad (87)$$

Thus the bulk sign gate is reduced to a single constitutive sign.

XXVIII. 6. FINAL PRACTICAL THEOREM

We now combine (73), (79), (83), and (87).

[Practical one-defect acceptance on the explicit pure-even canonical branch] Assume:

1. the working branch is

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0, \quad C_\parallel = C_E = C_O = 1;$$

2. the asymptotic constitutive branch is the simplest vacuum branch $\chi_\infty = 0$;
3. $z_0(\sigma_\infty) > 0$;
4. the projected low-slot structure is the practical lowest-channel case, with a unique signed lowest channel and the standard competitor pattern;
5. the remote-gap condition

$$\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}$$

holds.

Then a sufficient practical acceptance criterion is

$$\boxed{m(\sigma_\infty) > 0, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad \left| \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2 \right| < 1, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.} \quad (88)$$

The operational criterion requires

$$-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

Using (77),

$$\alpha > 0 \iff \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0.$$

Using (83),

$$-1 < R_c < 1 \iff \left| \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2 \right| < 1.$$

Using (87),

$$\mu_\infty > 0 \iff m(\sigma_\infty) > 0.$$

Combining these with the remote-gap condition yields (88).

XXIX. 7. SMALL-DEFECT COROLLARY

If

$$\left| \frac{\xi_c}{\xi_E} \alpha_X \right|$$

is $O(1)$ and $|\theta_0 \kappa| \ll 1$, then

$$\boxed{R_c = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2} \quad (89)$$

is automatically small, hence the R_c -window is naturally satisfied.

So in the small-defect regime the genuinely nontrivial constitutive tests are primarily:

$$\boxed{m(\sigma_\infty) > 0, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0,} \quad (90)$$

while the longitudinal burden inequality becomes mild.

XXX. 8. SHARPENED SIGN CASES FOR R_c

Because $(\theta_0\kappa)^2 \geq 0$, the sign of R_c is the sign of

$$\frac{\xi_c}{\xi_E} \alpha_X.$$

Hence:

a. Case A: $\frac{\xi_c}{\xi_E} \alpha_X > 0$. Then $R_c \geq 0$, so the lower bound $R_c > -1$ is automatic. Only the upper bound matters:

$$\boxed{\frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0\kappa)^2 < 1.}$$

b. Case B: $\frac{\xi_c}{\xi_E} \alpha_X < 0$. Then $R_c \leq 0$, so the upper bound $R_c < 1$ is automatic. Only the lower bound matters:

$$\boxed{\left| \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0\kappa)^2 \right| < 1.}$$

Thus the absolute-value form is the cleanest universal formulation.

XXXI. 9. HONEST STATUS

What is now closed structurally and branch-explicitly:

1. $g_O = 0$ on the explicit pure-even canonical branch;
2. g_E -positivity is exactly the sign of $(\beta_X/M_X^2)\theta_0\kappa$;
3. g_c enters only through the explicit burden ratio

$$R_c = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0\kappa)^2;$$

4. on the simplest vacuum branch, the bulk sign is exactly the sign of $m(\sigma_\infty)$.

Therefore the one-defect acceptance problem is no longer structurally open on this branch. What remains is only the insertion of actual asymptotic constitutive data and the remote-gap bound.

XXXII. FINAL SUMMARY

On the explicit pure-even canonical branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0, \quad C_\parallel = C_E = C_O = 1,$$

the transport coefficients are

$$g_c = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3, \quad g_E = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad g_O = 0.$$

Hence the transverse support sign is exactly

$$g_E > 0 \iff \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0.$$

The longitudinal burden ratio is explicitly

$$R_c = \frac{\xi_c g_c}{\xi_E g_E} = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0\kappa)^2.$$

On the simplest vacuum branch $\chi_\infty = 0$, one has

$$\mu_\infty = \frac{m(\sigma_\infty)}{z_0(\sigma_\infty)},$$

so, under $z_0(\sigma_\infty) > 0$,

$$\mu_\infty > 0 \iff m(\sigma_\infty) > 0.$$

Therefore the practical one-defect acceptance criterion reduces to

$$\boxed{m(\sigma_\infty) > 0, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad \left| \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2 \right| < 1, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho}.$$

So the explicit branch has now collapsed to:

- three constitutive checks,
- plus one spectral isolation check.

No additional structural derivation is required at this stage. The remaining frontier is only the finite insertion of actual branch data.

XXXIII. EXECUTABLE ACCEPT/REJECT SHEET FOR THE EXPLICIT ONE-DEFECT BRANCH

XXXIV. GOAL

At the current stage, the one-defect acceptance problem is no longer conceptual. It has already been reduced to a finite audit of

$$(g_c, g_E, g_O, \mu_\infty, \kappa_*).$$

The remaining task is therefore to write the final accept/reject sheet explicitly.

XXXV. 1. LOWEST-CHANNEL PRACTICAL CRITERION

In the practically relevant lowest-channel case, the sufficient one-defect criterion is

$$\boxed{-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho}.} \quad (91)$$

Equivalently, using

$$R_c = \frac{\xi_c g_c}{\xi_E g_E},$$

this is the reduced finite scalar form of the original core inequalities.

XXXVI. 2. EXPLICIT CANONICAL BRANCH

Fix the explicit canonical branch

$$\boxed{T_* = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0.} \quad (92)$$

Then

$$\boxed{g_c = \lambda_\parallel, \quad g_E = \alpha, \quad g_O = 0.} \quad (93)$$

The explicit coefficient sheet gives

$$g_c = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3, \quad (94)$$

$$g_E = \alpha = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad g_O = 0. \quad (95)$$

Hence

$$R_c = \frac{\xi_c}{\xi_E} \frac{g_c}{g_E} = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2. \quad (96)$$

XXXVII. 3. BULK SIGN

The exact bulk parameter is

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}. \quad (97)$$

Under the standard convention

$$z_0(\sigma_\infty) > 0,$$

one has

$$\mu_\infty > 0 \iff m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) > 0. \quad (98)$$

If one further specializes to the simplest vacuum branch

$$\chi_\infty = 0,$$

then this simplifies to

$$\mu_\infty > 0 \iff m(\sigma_\infty) > 0. \quad (99)$$

XXXVIII. 4. FINAL EXECUTABLE DECISION SHEET

Combining (91), (95), (96), and (99), one gets:

[Executable accept/reject sheet for the simplest explicit branch] Assume:

1. the explicit canonical branch (92);
2. the simplest vacuum asymptotic branch $\chi_\infty = 0$;
3. $z_0(\sigma_\infty) > 0$;
4. the lowest-channel reduction is valid.

Then a sufficient acceptance criterion is

$$m(\sigma_\infty) > 0, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad \left| \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2 \right| < 1, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho}. \quad (100)$$

The four checks in (91) are:

- $\mu_\infty > 0$, which becomes $m(\sigma_\infty) > 0$ on the simplest vacuum branch;
- $\alpha > 0$, which becomes $(\beta_X/M_X^2)\theta_0\kappa > 0$;
- $-1 < R_c < 1$, which becomes the absolute-value inequality in (100);
- the remote-gap inequality.

This proves the result.

XXXIX. 5. IMMEDIATE REJECT RULES

The same theorem yields the following immediate sufficient rejection rules.

a. Bulk reject. If

$$m(\sigma_\infty) \leq 0$$

(on the simplest vacuum branch), then

$$\mu_\infty \leq 0,$$

so the branch fails before any transport-side test.

b. Transverse-support reject. If

$$\frac{\beta_X}{M_X^2} \theta_0 \kappa \leq 0,$$

then

$$g_E \leq 0,$$

so the lowest transverse support test fails.

c. Longitudinal-overload reject. If

$$\left| \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2 \right| \geq 1,$$

then the longitudinal burden exceeds the allowed window.

d. Remote-sector reject. If

$$\Delta_{\text{bench},\rho}^{\text{full}} \leq \eta_{\text{rem},\rho},$$

then the enriched low slot is not spectrally isolated from the remote sector.

XL. 6. UNIT-BENCHMARK COROLLARY

For the benchmark choice

$$\alpha_X = \beta_X = M_X = \xi_E = \xi_c = z_{0,0} = 1,$$

one obtains

$$\boxed{R_c = 0.09375 (\theta_0 \kappa)^2.} \tag{101}$$

Hence:

$$\theta_0 \kappa = 0.1 \Rightarrow R_c = 9.375 \times 10^{-4},$$

$$\theta_0 \kappa = 0.3 \Rightarrow R_c = 8.4375 \times 10^{-3},$$

$$\theta_0 \kappa = 1 \Rightarrow R_c = 9.375 \times 10^{-2}.$$

Therefore, throughout the natural small-defect regime, the R_c -window is automatically mild.

XLI. 7. HONEST STRUCTURAL CAVEAT

The preceding decision sheet uses the explicit pure-even coefficient identification

$$g_E = \alpha = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa.$$

However, the broader Schur-reduced working-branch formula can also be written as

$$\alpha = I_\theta \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}.$$

Therefore the sign of g_E is fully branch-explicit only after fixing which branch normalization and coefficient dictionary is being used. This does not reopen the structure of the proof, but it does mean that the final numerical sign insertion is still branch-specific.

XLII. 8. HONEST CONCLUSION

What is now fully closed:

1. the one-defect acceptance problem is reduced to four scalar checks;
2. the explicit decision sheet (100) is complete;
3. the reject modes are fully classified.

What is still not fixed by the uploaded notes:

1. the actual branch values of $m(\sigma_\infty)$;
2. the actual matching constants ξ_E, ξ_c ;
3. the actual remote-gap margin

$$\Delta_{\text{bench},\rho}^{\text{full}} - \eta_{\text{rem},\rho};$$

4. the final branch-normalization convention entering the sign of g_E .

So the remaining frontier is not theorem-building. It is finite constitutive insertion.

XLIII. FINAL SUMMARY

The one-defect acceptance problem has already been reduced to the finite data

$$(g_c, g_E, g_O, \mu_\infty, \kappa_*).$$

On the explicit canonical branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the lowest-channel practical criterion is

$$-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

Using the explicit coefficient sheet,

$$g_c = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3, \quad g_E = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad g_O = 0,$$

so

$$R_c = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2.$$

Moreover,

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)},$$

and on the simplest vacuum branch $\chi_\infty = 0$ with $z_0(\sigma_\infty) > 0$,

$$\mu_\infty > 0 \iff m(\sigma_\infty) > 0.$$

Therefore the explicit accept/reject sheet is

$$\boxed{m(\sigma_\infty) > 0, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad \left| \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2 \right| < 1, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho}.$$

This is the fully reduced practical criterion.

However, the uploaded notes still do not fix the actual branch values of

$$m(\sigma_\infty), \quad \xi_E, \quad \xi_c, \quad \Delta_{\text{bench}, \rho}^{\text{full}} - \eta_{\text{rem}, \rho},$$

nor do they completely eliminate the branch-normalization ambiguity in the final sign dictionary for g_E .

Hence the remaining frontier is finite constitutive insertion, not new theorem-building.

XLIV. NORMALIZATION-AWARE EXACT PASS CERTIFICATE FOR GATE 5

XLV. GOAL

We continue the current line up to its actual termination point.

The correct endpoint is not another auxiliary reformulation. It is the exact normalization-aware reduction of the post-acceptance one-defect test to a finite set of branch scalars that can be directly inserted on the intended Class II branch.

XLVI. 1. EXACT ONE-DEFECT CORE CRITERION

The one-defect theorem is controlled by

$$(g_c, g_E, g_O, \mu_\infty, \kappa_*),$$

with core coefficients

$$\boxed{a_0 = \frac{\xi_E g_E + \xi_c g_c}{z_{0,0}}, \quad b_0 = \frac{\xi_E g_E}{z_{0,0}}, \quad c_0 = \frac{\xi_O g_O}{z_{0,0}}.} \quad (102)$$

The sufficient one-defect/one-mode criterion is

$$\boxed{\mu_\infty > 0, \quad a_0 > 0, \quad b_0 > 0, \quad \frac{|\kappa_*| b_0}{a_0} > \frac{1}{2}.} \quad (103)$$

In the practically most relevant lowest-channel case

$$\boxed{|\kappa_*| = 1,} \quad (104)$$

the decisive inequality becomes

$$\boxed{\xi_E g_E > \xi_c g_c.} \quad (105)$$

Assuming $z_{0,0} > 0$, (103) is equivalent to

$$\boxed{\mu_\infty > 0, \quad \xi_E g_E > 0, \quad -1 < R_{\text{exact}} < 1,} \quad (106)$$

where

$$\boxed{R_{\text{exact}} := \frac{\xi_c g_c}{\xi_E g_E}.} \quad (107)$$

Finally, Gate 5 must be supported by the remote-gap support theorem:

$$\boxed{\Delta_{\text{gap}} := \Delta_{\text{bench}, \rho}^{\text{fin}} - \eta_{R, \rho} > 0.} \quad (108)$$

Thus the exact post-acceptance Gate-5 problem is already reduced to four scalars:

$$\boxed{\Delta_\mu, \quad S_E, \quad R_{\text{exact}}, \quad \Delta_{\text{gap}},} \quad (109)$$

where

$$\Delta_\mu := m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty), \quad S_E := \xi_E g_E.$$

XLVII. 2. EXACT NORMALIZATION LAYER

The exact reduced coefficient dictionary is

$$\boxed{\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O.} \quad (110)$$

Together with the exact overlap-normalization law

$$\boxed{g_c = \frac{\lambda_\parallel}{C_c}, \quad g_E = \frac{\alpha}{C_E}, \quad g_O = \frac{\beta}{C_O},} \quad (111)$$

this gives

$$\boxed{g_c = \frac{I_{\theta'^3}}{2C_c} \lambda_c, \quad g_E = \frac{I_\theta}{C_E} \Lambda_E, \quad g_O = \frac{I_\theta}{C_O} \Lambda_O.} \quad (112)$$

Therefore the transverse sign scalar is exactly

$$\boxed{S_E = \xi_E g_E = \xi_E \frac{I_\theta}{C_E} \Lambda_E.} \quad (113)$$

Likewise,

$$R_{\text{exact}} = \frac{\xi_c g_c}{\xi_E g_E} = \frac{\xi_c}{\xi_E} \frac{C_E}{C_c} \frac{I_{\theta'^3}}{2I_\theta} \frac{\lambda_c}{\Lambda_E}. \quad (114)$$

Using

$$\boxed{I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3,} \quad (115)$$

one obtains

$$\boxed{\frac{I_{\theta'^3}}{2I_\theta} = \frac{3}{32} (\theta_0 \kappa)^2.} \quad (116)$$

Hence

$$\boxed{R_{\text{exact}} = \frac{\xi_c}{\xi_E} \frac{C_E}{C_c} \frac{3}{32} (\theta_0 \kappa)^2 \frac{\lambda_c}{\Lambda_E}.} \quad (117)$$

This is the exact normalization-aware longitudinal burden ratio. It is the correct endpoint of the current line, because it removes all earlier convention-dependent ambiguity.

XLVIII. 3. BULK SCALAR

The exact bulk mass parameter is

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)} = \frac{\Delta_\mu}{z_0(\sigma_\infty)}. \quad (118)$$

Thus, assuming

$$z_0(\sigma_\infty) > 0, \quad (119)$$

one has

$$\mu_\infty > 0 \iff \Delta_\mu > 0. \quad (120)$$

If the intended branch is further the simplest vacuum branch, then $\chi_\infty = 0$ and

$$\Delta_\mu = m(\sigma_\infty). \quad (121)$$

XLIX. 4. WORKING-BRANCH SOURCE PACKAGES

The current working-branch scalar reductions are

$$\Lambda_E = \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_E, \quad \Lambda_O = \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_O, \quad (122)$$

and longitudinally

$$\lambda_c^{(JJ)} \sim \frac{\alpha_X^2}{M_X^2} \mathcal{N}_{JJ} m_\parallel u_\parallel |G_*| A_{G_*}. \quad (123)$$

Therefore

$$S_E = \xi_E \frac{I_\theta}{C_E} \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_E. \quad (124)$$

And, at the same working-branch leading level,

$$R_{\text{exact}} \sim \frac{\xi_c}{\xi_E} \frac{C_E}{C_c} \frac{3}{32} (\theta_0 \kappa)^2 \frac{\alpha_X}{\beta_X} \frac{\mathcal{N}_{JJ} m_\parallel u_\parallel |G_*| A_{G_*}}{\mathfrak{T}_E}. \quad (125)$$

Equation (125) is not a new theorem; it is the honest working-branch insertion layer obtained by combining the already derived finite scalar packages.

L. 5. EXACT PASS CERTIFICATE

We now combine (106), (108), (113), (117), and (120).

[Exact post-acceptance Gate-5 certificate on the intended branch] Assume:

1. the current intended branch has already passed the carrier-level Gates 2–4;
2. the low slot is the practically relevant lowest-channel case $|\kappa_*| = 1$;
3. $z_0(\sigma_\infty) > 0$.

Then Gate 5 passes if and only if the following four exact scalar conditions hold:

$$\boxed{\Delta_\mu > 0,} \quad (126)$$

$$\boxed{S_E = \xi_E \frac{I_\theta}{C_E} \Lambda_E > 0,} \quad (127)$$

$$\boxed{-1 < R_{\text{exact}} = \frac{\xi_c}{\xi_E} \frac{C_E}{C_c} \frac{3}{32} (\theta_0 \kappa)^2 \frac{\lambda_c}{\Lambda_E} < 1,} \quad (128)$$

$$\boxed{\Delta_{\text{gap}} = \Delta_{\text{bench},\rho}^{\text{fin}} - \eta_{R,\rho} > 0.} \quad (129)$$

Equivalently, the current line ends at the four-scalar exact insertion problem

$$(\Delta_\mu, S_E, R_{\text{exact}}, \Delta_{\text{gap}}).$$

Under $z_0(\sigma_\infty) > 0$, $\mu_\infty > 0 \iff \Delta_\mu > 0$. The lowest-channel one-defect criterion is equivalent to

$$\mu_\infty > 0, \quad \xi_E g_E > 0, \quad -1 < R_{\text{exact}} < 1.$$

The exact normalization layer gives

$$\xi_E g_E = S_E, \quad \frac{\xi_c g_c}{\xi_E g_E} = R_{\text{exact}}.$$

Finally the reduced carrier statement must be supported by remote-gap isolation, namely $\Delta_{\text{gap}} > 0$. This proves the claim.

LI. 6. SIMPLEST VACUUM + PURE-EVEN SPECIALIZATION

If the intended branch is also:

$$\chi_\infty = 0, \quad \Lambda_O = 0,$$

then

$$\boxed{\Delta_\mu = m(\sigma_\infty), \quad g_O = 0.} \quad (130)$$

Hence the exact pass certificate becomes

$$\boxed{m(\sigma_\infty) > 0, \quad \xi_E \frac{I_\theta}{C_E} \Lambda_E > 0, \quad -1 < \frac{\xi_c}{\xi_E} \frac{C_E}{C_c} \frac{3}{32} (\theta_0 \kappa)^2 \frac{\lambda_c}{\Lambda_E} < 1, \quad \Delta_{\text{gap}} > 0.} \quad (131)$$

If one further inserts the working-branch reduced packages,

$$\boxed{m(\sigma_\infty) > 0,} \quad (132)$$

$$\boxed{\xi_E \frac{I_\theta}{C_E} \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_E > 0,} \quad (133)$$

$$\boxed{-1 < \frac{\xi_c}{\xi_E} \frac{C_E}{C_c} \frac{3}{32} (\theta_0 \kappa)^2 \frac{\alpha_X}{\beta_X} \frac{\mathcal{N}_{JJ} m_{\parallel} u_{\parallel}}{\mathfrak{T}_E} |G_*| A_{G_*} < 1,} \quad (134)$$

$$\boxed{\Delta_{\text{bench},\rho}^{\text{fin}} > \eta_{R,\rho}.} \quad (135)$$

This is the actual end of the current line. Nothing structural remains beyond finite branch insertion.

LII. 7. HONEST CLOSURE STATUS

What is now closed:

1. the exact normalization-aware Gate-5 algebra;
2. the exact burden ratio R_{exact} ;
3. the exact transverse sign scalar S_E ;
4. the exact four-scalar pass certificate.

What still remains open explicitly:

1. the actual numerical or sign value of Δ_μ ;
2. the actual numerical or sign value of S_E ;
3. the actual numerical value of R_{exact} ;
4. the actual remote-gap margin Δ_{gap} .

Thus the present line is genuinely finished at the level of theory reduction. The only remaining work inside this line is direct finite branch insertion.

LIII. FINAL SUMMARY

The current one-defect acceptance line ends at an exact normalization-aware four-scalar certificate. The exact core criterion is

$$\mu_\infty > 0, \quad a_0 > 0, \quad b_0 > 0, \quad \frac{|\kappa_*|b_0}{a_0} > \frac{1}{2},$$

with

$$a_0 = \frac{\xi_E g_E + \xi_c g_c}{z_{0,0}}, \quad b_0 = \frac{\xi_E g_E}{z_{0,0}}.$$

In the lowest-channel case $|\kappa_*| = 1$, this reduces to the exact four-scalar problem

$$\boxed{\Delta_\mu, \quad S_E, \quad R_{\text{exact}}, \quad \Delta_{\text{gap}},}$$

where

$$\Delta_\mu = m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty),$$

$$S_E = \xi_E g_E = \xi_E \frac{I_\theta}{C_E} \Lambda_E,$$

$$R_{\text{exact}} = \frac{\xi_c g_c}{\xi_E g_E} = \frac{\xi_c}{\xi_E} \frac{C_E}{C_c} \frac{3}{32} (\theta_0 \kappa)^2 \frac{\lambda_c}{\Lambda_E},$$

$$\Delta_{\text{gap}} = \Delta_{\text{bench},\rho}^{\text{fin}} - \eta_{R,\rho}.$$

Therefore Gate 5 passes exactly when

$$\boxed{\Delta_\mu > 0, \quad S_E > 0, \quad -1 < R_{\text{exact}} < 1, \quad \Delta_{\text{gap}} > 0.}$$

On the simplest vacuum + pure-even branch, this becomes

$$\boxed{m(\sigma_\infty) > 0, \quad \xi_E \frac{I_\theta}{C_E} \Lambda_E > 0, \quad -1 < \frac{\xi_c}{\xi_E} \frac{C_E}{C_c} \frac{3}{32} (\theta_0 \kappa)^2 \frac{\lambda_c}{\Lambda_E} < 1, \quad \Delta_{\text{bench},\rho}^{\text{fin}} > \eta_{R,\rho}.}$$

This is the actual termination point of the current line: only finite branch insertion remains.

LIV. EXACT SIGN AUDIT FOR Δ_μ AND S_E

LV. GOAL

Inside the current one-defect acceptance line, the exact finite audit has already been reduced to

$$(\Delta_\mu, S_E, R_{\text{exact}}, \Delta_{\text{gap}}).$$

In this section we close the first two sign problems:

$$\Delta_\mu = m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty), \quad S_E = \xi_E g_E.$$

LVI. 1. BULK SIGN SCALAR Δ_μ

The exact bulk mass parameter is

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)} = \frac{\Delta_\mu}{z_0(\sigma_\infty)}. \quad (136)$$

Assume the standard sign convention

$$z_0(\sigma_\infty) > 0. \quad (137)$$

Then

$$\mu_\infty > 0 \iff \Delta_\mu > 0. \quad (138)$$

Therefore the exact bulk sign table is:

$$\Delta_\mu > 0 \iff \mu_\infty > 0, \quad \Delta_\mu = 0 \iff \mu_\infty = 0, \quad \Delta_\mu < 0 \iff \mu_\infty < 0. \quad (139)$$

A. Branch-resolved cases

Because

$$\Delta_\mu = m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty),$$

there are three honest asymptotic sign patterns.

a. Case A: $u_X(\sigma_\infty) \geq 0$, $\chi_\infty \geq 0$. Then the Class-II correction reduces the bulk gap, and one needs

$$m(\sigma_\infty) > 3\chi_\infty u_X(\sigma_\infty) \geq 0. \quad (140)$$

b. Case B: $u_X(\sigma_\infty) \leq 0$, $\chi_\infty \geq 0$. Then the correction helps the bulk gap, so the exact condition is only

$$m(\sigma_\infty) > 3\chi_\infty u_X(\sigma_\infty), \quad (141)$$

which is weaker than $m(\sigma_\infty) > 0$.

c. Case C: sign-indefinite asymptotic branch. Then there is no further universal simplification:

$$\text{sgn } \Delta_\mu = \text{sgn} \left(m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) \right). \quad (142)$$

B. Simplest vacuum branch

If the intended asymptotic branch is the simplest vacuum branch,

$$\boxed{\chi_\infty = 0,} \quad (143)$$

then

$$\boxed{\Delta_\mu = m(\sigma_\infty), \quad \mu_\infty > 0 \iff m(\sigma_\infty) > 0.} \quad (144)$$

So on that branch the bulk gate is no longer a balance problem. It is a one-sign problem.

LVII. 2. TRANSVERSE SUPPORT SCALAR S_E

The exact normalization layer is

$$\boxed{\alpha = I_\theta \Lambda_E, \quad g_E = \frac{\alpha}{C_E}.} \quad (145)$$

Hence

$$\boxed{S_E := \xi_E g_E = \xi_E \frac{I_\theta}{C_E} \Lambda_E.} \quad (146)$$

For Branch B the exact overlap coefficient is

$$\boxed{I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa.} \quad (147)$$

Therefore

$$\boxed{S_E = -\frac{\sqrt{3}\pi}{16} \frac{\xi_E}{C_E} \theta_0 \kappa \Lambda_E.} \quad (148)$$

Since $\sqrt{3}\pi/16 > 0$, the exact sign law is

$$\boxed{\text{sgn } S_E = -\text{sgn}\left(\frac{\xi_E}{C_E} \theta_0 \kappa \Lambda_E\right).} \quad (149)$$

Thus the exact sign table is:

$$\boxed{\frac{\xi_E}{C_E} \theta_0 \kappa \Lambda_E < 0 \iff S_E > 0,} \quad (150)$$

$$\boxed{\frac{\xi_E}{C_E} \theta_0 \kappa \Lambda_E = 0 \iff S_E = 0,} \quad (151)$$

$$\boxed{\frac{\xi_E}{C_E} \theta_0 \kappa \Lambda_E > 0 \iff S_E < 0.} \quad (152)$$

LVIII. 3. WORKING-BRANCH PACKAGE FOR S_E

The transverse branch-level reduced package is

$$\Lambda_E = \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_E. \quad (153)$$

Substituting (153) into (148) gives

$$S_E = -\frac{\sqrt{3}\pi}{16} \frac{\xi_E}{C_E} \theta_0 \kappa \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_E. \quad (154)$$

Hence

$$\text{sgn } S_E = -\text{sgn}\left(\frac{\xi_E}{C_E} \theta_0 \kappa \alpha_X \beta_X \mathfrak{T}_E\right), \quad (155)$$

since $M_X^2 > 0$ does not affect the sign.

Therefore the transverse sign problem is fully reduced to the finite sign audit of

$$\xi_E, \quad C_E, \quad \theta_0 \kappa, \quad \alpha_X, \quad \beta_X, \quad \mathfrak{T}_E.$$

LIX. 4. EXPLICIT PURE-EVEN WORKING REALIZATION

There is also a sharper practical witness realization:

$$\alpha = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad \beta = 0. \quad (156)$$

Therefore

$$g_E = \frac{\alpha}{C_E} = \frac{1}{C_E} \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa. \quad (157)$$

and

$$S_E = \frac{\xi_E}{C_E} \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa. \quad (158)$$

Hence the practical sign law on this explicit pure-even realization is

$$\text{sgn } S_E = \text{sgn}\left(\frac{\xi_E}{C_E} \beta_X \theta_0 \kappa\right). \quad (159)$$

In the canonical practical choice

$$C_E = 1, \quad \xi_E > 0, \quad (160)$$

this collapses to

$$S_E > 0 \iff \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0. \quad (161)$$

LX. 5. HONEST RELATION BETWEEN THE TWO SIGN FORMULAS

The formulas (149) and (159) are not contradictory. They correspond to two different insertion levels:

1. the exact normalization-aware overlap formula, written in terms of Λ_E ;
2. the practical explicit pure-even witness realization, where the branch-specific sign of Λ_E has already been absorbed into the chosen realization.

Therefore the exact sign driver is always Λ_E , but the practical explicit pure-even witness may reduce this further to the simpler $\beta_X \theta_0 \kappa$ sign test.

LXI. 6. PARTIAL GATE-5 CERTIFICATE NOW OBTAINED

At this stage, the first two exact sign audits are completely reduced:

$$\boxed{\Delta_\mu > 0 \iff m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) > 0 \quad (z_0(\sigma_\infty) > 0),} \quad (162)$$

$$\boxed{S_E > 0 \iff -\frac{\xi_E}{C_E} \theta_0 \kappa \Lambda_E > 0,} \quad (163)$$

or at the working-branch package level,

$$\boxed{S_E > 0 \iff -\frac{\xi_E}{C_E} \theta_0 \kappa \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_E > 0.} \quad (164)$$

On the simplest vacuum + explicit pure-even practical realization, these become

$$\boxed{m(\sigma_\infty) > 0, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0 \quad (C_E = 1, \xi_E > 0).} \quad (165)$$

Thus the remaining Gate-5 work inside the current line is only:

$$R_{\text{exact}} \quad \text{and} \quad \Delta_{\text{gap}}.$$

LXII. FINAL SUMMARY

The first two exact sign variables of the current Gate-5 finite audit are

$$\Delta_\mu = m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty), \quad S_E = \xi_E g_E.$$

The bulk sign is exact:

$$\mu_\infty = \frac{\Delta_\mu}{z_0(\sigma_\infty)}.$$

Hence, under $z_0(\sigma_\infty) > 0$,

$$\boxed{\mu_\infty > 0 \iff \Delta_\mu > 0.}$$

On the simplest vacuum branch $\chi_\infty = 0$,

$$\boxed{\Delta_\mu = m(\sigma_\infty), \quad \mu_\infty > 0 \iff m(\sigma_\infty) > 0.}$$

For the transverse support scalar,

$$S_E = \xi_E \frac{I_\theta}{C_E} \Lambda_E, \quad I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa,$$

so

$$S_E = -\frac{\sqrt{3}\pi}{16} \frac{\xi_E}{C_E} \theta_0 \kappa \Lambda_E.$$

Therefore

$$S_E > 0 \iff \frac{\xi_E}{C_E} \theta_0 \kappa \Lambda_E < 0.$$

At the working-branch package level,

$$\Lambda_E = \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_E,$$

hence

$$S_E > 0 \iff \frac{\xi_E}{C_E} \theta_0 \kappa \alpha_X \beta_X \mathfrak{T}_E < 0.$$

On the explicit pure-even practical realization,

$$\alpha = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad \beta = 0,$$

so under $C_E = 1$ and $\xi_E > 0$,

$$S_E > 0 \iff \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0.$$

Thus the current line has now reduced the first two Gate-5 sign problems completely. Only the exact burden ratio R_{exact} and the remote-gap margin Δ_{gap} remain.

LXIII. EXACT SIGN/SIZE AUDIT FOR R_{exact}

LXIV. GOAL

Inside the current Gate-5 finite audit, only two items remain after the sign audits of

$$\Delta_\mu \quad \text{and} \quad S_E.$$

The next one is the exact longitudinal burden ratio

$$R_{\text{exact}} := \frac{\xi_c g_c}{\xi_E g_E}.$$

We now reduce it completely to a finite branch scalar audit.

LXV. 1. EXACT NORMALIZATION-AWARE FORMULA

The exact overlap dictionary is

$$\lambda_{\parallel} = \frac{1}{2} I_{\theta^3} \lambda_c, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O, \quad (166)$$

together with

$$g_c = \frac{\lambda_{\parallel}}{C_c}, \quad g_E = \frac{\alpha}{C_E}, \quad g_O = \frac{\beta}{C_O}. \quad (167)$$

Hence

$$g_c = \frac{I_{\theta'^3}}{2C_c} \lambda_c, \quad g_E = \frac{I_{\theta}}{C_E} \Lambda_E. \quad (168)$$

Therefore

$$\begin{aligned} R_{\text{exact}} &= \frac{\xi_c g_c}{\xi_E g_E} \\ &= \frac{\xi_c}{\xi_E} \frac{\frac{I_{\theta'^3}}{2C_c} \lambda_c}{\frac{I_{\theta}}{C_E} \Lambda_E} \\ &= \frac{\xi_c}{\xi_E} \frac{C_E}{C_c} \frac{I_{\theta'^3}}{2I_{\theta}} \frac{\lambda_c}{\Lambda_E}. \end{aligned} \quad (169)$$

Using the explicit defect overlaps

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad (170)$$

one gets

$$\frac{I_{\theta'^3}}{2I_{\theta}} = \frac{3}{32} (\theta_0 \kappa)^2. \quad (171)$$

Substituting this into (169),

$$R_{\text{exact}} = \frac{\xi_c}{\xi_E} \frac{C_E}{C_c} \frac{3}{32} (\theta_0 \kappa)^2 \frac{\lambda_c}{\Lambda_E}. \quad (172)$$

This is the exact normalization-aware burden ratio.

LXVI. 2. EXACT SIGN LAW

We work inside the already surviving transverse-support sector, so

$$S_E = \xi_E g_E \neq 0 \quad \implies \quad \Lambda_E \neq 0.$$

Then the sign of R_{exact} is well-defined.

Since

$$\frac{3}{32} > 0, \quad (\theta_0 \kappa)^2 \geq 0,$$

the defect sign does not enter the sign of R_{exact} . More precisely, for $\theta_0 \kappa \neq 0$,

$$\text{sgn } R_{\text{exact}} = \text{sgn} \left(\frac{\xi_c}{\xi_E} \frac{C_E}{C_c} \frac{\lambda_c}{\Lambda_E} \right). \quad (173)$$

Hence the exact sign table is:

$$\frac{\xi_c}{\xi_E} \frac{C_E}{C_c} \frac{\lambda_c}{\Lambda_E} > 0 \iff R_{\text{exact}} > 0, \quad (174)$$

$$\boxed{\lambda_c = 0 \iff R_{\text{exact}} = 0 \quad (\Lambda_E \neq 0),} \quad (175)$$

$$\boxed{\frac{\xi_c}{\xi_E} \frac{C_E}{C_c} \frac{\lambda_c}{\Lambda_E} < 0 \iff R_{\text{exact}} < 0.} \quad (176)$$

Thus the burden sign is controlled only by the relative sign of the longitudinal and transverse source packages, dressed by the matching/normalization factors.

LXVII. 3. EXACT SIZE WINDOW

In the practically relevant lowest-channel case, Gate 5 requires

$$-1 < R_{\text{exact}} < 1.$$

Using (172), this is equivalent to

$$\boxed{\left| \frac{\xi_c}{\xi_E} \frac{C_E}{C_c} \frac{3}{32} (\theta_0 \kappa)^2 \frac{\lambda_c}{\Lambda_E} \right| < 1.} \quad (177)$$

Equivalently,

$$\boxed{\left| \frac{\lambda_c}{\Lambda_E} \right| < \frac{32}{3} \left| \frac{\xi_E}{\xi_c} \right| \left| \frac{C_c}{C_E} \right| \frac{1}{(\theta_0 \kappa)^2}.} \quad (178)$$

So the exact burden window is a finite upper bound on the source ratio $|\lambda_c/\Lambda_E|$.

LXVIII. 4. AUTOMATIC ONE-SIDED PASSES

Equation (177) immediately gives two useful one-sided reductions.

a. Case A: $R_{\text{exact}} \geq 0$. Then the lower bound $R_{\text{exact}} > -1$ is automatic, and Gate 5 only requires

$$\boxed{R_{\text{exact}} < 1.} \quad (179)$$

b. Case B: $R_{\text{exact}} \leq 0$. Then the upper bound $R_{\text{exact}} < 1$ is automatic, and Gate 5 only requires

$$\boxed{R_{\text{exact}} > -1.} \quad (180)$$

Thus the sign audit tells us which side of the window is nontrivial.

LXIX. 5. WORKING-BRANCH SOURCE INSERTION

The current longitudinal finite scalar reduction gives

$$\boxed{\lambda_c^{(JJ)} \sim \frac{\alpha_X^2}{M_X^2} \mathcal{N}_{JJ} m_{\parallel} u_{\parallel} |G_*| A_{G_*},} \quad (181)$$

while the transverse branch-level package is

$$\boxed{\Lambda_E = \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_E.} \quad (182)$$

Therefore, at the current leading working-branch level,

$$\boxed{\frac{\lambda_c}{\Lambda_E} \sim \frac{\alpha_X}{\beta_X} \frac{\mathcal{N}_{JJ} m_{\parallel} u_{\parallel} |G_*| A_{G_*}}{\mathfrak{T}_E}.} \quad (183)$$

Substituting into (172),

$$R_{\text{exact}} \sim \frac{\xi_c}{\xi_E} \frac{C_E}{C_c} \frac{3}{32} (\theta_0 \kappa)^2 \frac{\alpha_X}{\beta_X} \frac{\mathcal{N}_{JJ} m_{\parallel} u_{\parallel}}{\mathfrak{T}_E} \frac{|G_*| A_{G_*}}{\mathfrak{T}_E}. \quad (184)$$

Hence the exact sign law becomes

$$\text{sgn } R_{\text{exact}} = \text{sgn} \left(\frac{\xi_c}{\xi_E} \frac{C_E}{C_c} \frac{\alpha_X}{\beta_X} \frac{\mathcal{N}_{JJ} m_{\parallel} u_{\parallel}}{\mathfrak{T}_E} \frac{|G_*| A_{G_*}}{\mathfrak{T}_E} \right), \quad (185)$$

and the exact size window becomes

$$\left| \frac{\xi_c}{\xi_E} \frac{C_E}{C_c} \frac{3}{32} (\theta_0 \kappa)^2 \frac{\alpha_X}{\beta_X} \frac{\mathcal{N}_{JJ} m_{\parallel} u_{\parallel}}{\mathfrak{T}_E} \frac{|G_*| A_{G_*}}{\mathfrak{T}_E} \right| < 1. \quad (186)$$

So the longitudinal frontier is fully reduced to the finite sign/size audit of

$$\mathcal{N}_{JJ}, \quad m_{\parallel}, \quad u_{\parallel}, \quad A_{G_*}, \quad \mathfrak{T}_E, \quad \text{and the calibration constants } C_c, C_E, \xi_c, \xi_E.$$

LXX. 6. EXPLICIT PRACTICAL SPECIALIZATION

In the practical explicit branch summary, the burden was already reduced to

$$R_c = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2, \quad (187)$$

with the acceptance window

$$\left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3 |\alpha_X| (\theta_0 \kappa)^2}. \quad (188)$$

This is precisely the specialization of the exact formula (172) to the explicit branch-level normalization choice in which the source ratio

$$\frac{C_E}{C_c} \frac{\lambda_c}{\Lambda_E}$$

collapses to α_X .

LXXI. 7. PARTIAL GATE-5 CERTIFICATE AFTER THE THIRD AUDIT

At this point, three of the four Gate-5 slots are completely reduced:

$$\Delta_{\mu} > 0 \iff m(\sigma_{\infty}) - 3\chi_{\infty} u_X(\sigma_{\infty}) > 0 \quad (z_0(\sigma_{\infty}) > 0), \quad (189)$$

$$S_E > 0 \iff -\frac{\xi_E}{C_E} \theta_0 \kappa \Lambda_E > 0, \quad (190)$$

$$-1 < R_{\text{exact}} < 1 \iff \left| \frac{\xi_c}{\xi_E} \frac{C_E}{C_c} \frac{3}{32} (\theta_0 \kappa)^2 \frac{\lambda_c}{\Lambda_E} \right| < 1. \quad (191)$$

Therefore only the support-theorem side remains:

$$\Delta_{\text{gap}} = \Delta_{\text{bench}, \rho}^{\text{fin}} - \eta_{R, \rho}.$$

No additional structural work remains inside the present line beyond that final support insertion.

LXXII. FINAL SUMMARY

The exact normalization-aware longitudinal burden ratio is

$$R_{\text{exact}} = \frac{\xi_c g_c}{\xi_E g_E} = \frac{\xi_c}{\xi_E} \frac{C_E}{C_c} \frac{3}{32} (\theta_0 \kappa)^2 \frac{\lambda_c}{\Lambda_E}.$$

Hence, inside the already surviving transverse-support sector $\Lambda_E \neq 0$,

$$\text{sgn } R_{\text{exact}} = \text{sgn} \left(\frac{\xi_c}{\xi_E} \frac{C_E}{C_c} \frac{\lambda_c}{\Lambda_E} \right).$$

So the defect-sign information drops out: only the relative sign of the longitudinal and transverse source packages matters.

The exact lowest-channel burden window is

$$-1 < R_{\text{exact}} < 1 \iff \left| \frac{\xi_c}{\xi_E} \frac{C_E}{C_c} \frac{3}{32} (\theta_0 \kappa)^2 \frac{\lambda_c}{\Lambda_E} \right| < 1.$$

Equivalently,

$$\left| \frac{\lambda_c}{\Lambda_E} \right| < \frac{32}{3} \left| \frac{\xi_E}{\xi_c} \right| \frac{C_c}{C_E} \frac{1}{(\theta_0 \kappa)^2}.$$

At the working-branch leading level,

$$\lambda_c^{(JJ)} \sim \frac{\alpha_X^2}{M_X^2} \mathcal{N}_{JJ} m_{\parallel} u_{\parallel} |G_*| A_{G_*}, \quad \Lambda_E = \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_E,$$

so

$$R_{\text{exact}} \sim \frac{\xi_c}{\xi_E} \frac{C_E}{C_c} \frac{3}{32} (\theta_0 \kappa)^2 \frac{\alpha_X}{\beta_X} \frac{\mathcal{N}_{JJ} m_{\parallel} u_{\parallel} |G_*| A_{G_*}}{\mathfrak{T}_E}.$$

Thus the third Gate-5 slot is fully reduced to a finite sign/size audit.

After this, only the remote-gap support scalar remains:

$$\Delta_{\text{gap}} = \Delta_{\text{bench}, \rho}^{\text{fin}} - \eta_{R, \rho}.$$

LXXIII. EXACT PASS/CERTIFICATE TABLE FOR Δ_{gap}

LXXIV. GOAL

We now close the last remaining scalar in the current one-defect acceptance line:

$$\Delta_{\text{gap}} := \Delta_{\text{bench}, \rho}^{\text{fin}} - \eta_{R, \rho}.$$

The purpose is not to restate the support theorem, but to convert it into a finite pass/certificate table for the intended explicit branch.

LXXV. 1. EXACT SUPPORT THEOREM FORM

The remote-gap support theorem has the fixed form

$$\Delta_{\text{bench}, \rho}^{\text{fin}} > \eta_{R, \rho} \implies \Delta_{\rho} \geq \Delta_{\text{bench}, \rho}^{\text{fin}} - \eta_{R, \rho} > 0. \quad (192)$$

Equivalently,

$$\Delta_{\text{gap}} > 0 \implies \Delta_{\rho} > 0. \quad (193)$$

Thus the final scalar in the present line is

$$\Delta_{\text{gap}} = \Delta_{\text{bench}, \rho}^{\text{fin}} - \eta_{R, \rho}. \quad (194)$$

LXXVI. 2. SECTORWISE BENCHMARK DECOMPOSITION

The working finite benchmark is

$$\Delta_{\text{bench},\rho}^{\text{fin}} = \min \left\{ 4K - |r_*|, \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2}, q_* \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)} \right\}. \quad (195)$$

Define the four benchmark slots

$$B_{\text{high}} := 4K - |r_*|, \quad B_{\text{vol}} := \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2}, \quad (196)$$

$$B_{\text{int}} := q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \quad B_{\text{non}} := \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)}. \quad (197)$$

Then

$$\Delta_{\text{bench},\rho}^{\text{fin}} = \min\{B_{\text{high}}, B_{\text{vol}}, B_{\text{int}}, B_{\text{non}}\}. \quad (198)$$

LXXVII. 3. RESIDUAL BUDGET DECOMPOSITION

The residual side satisfies

$$\eta_{R,\rho} \leq \eta_{R,\rho}^{\text{tr}} + \eta_{\text{tail},\rho} + \eta_{\text{diag},\rho}. \quad (199)$$

Hence any explicit upper budget

$$E_{\text{rem}} \geq \eta_{R,\rho} \quad (200)$$

yields the sufficient implication

$$\min\{B_{\text{high}}, B_{\text{vol}}, B_{\text{int}}, B_{\text{non}}\} > E_{\text{rem}} \implies \Delta_{\text{gap}} > 0. \quad (201)$$

This is the correct certificate form because E_{rem} is an upper bound, not the exact remote norm in general.

LXXVIII. 4. CANONICAL PURE-EVEN EXPLICIT BRANCH

For the explicit canonical pure-even branch, the benchmark and residual sides have already been assembled into one fully explicit sufficient inequality:

$$\begin{aligned} & \min \left\{ 4K - |r_*|, \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2}, q_* \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Xi_{\text{non}}^{(1)}(\rho) \right\} \\ & > 2 \left[\kappa \rho (|C_E| + |C_O|) + \frac{1}{M_X^2} (\alpha_X^2 + |\alpha_X \beta_X| + \beta_X^2) \right] + 2C_{\text{high},\text{non}}\rho + \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a} + \eta_{\text{diag},\rho}. \end{aligned} \quad (202)$$

Here the same-shell benchmark slot is explicitly

$$\Xi_{\text{non}}^{(1)}(\rho) := \left| -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16} + 12uA^2 \right| - 4uA^2 - 2a_K \mathcal{S}_{\text{diag}}^{EE} - \varepsilon_{\text{small-}p,\rho}^{(1)} - \varepsilon_{\text{non-enr},\rho}^{(1)}. \quad (203)$$

If the pure-even branch is also enforced on the residual transport side, one may sharpen

$$|C_E| + |C_O| \longrightarrow |C_E|. \quad (204)$$

LXXIX. 5. EXACT PASS/CERTIFICATE TABLE

We now rewrite (202) into a direct decision table.
Define

$$E_{\text{rem}}^{(\text{even})} := 2 \left[\kappa \rho (|C_E| + |C_O|) + \frac{1}{M_X^2} (\alpha_X^2 + |\alpha_X \beta_X| + \beta_X^2) \right] + 2C_{\text{high,non}} \rho + \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a} + \eta_{\text{diag}, \rho}. \quad (205)$$

Then the explicit branch certificate is

$$\min\{B_{\text{high}}, B_{\text{vol}}, B_{\text{int}}, \Xi_{\text{non}}^{(1)}(\rho)\} > E_{\text{rem}}^{(\text{even})}. \quad (206)$$

Since

$$\min\{x_1, x_2, x_3, x_4\} > y \iff x_i > y \text{ for all } i,$$

(206) is equivalent to the four simultaneous inequalities

$$B_{\text{high}} > E_{\text{rem}}^{(\text{even})}, \quad (207)$$

$$B_{\text{vol}} > E_{\text{rem}}^{(\text{even})}, \quad (208)$$

$$B_{\text{int}} > E_{\text{rem}}^{(\text{even})}, \quad (209)$$

$$\Xi_{\text{non}}^{(1)}(\rho) > E_{\text{rem}}^{(\text{even})}. \quad (210)$$

Therefore the exact explicit pass table is:

a. Certified pass. If (207)–(210) all hold, then

$$\Delta_{\text{gap}} > 0, \quad \Delta_{\rho} > 0.$$

b. Certificate failure / inconclusive. If one or more of (207)–(210) is not verified, then the support certificate is not established from this bound.

Importantly, this is *not* yet a proof of actual Gate-1 failure, because

$$E_{\text{rem}}^{(\text{even})}$$

is only an upper bound on the true remote norm.

c. Definite failure. A definite failure requires either:

1. the actual norm $\eta_{R,\rho}$ to be computed and found to satisfy

$$\Delta_{\text{bench}, \rho}^{\text{fin}} \leq \eta_{R,\rho},$$

or

2. an alternative spectral obstruction, e.g. a negative/nonpositive Gershgorin lower bound at the actual remote block level.

LXXX. 6. SECTORWISE INTERPRETATION

The four explicit slots have distinct meanings:

1. B_{high} : high-shell separation;
2. B_{vol} : heavy volumetric isolation;
3. B_{int} : wrong-internal exclusion;
4. $\Xi_{\text{non}}^{(1)}(\rho)$: same-shell non-enriched exclusion with small- p drift and same-shell correction budgets included.

Hence the support gate passes only if *all* four remote threats stay above the same residual budget.

LXXXI. 7. FINAL TERMINATION OF THE CURRENT LINE

At this point the current one-defect line is fully reduced to the four scalar tests

$$\boxed{\Delta_\mu > 0, \quad S_E > 0, \quad -1 < R_{\text{exact}} < 1, \quad \Delta_{\text{gap}} > 0.} \quad (211)$$

The first three have already been reduced to finite sign/size audits. The fourth is now reduced to the explicit certificate table (207)–(210).

Therefore no structural theorem-building remains inside the present line. Only explicit finite insertion of the intended branch constants remains.

LXXXII. 8. HONEST STATUS

The current files already state that:

1. the first three benchmark entries are explicit or nearly explicit;
2. the genuinely model-specific unresolved pieces are the same-shell non-enriched benchmark and the remote-mixing norm budget.

So the present line is genuinely finished at the level of exact reduction. What remains is:

$$\Xi_{\text{non}}^{(1)}(\rho) \quad \text{and} \quad E_{\text{rem}}^{(\text{even})}$$

evaluated on the intended branch.

LXXXIII. FINAL SUMMARY

The final support scalar in the current one-defect line is

$$\Delta_{\text{gap}} = \Delta_{\text{bench},\rho}^{\text{fin}} - \eta_{R,\rho}.$$

The exact support theorem is

$$\Delta_{\text{bench},\rho}^{\text{fin}} > \eta_{R,\rho} \implies \Delta_\rho \geq \Delta_{\text{bench},\rho}^{\text{fin}} - \eta_{R,\rho} > 0.$$

The finite benchmark side is

$$\Delta_{\text{bench},\rho}^{\text{fin}} = \min \left\{ 4K - |r_*|, \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2}, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)} \right\}.$$

For the explicit canonical pure-even branch, this becomes the fully assembled sufficient inequality

$$\min \left\{ B_{\text{high}}, B_{\text{vol}}, B_{\text{int}}, \Xi_{\text{non}}^{(1)}(\rho) \right\} > E_{\text{rem}}^{(\text{even})},$$

where

$$E_{\text{rem}}^{(\text{even})} = 2 \left[\kappa \rho (|C_E| + |C_O|) + \frac{1}{M_X^2} (\alpha_X^2 + |\alpha_X \beta_X| + \beta_X^2) \right] + 2C_{\text{high,non}}\rho + \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a} + \eta_{\text{diag},\rho},$$

and the same-shell slot is

$$\Xi_{\text{non}}^{(1)}(\rho) = \left| -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16} + 12uA^2 \right| - 4uA^2 - 2a_K \mathcal{S}_{\text{diag}}^{EE} - \varepsilon_{\text{small-}p,\rho}^{(1)} - \varepsilon_{\text{non-enr},\rho}^{(1)}.$$

Equivalently, the support certificate passes exactly when all four sectorwise inequalities hold:

$$B_{\text{high}} > E_{\text{rem}}^{(\text{even})}, \quad B_{\text{vol}} > E_{\text{rem}}^{(\text{even})}, \quad B_{\text{int}} > E_{\text{rem}}^{(\text{even})}, \quad \Xi_{\text{non}}^{(1)}(\rho) > E_{\text{rem}}^{(\text{even})}.$$

This is a certified pass criterion.

If these inequalities are not verified, the present support theorem becomes inconclusive, but actual Gate-1 failure is not yet proved unless the true remote norm is computed or an alternative spectral obstruction is established.

Therefore the current one-defect line is now fully reduced to the finite four-scalar audit

$$\Delta_\mu, \quad S_E, \quad R_{\text{exact}}, \quad \Delta_{\text{gap}},$$

with the last slot now converted into an explicit sectorwise certificate table.

LXXXIV. CARRIER-LEVEL EXPLICIT BRANCH ADMISSIBILITY CERTIFICATE

LXXXV. 1. WHAT COUNTS AS BRANCH ADMISSIBILITY

The first correction is logical.

Post-acceptance Gate 5 is *not* the same as carrier-level branch admissibility. The working notes state explicitly that only after Gates 1–4 pass does it make sense to test the downstream one-defect / one-mode realization.

Therefore the correct notion of explicit branch admissibility is the carrier-level acceptance problem.

$$\boxed{\text{explicit branch admissibility} \equiv \text{carrier-level acceptance.}} \quad (212)$$

The integrated acceptance logic is already fixed:

$$\boxed{\text{G2 passes} \wedge \text{G3 passes} \wedge \text{G4 passes.}} \quad (213)$$

Equivalently,

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0), \quad A_{\text{res}} \cap \mathcal{M}_D = \{0\}.} \quad (214)$$

LXXXVI. 2. GATE 2: LONGITUDINAL SURVIVAL

Gate 2 is already reduced to a finite executable audit.

The basis-level overlap criterion is

$$\boxed{\ell_{\parallel A} = E_{A,ij}^3 \hat{n}_i \hat{n}_j, \quad \exists A : \ell_{\parallel A} \neq 0.} \quad (215)$$

This is equivalent to the microscopic seed-survival statement

$$\boxed{c_{\parallel}^{(\text{mic})} \neq 0.} \quad (216)$$

At the graded/radial level, the survival test becomes

$$\boxed{J_{\parallel}^{(3)} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) s_3(\xi) \neq 0.} \quad (217)$$

Finally, after Bloch calibration,

$$\boxed{\lambda_{\parallel} = C_c g_c, \quad C_c = \frac{1}{2} \text{Tr}(\sigma_3 M_{(c)}^{\parallel}),} \quad (218)$$

so the final calibration condition is

$$\boxed{C_c \neq 0.} \quad (219)$$

Therefore the complete Gate-2 pass condition is

$$\boxed{\exists A : \ell_{\parallel A} \neq 0 \iff c_{\parallel}^{(\text{mic})} \neq 0, \quad J_{\parallel}^{(3)} \neq 0, \quad C_c \neq 0.} \quad (220)$$

LXXXVII. 3. GATE 3: TRANSVERSE SURVIVAL

At the working-branch level, Gate 3 is also reduced to a finite explicit audit. The strongest practical explicit Class II branch test is

$$\boxed{A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0.} \quad (221)$$

Transport survival alone is not enough. The physical coefficients still require Bloch calibration:

$$\boxed{\alpha = C_E g_E, \quad \beta = C_O g_O.} \quad (222)$$

Hence the final physical pass condition is

$$\boxed{(C_E, C_O) \neq (0, 0).} \quad (223)$$

Equivalently,

$$\boxed{(c_E, c_O) \neq (0, 0), \quad (C_E, C_O) \neq (0, 0) \implies (\alpha, \beta) \neq (0, 0).} \quad (224)$$

So the practical Gate-3 pass certificate is

$$\boxed{A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0, \quad (C_E, C_O) \neq (0, 0).} \quad (225)$$

LXXXVIII. 4. GATE 4: MASS PROTECTION

The exact constant Dirac-mass space is

$$\boxed{\mathcal{M}_D = \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}.} \quad (226)$$

The exact Gate-4 condition is

$$\boxed{A_{\text{res}} \cap \mathcal{M}_D = \{0\}.} \quad (227)$$

A sufficient mechanism is exact residual valley symmetry:

$$\boxed{U(1)_V \quad \text{or} \quad \mathbb{Z}_2^V.} \quad (228)$$

Therefore Gate 4 passes iff either:

1. exact residual $U(1)_V$ survives;
2. exact residual $\mathbb{Z}_2^V$ survives;
3. or explicit computation proves $A_{\text{res}} \cap \mathcal{M}_D = \{0\}$.

LXXXIX. 5. CARRIER-LEVEL EXPLICIT BRANCH ADMISSIBILITY THEOREM

Combining (213), (220), (225), and (227), we obtain:

[Explicit branch admissibility certificate] For the refined locked + Class II explicit working branch, a sufficient carrier-level admissibility certificate is:

$$\boxed{\exists A : \ell_{\parallel A} \neq 0, \quad J_{\parallel}^{(3)} \neq 0, \quad C_c \neq 0,} \quad (229)$$

$$\boxed{A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0, \quad (C_E, C_O) \neq (0, 0),} \quad (230)$$

$$\boxed{A_{\text{res}} \cap \mathcal{M}_D = \{0\}}, \quad (231)$$

or, sufficiently,

$$\boxed{U(1)_V \text{ survives} \quad \text{or} \quad \mathbb{Z}_2^V \text{ survives.}} \quad (232)$$

Under these conditions,

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0), \quad A_{\text{res}} \cap \mathcal{M}_D = \{0\}}, \quad (233)$$

hence the branch is accepted as a viable microscopic massless Dirac carrier.

XC. 6. RELATION TO GATE 1 AND GATE 5

Gate 1 is a support theorem in the same parameter regime:

$$\boxed{\Delta_{\text{bench}, \rho}^{\text{fin}} > \eta_{R, \rho}}. \quad (234)$$

It is already structurally closed and reduced to finite explicit verification.

Gate 5 is downstream:

$$\boxed{-1 < R_c < 1, \quad \alpha > 0, \quad \mu_{\infty} > 0, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho}}. \quad (235)$$

So Gate 5 is not part of the carrier-level admissibility certificate. It is the one-defect realization test after branch acceptance.

XCI. 7. HONEST ENDPOINT

What is now closed:

1. the exact logical meaning of explicit branch admissibility;
2. the finite carrier-level certificate (G2, G3, G4);
3. the separation between carrier-level admissibility and post-acceptance Gate 5.

What is still not explicitly inserted on the intended branch:

$$\boxed{C_c, \quad C_E, \quad C_O, \quad J_{\parallel}^{(3)}, \quad A_{G_*}(C_1 \cdot G_*), \quad A_{G_*}(C_2 \cdot G_*)}, \quad (236)$$

and the residual-symmetry computation for A_{res} .

Therefore the current honest status is:

$$\boxed{\text{explicit branch admissibility is reduced to a finite executable certificate,}} \quad (237)$$

but

$$\boxed{\text{the actual pass stamp still requires finite branch insertion.}} \quad (238)$$

XCII. FINAL SUMMARY

Explicit branch admissibility is not Gate 5. It is the carrier-level acceptance problem.
Its exact logic is

$$\boxed{\text{G2 passes} \wedge \text{G3 passes} \wedge \text{G4 passes.}}$$

Equivalently,

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0), \quad A_{\text{res}} \cap \mathcal{M}_D = \{0\}.}$$

For the refined locked + Class II explicit working branch, this reduces to the finite executable certificate:

$$\boxed{\exists A : \ell_{\parallel A} \neq 0, \quad J_{\parallel}^{(3)} \neq 0, \quad C_c \neq 0,}$$

$$\boxed{A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0, \quad (C_E, C_O) \neq (0, 0),}$$

$$\boxed{A_{\text{res}} \cap \mathcal{M}_D = \{0\},}$$

or sufficiently,

$$\boxed{U(1)_V \text{ survives} \quad \text{or} \quad \mathbb{Z}_2^V \text{ survives.}}$$

Gate 1 remains a support theorem in the same regime, while Gate 5 is the downstream one-defect realization test after branch acceptance.

Therefore:

$$\boxed{\text{the exact admissibility certificate is now known,}}$$

but

$$\boxed{\text{the actual explicit branch pass still requires finite insertion of branch data.}}$$

XCIII. ACTUAL RESIDUAL-SYMMETRY CLOSURE OF GATE 4

XCIV. GOAL

We now push Gate 4 to its actual endpoint inside the current line.

The theorem-level reduction was already known:

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\}$$

is the exact mass-protection criterion, and exact residual valley $U(1)_V$ or already exact $\mathbb{Z}_2^V$ is sufficient.

The only remaining question is whether the *actual corrected microscopic branch* preserves the same residual generator through all constant corrections and remote elimination.

XCV. 1. ACTUAL PROJECTED SHELL BLOCK AND ITS RESIDUAL GENERATOR

For the actual projected massless shell block, the principal-symbol core has the form

$$\boxed{H_{\text{core}}^{\text{lin}}(p) = v_{\parallel} p_{\parallel} \tau_3 \otimes \mathbf{1}_2 + \nu_1 p_1 \tau_1 \otimes s_1 + \nu_2 p_2 \tau_1 \otimes s_2.} \quad (239)$$

Its commutant is

$$\text{Comm}(\Gamma_1, \Gamma_2, \Gamma_3) = \text{span}_{\mathbb{R}}\{\mathbf{1}_4, Q_*\}, \quad Q_* = \tau_3 \otimes s_3. \quad (240)$$

Therefore the actual projected shell block carries an exact continuous residual symmetry

$$U(1)_V^{(\text{actual})} : \quad U_\theta = e^{i\theta Q_*/2}. \quad (241)$$

The weaker remnant

$$\mathbb{Z}_2^V : \quad S_V := Q_*, \quad S_V^2 = \mathbf{1}_4, \quad (242)$$

is automatically present as a subgroup.

XCVI. 2. ACTUAL CONSTANT MASS SPACE

The actual constant Dirac-mass space of this projected block is

$$\mathcal{M}_{\text{actual}} = \text{span}_{\mathbb{R}}\{M_1, M_2\}, \quad M_1 = \tau_2 \otimes \mathbf{1}_2, \quad M_2 = \tau_1 \otimes s_3. \quad (243)$$

Under the exact residual $U(1)_V^{(\text{actual})}$, these rotate as

$$U_\theta M_1 U_\theta^{-1} = \cos \theta M_1 + \sin \theta M_2, \quad (244)$$

$$U_\theta M_2 U_\theta^{-1} = -\sin \theta M_1 + \cos \theta M_2. \quad (245)$$

Hence no nonzero element of $\mathcal{M}_{\text{actual}}$ is invariant under the full $U(1)$:

$$A_{U(1)_V^{(\text{actual})}} \cap \mathcal{M}_{\text{actual}} = \{0\}. \quad (246)$$

Even the weaker discrete remnant already suffices, since

$$S_V M_1 S_V^{-1} = -M_1, \quad S_V M_2 S_V^{-1} = -M_2, \quad (247)$$

so

$$A_{\mathbb{Z}_2^V} \cap \mathcal{M}_{\text{actual}} = \{0\}. \quad (248)$$

Therefore the actual projected shell block is already mass-protected at the principal-symbol level.

XCVII. 3. HONEST REMAINING FRONTIER

The honest remaining frontier is not to discover a new symmetry. The symmetry has already been identified explicitly:

$$Q_* = \tau_3 \otimes s_3.$$

What remains is only whether the *full corrected microscopic block*, including all constant corrections and remote-sector elimination, preserves the same Q_* .

This is the precise meaning of the residual-symmetry audit.

XCVIII. 4. FULL CORRECTED CONSTANT BLOCK

Let the final corrected constant block be written schematically as

$$H_{\text{full}}^{\text{const}} = H_{\Psi\Psi}^{\text{const}} + \delta H_{\text{Class II}}^{\text{const}} + \Sigma(0, 0), \quad (249)$$

where:

- $H_{\Psi\Psi}^{\text{const}}$ is the bare projected constant block,
- $\delta H_{\text{Class II}}^{\text{const}}$ is the Class-II induced constant correction,
- $\Sigma(0, 0)$ is the remote Schur self-energy at zero energy and momentum.

The practical full-basis Gate-4 theorem states that if the same Q_* commutes with all three pieces, namely

$$[Q_*, H_{\Psi\Psi}^{\text{const}}] = 0, \quad (250)$$

$$[Q_*, \delta H_{\text{Class II}}^{\text{const}}] = 0, \quad (251)$$

$$[Q_*, \Sigma(0, 0)] = 0, \quad (252)$$

then

$$A_{\text{res}}^{(\text{full})} \cap \mathcal{M}_{\text{actual}} = \{0\}. \quad (253)$$

So the exact practical Gate-4 pass certificate is the finite commutator audit

$$[Q_*, H_{\Psi\Psi}^{\text{const}}] = 0, \quad [Q_*, \delta H_{\text{Class II}}^{\text{const}}] = 0, \quad [Q_*, \Sigma(0, 0)] = 0.$$

XCIX. 5. WHY THIS IS ENOUGH

If (250)–(252) hold, then the full corrected constant block is invariant under the same exact residual $U(1)_V^{(\text{actual})}$ generated by Q_* . Therefore every residual-symmetry-allowed constant operator belongs to the Q_* -commutant.

But the actual mass space $\mathcal{M}_{\text{actual}}$ transforms nontrivially under this $U(1)$, equivalently is odd under the $\mathbb{Z}_2$ remnant. Hence no nonzero element of $\mathcal{M}_{\text{actual}}$ can belong to the residual allowed algebra, and

$$A_{\text{res}}^{(\text{full})} \cap \mathcal{M}_{\text{actual}} = \{0\}.$$

Thus Gate 4 passes.

C. 6. THE SHARP LOGICAL STATUS

The exact logical status is therefore:

1. theorem-level Gate 4 was already reduced to the finite test

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\};$$

2. the actual principal-symbol block already exhibits the correct protecting symmetry Q_* ;
3. the only remaining microscopic work is to verify that the same Q_* survives all constant corrections and remote elimination.

So Gate 4 is no longer conceptually open. It is a finite commutator audit on the intended branch.

CI. 7. FINAL ACTUAL-BASIS GATE-4 THEOREM

[Actual-basis Gate-4 pass certificate] Assume the actual projected shell carrier block has principal-symbol generator

$$Q_* = \tau_3 \otimes s_3$$

and actual mass space

$$\mathcal{M}_{\text{actual}} = \text{span}_{\mathbb{R}}\{M_1, M_2\}.$$

If the full corrected constant block satisfies

$$[Q_*, H_{\Psi\Psi}^{\text{const}}] = 0, \quad [Q_*, \delta H_{\text{Class II}}^{\text{const}}] = 0, \quad [Q_*, \Sigma(0, 0)] = 0,$$

then

$$A_{\text{res}}^{(\text{full})} \cap \mathcal{M}_{\text{actual}} = \{0\},$$

and therefore Gate 4 passes.

The principal-symbol block has exact residual $U(1)_V^{(\text{actual})}$ generated by Q_* . The full corrected constant block preserves the same symmetry if all three commutators vanish. The actual mass space transforms as a nontrivial charged doublet under this $U(1)$, equivalently is odd under the $\mathbb{Z}_2$ remnant. Hence no nonzero constant mass lies in the residual allowed algebra. Therefore

$$A_{\text{res}}^{(\text{full})} \cap \mathcal{M}_{\text{actual}} = \{0\}.$$

CII. FINAL SUMMARY

Gate 4 is now reduced to an actual-basis finite commutator audit.

The actual projected massless shell block already carries an exact residual valley symmetry

$$Q_* = \tau_3 \otimes s_3, \quad U_\theta = e^{i\theta Q_*/2},$$

and its actual constant mass space is

$$\mathcal{M}_{\text{actual}} = \text{span}_{\mathbb{R}}\{M_1, M_2\}, \quad M_1 = \tau_2 \otimes \mathbf{1}_2, \quad M_2 = \tau_1 \otimes s_3.$$

Under this exact residual $U(1)$, the two mass generators rotate as a charged doublet, so

$$A_{U(1)_V^{(\text{actual})}} \cap \mathcal{M}_{\text{actual}} = \{0\}.$$

Already the weaker remnant $\mathbb{Z}_2^V$ suffices.

Therefore the actual projected shell block itself is mass-protected at the principal-symbol level.

The only remaining microscopic question is whether the same Q_* survives all constant corrections and remote elimination.

If

$$[Q_*, H_{\Psi\Psi}^{\text{const}}] = 0, \quad [Q_*, \delta H_{\text{Class II}}^{\text{const}}] = 0, \quad [Q_*, \Sigma(0, 0)] = 0,$$

then

$$A_{\text{res}}^{(\text{full})} \cap \mathcal{M}_{\text{actual}} = \{0\},$$

and Gate 4 passes.

So Gate 4 is no longer a symmetry-discovery problem. It is a finite verification problem for the preservation of an already identified residual generator.

CIII. CLOSURE OF THE REMOTE SCHUR PART OF GATE 4 VIA THE Q_R -INTERTWINER ROUTE

CIV. GOAL

Inside the current branch, Gate 4 has already been reduced to the triple commutator test

$$\boxed{[Q_*, H_{\Psi\Psi}^{\text{const}}] = 0, \quad [Q_*, \delta H_{\text{Class II}}^{\text{const}}] = 0, \quad [Q_*, \Sigma(0, 0)] = 0.} \quad (254)$$

For the present refined locked + Class II branch, the second commutator is already closed,

$$\boxed{[Q_*, \delta H_{\text{Class II}}^{\text{const}}] = 0,} \quad (255)$$

and after removal of the dangerous scalar identity-channel obstruction, the allowed low-block constant sector is

$$\boxed{H_{\Psi\Psi}^{\text{const}} = a_0 \mathbf{1}_4 + a_* Q_*,} \quad (256)$$

hence

$$\boxed{[Q_*, H_{\Psi\Psi}^{\text{const}}] = 0.} \quad (257)$$

Therefore the only remaining microscopic Gate-4 frontier is

$$\boxed{[Q_*, \Sigma(0, 0)] = 0.} \quad (258)$$

CV. 1. ACTUAL CANONICAL BASIS DATA

In the actual canonical shell basis,

$$\boxed{Q_* = \tau_3 \otimes s_3,} \quad (259)$$

and the actual constant mass space is

$$\boxed{\mathcal{M}_{\text{actual}} = \text{span}_{\mathbb{R}}\{M_1, M_2\}, \quad M_1 = \tau_2 \otimes \mathbf{1}_2, \quad M_2 = \tau_1 \otimes s_3.} \quad (260)$$

Let

$$\mathcal{C}_{Q_*} = \{K \in \text{Herm}(4) : [Q_*, K] = 0\}$$

be the Hermitian Q_* -commutant. Then

$$\boxed{\mathcal{C}_{Q_*} \cap \mathcal{M}_{\text{actual}} = \{0\}.} \quad (261)$$

Therefore every exact constant operator commuting with Q_* is automatically mass-protecting.

CVI. 2. BLOCK FORM OF THE FULL HAMILTONIAN

Write the full zero-momentum block Hamiltonian as

$$\boxed{H_{\text{full}}(0) = \begin{pmatrix} H_L(0) & V(0) \\ V(0)^\dagger & H_R(0) \end{pmatrix},} \quad (262)$$

where:

- $H_L(0)$ is the isolated low shell carrier block,
- $H_R(0)$ is the remote block,
- $V(0)$ is the constant low-remote mixing block.

The remote Schur self-energy is

$$\boxed{\Sigma(0, 0) = V(0)(-H_R(0))^{-1}V(0)^\dagger.} \quad (263)$$

CVII. 3. EXACT RESIDUAL INTERTWINER ROUTE

A sufficient executable Route II is the existence of a Hermitian generator Q_R on the remote sector such that

$$\boxed{[Q_* \oplus Q_R, H_{\text{full}}(0)] = 0.} \quad (264)$$

Blockwise this is equivalent to

$$\boxed{[Q_*, H_L(0)] = 0, \quad [Q_R, H_R(0)] = 0, \quad Q_* V(0) = V(0) Q_R.} \quad (265)$$

The practical statement in the actual-basis attachment is identical:

$$\boxed{[Q_*, H_L] = 0, \quad [Q_R, H_R] = 0, \quad Q_* W = W Q_R \implies [Q_*, \Sigma(0, 0)] = 0.} \quad (266)$$

CVIII. 4. DIRECT PROOF THAT $[Q_*, \Sigma(0, 0)] = 0$

Since $[Q_R, H_R(0)] = 0$, the remote resolvent also commutes with Q_R :

$$\boxed{Q_R H_R(0)^{-1} = H_R(0)^{-1} Q_R.} \quad (267)$$

Now compute:

$$\begin{aligned} Q_* \Sigma(0, 0) &= Q_* V(0) (-H_R(0))^{-1} V(0)^\dagger \\ &= V(0) Q_R (-H_R(0))^{-1} V(0)^\dagger \\ &= V(0) (-H_R(0))^{-1} Q_R V(0)^\dagger \\ &= V(0) (-H_R(0))^{-1} V(0)^\dagger Q_* \\ &= \Sigma(0, 0) Q_*. \end{aligned} \quad (268)$$

Therefore

$$\boxed{[Q_*, \Sigma(0, 0)] = 0.} \quad (269)$$

Hence

$$\boxed{\Sigma(0, 0) \in \mathcal{C}_{Q_*}.} \quad (270)$$

Using (261), this implies

$$\boxed{\Sigma(0, 0) \cap \mathcal{M}_{\text{actual}} = \{0\}.} \quad (271)$$

So the remote Schur part cannot generate an exact constant Dirac mass.

CIX. 5. CHARGE-SECTOR FORMULATION

The most practical implementation of Route II is to decompose the remote sector into Q_R -charge sectors:

$$\boxed{\mathcal{H}_R = \bigoplus_{q \in \mathcal{Q}} \mathcal{H}_R^{(q)}, \quad Q_R|_{\mathcal{H}_R^{(q)}} = q \mathbf{1}.} \quad (272)$$

Similarly the low shell carrier decomposes under Q_* into

$$\boxed{\mathcal{H}_D = \mathcal{H}_D^{(+)} \oplus \mathcal{H}_D^{(-)}.} \quad (273)$$

Then the intertwiner condition

$$Q_* V(0) = V(0) Q_R$$

is equivalent to the charge-preserving selection rule

$$\boxed{V(0) : \mathcal{H}_R^{(q)} \longrightarrow \mathcal{H}_D^{(q)} \quad \text{only.}} \quad (274)$$

Thus the remote part of Gate 4 is reduced to the finite audit:

1. decompose $\mathcal{H}_R$ into exact Q_R -charge sectors;
2. verify $[Q_R, H_R(0)] = 0$;
3. verify that every nonzero block of $V(0)$ preserves the charge;
4. conclude $[Q_*, \Sigma(0, 0)] = 0$.

CX. 6. FINAL GATE-4 CLOSURE FOR THE CURRENT BRANCH

For the current refined locked + Class II branch:

$$\boxed{[Q_*, H_{\Psi\Psi}^{\text{const}}] = 0, \quad [Q_*, \delta H_{\text{Class II}}^{\text{const}}] = 0} \quad (275)$$

are already the natural working conclusions.

If, in addition, there exists a Hermitian Q_R satisfying

$$\boxed{[Q_R, H_R] = 0, \quad Q_* W = W Q_R,} \quad (276)$$

then

$$\boxed{[Q_*, \Sigma(0, 0)] = 0.} \quad (277)$$

Therefore the full corrected constant sector

$$\boxed{\delta H_{\text{const}} = H_{\Psi\Psi}^{\text{const}} + \delta H_{\text{Class II}}^{\text{const}} + \Sigma(0, 0)} \quad (278)$$

belongs to the Q_* -commutant,

$$\delta H_{\text{const}} \in \mathcal{C}_{Q_*},$$

and thus

$$\boxed{A_{\text{res}}^{(\text{full})} \cap \mathcal{M}_{\text{actual}} = \{0\}.} \quad (279)$$

So Gate 4 passes.

CXI. 7. HONEST REMAINING FRONTIER

At this point Gate 4 is no longer conceptually open.

The remaining microscopic work is only one of the following finite audits:

1. Route I: prove strict remote mismatch

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0,$$

or

2. Route II: explicitly construct Q_R and verify

$$[Q_R, H_R] = 0, \quad Q_* W = W Q_R.$$

Therefore the current branch has reduced Gate 4 completely to a finite symmetry/intertwiner verification problem.

CXII. FINAL SUMMARY

The actual canonical basis already provides

$$Q_* = \tau_3 \otimes s_3, \quad \mathcal{M}_{\text{actual}} = \text{span}_{\mathbb{R}}\{\tau_2 \otimes \mathbf{1}_2, \tau_1 \otimes s_3\},$$

with

$$\mathcal{C}_{Q_*} \cap \mathcal{M}_{\text{actual}} = \{0\}.$$

Hence every exact constant operator commuting with Q_* is automatically mass-protecting. :contentReference[oaicite:5]index=5

For the current refined locked + Class II branch,

$$[Q_*, H_{\Psi\Psi}^{\text{const}}] = 0$$

holds because the allowed low-block constant sector is

$$H_{\Psi\Psi}^{\text{const}} = a_0 \mathbf{1}_4 + a_* Q_*,$$

and

$$[Q_*, \delta H_{\text{Class II}}^{\text{const}}] = 0$$

is already the natural working conclusion.

Therefore the only remaining Gate-4 frontier is the remote Schur part

$$[Q_*, \Sigma(0, 0)] = 0.$$

A sufficient executable Route II is the existence of a Hermitian remote generator Q_R such that

$$[Q_* \oplus Q_R, H_{\text{full}}(0)] = 0,$$

equivalently

$$[Q_*, H_L(0)] = 0, \quad [Q_R, H_R(0)] = 0, \quad Q_* V(0) = V(0) Q_R.$$

Under these conditions,

$$[Q_*, \Sigma(0, 0)] = 0,$$

so

$$\Sigma(0, 0) \in \mathcal{C}_{Q_*}, \quad \Sigma(0, 0) \cap \mathcal{M}_{\text{actual}} = \{0\}.$$

Thus Gate 4 passes.

So the remaining Gate-4 work is no longer conceptual. It is a finite audit:

either prove strict remote mismatch, or explicitly construct Q_R and verify $[Q_R, H_R] = 0$, $Q_* W = W Q_R$.
--

CXIII. ROUTE-II CHARGE-TABLE AUDIT AND THE ACTUAL CLOSURE OF GATE 4

CXIV. 1. SHARP STATUS FOR THE CURRENT WORKING TRUNCATION

For the present working truncation, the remote sector is

$$\mathcal{H}_R = \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}}, \quad \mathcal{H}_{\text{extra}} = \emptyset.$$

Hence the strict shell-relevant mismatch route already gives

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0, \quad V(0) = 0, \quad \Sigma(0, 0) = 0.$$

Therefore the remote-sector part of Gate 4 is already closed by Route I in the current working truncation.

So, for the current branch,

Route II is not needed for closure.

CXV. 2. WHY ROUTE II IS STILL USEFUL

Route II is the constructive route needed only if the truncation is enlarged and a dangerous extra remote sector appears:

$$\mathcal{H}_{\text{extra}} \neq \emptyset.$$

Then the remaining Gate-4 task is the finite audit

$$\exists Q_R : [Q_R, H_R(0)] = 0, \quad Q_* V(0) = V(0) Q_R.$$

CXVI. 3. LOW-SHELL CHARGE DECOMPOSITION

In the actual canonical basis, the low-shell residual generator is

$$Q_* = \tau_3 \otimes s_3.$$

The low shell carrier decomposes into charge sectors as

$$\mathcal{H}_D^{(+)} = \text{span}\{|G_+, u_1\rangle, |G_-, u_2\rangle\}, \quad \mathcal{H}_D^{(-)} = \text{span}\{|G_+, u_2\rangle, |G_-, u_1\rangle\}.$$

So the low block is already split into Q_* -charges +1 and -1.

CXVII. 4. REMOTE-SECTOR CHARGE DECOMPOSITION

For an enlarged dangerous truncation, decompose the remote sector as

$$\mathcal{H}_R = \mathcal{H}_R^{(+)} \oplus \mathcal{H}_R^{(-)} \oplus \mathcal{H}_R^{(0)}.$$

Define the remote charge operator by

$$Q_R|_{\mathcal{H}_R^{(+)}} = +1, \quad Q_R|_{\mathcal{H}_R^{(-)}} = -1, \quad Q_R|_{\mathcal{H}_R^{(0)}} = 0.$$

Equivalently, in projector language,

$$R_+, \quad R_-, \quad R_0$$

are the corresponding remote projectors, and one may write

$$Q_R = R_+ - R_- + Q_0,$$

where Q_0 is any Hermitian involution on the neutral sector satisfying

$$Q_0^2 = R_0, \quad [Q_0, H_R^{(0)}] = 0.$$

Then Q_R is Hermitian and

$$[Q_R, H_R(0)] = 0$$

once the remote block is charge-block-diagonal.

CXVIII. 5. CHARGE-PRESERVING COUPLING ASSUMPTION

The exact Route-II intertwiner condition is

$$Q_* V(0) = V(0) Q_R.$$

In block form this is equivalent to the charge-selection rule

$$\boxed{V(0) : \mathcal{H}_R^{(q)} \longrightarrow \mathcal{H}_D^{(q)} \text{ only.}}$$

Equivalently,

$$V(0) = P_+ V(0) R_+ + P_- V(0) R_-,$$

with

$$P_+ V(0) R_- = 0, \quad P_- V(0) R_+ = 0, \quad V(0) R_0 = 0.$$

So the finite Route-II audit is exactly:

1. decompose $\mathcal{H}_R$ into Q_R -charge sectors;
2. verify $H_R(0)$ is block diagonal in $q = +1, -1, 0$, equivalently $[Q_R, H_R(0)] = 0$;
3. verify every nonzero low-remote coupling block preserves charge:

$$V(0) : \mathcal{H}_R^{(q)} \rightarrow \mathcal{H}_D^{(q)} \text{ only.}$$

CXIX. 6. SCHUR SELF-ENERGY INHERITANCE

Under

$$[Q_R, H_R(0)] = 0, \quad Q_* V(0) = V(0) Q_R,$$

the remote resolvent commutes with Q_R , and therefore

$$Q_* \Sigma(0, 0) = \Sigma(0, 0) Q_*.$$

Hence

$$\boxed{[Q_*, \Sigma(0, 0)] = 0.}$$

Since the actual constant mass space satisfies

$$\mathcal{C}_{Q_*} \cap \mathcal{M}_{\text{actual}} = \{0\},$$

this implies

$$\boxed{\Sigma(0, 0) \cap \mathcal{M}_{\text{actual}} = \{0\}.}$$

So the remote self-energy cannot generate a constant actual Dirac mass.

CXX. 7. FINAL EXECUTABLE GATE-4 AUDIT SHEET

Therefore the exact practical Gate-4 audit is:

a. Working truncation.

$$\boxed{\mathcal{H}_{\text{extra}} = \emptyset \implies V(0) = 0, \quad \Sigma(0, 0) = 0.}$$

Hence the remote-sector part of Gate 4 is already closed.

b. Enlarged dangerous truncation. Choose

$$Q_* = \tau_3 \otimes s_3, \quad \mathcal{H}_D = \mathcal{H}_D^{(+)} \oplus \mathcal{H}_D^{(-)},$$

decompose

$$\mathcal{H}_R = \mathcal{H}_R^{(+)} \oplus \mathcal{H}_R^{(-)} \oplus \mathcal{H}_R^{(0)},$$

define Q_R by the three charges $+1, -1, 0$, and verify:

$$\boxed{[Q_R, H_R(0)] = 0, \quad Q_* V(0) = V(0) Q_R.}$$

If yes, then

$$\boxed{[Q_*, \Sigma(0, 0)] = 0, \quad \Sigma(0, 0) \cap \mathcal{M}_{\text{actual}} = \{0\}.}$$

So Gate 4 is finite and executable in both cases.

CXXI. FINAL SUMMARY

For the remote-sector part of Gate 4 there are two exact routes.

a. Route I. If

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0,$$

then

$$V(0) = 0, \quad \Sigma(0, 0) = 0.$$

In the present working truncation,

$$\mathcal{H}_{\text{extra}} = \emptyset,$$

so this route already closes the remote-sector part of Gate 4.

b. Route II. For enlarged dangerous truncations, decompose the low carrier as

$$\mathcal{H}_D = \mathcal{H}_D^{(+)} \oplus \mathcal{H}_D^{(-)},$$

with

$$\mathcal{H}_D^{(+)} = \text{span}\{|G_+, u_1\rangle, |G_-, u_2\rangle\}, \quad \mathcal{H}_D^{(-)} = \text{span}\{|G_+, u_2\rangle, |G_-, u_1\rangle\},$$

under

$$Q_* = \tau_3 \otimes s_3.$$

Then decompose the dangerous remote sector into

$$\mathcal{H}_R = \mathcal{H}_R^{(+)} \oplus \mathcal{H}_R^{(-)} \oplus \mathcal{H}_R^{(0)},$$

and define

$$Q_R|_{\mathcal{H}_R^{(+)}} = +1, \quad Q_R|_{\mathcal{H}_R^{(-)}} = -1, \quad Q_R|_{\mathcal{H}_R^{(0)}} = 0.$$

The finite Route-II audit is:

$$[Q_R, H_R(0)] = 0, \quad Q_* V(0) = V(0) Q_R.$$

In block form, this is equivalent to:

$$V(0) : \mathcal{H}_R^{(q)} \rightarrow \mathcal{H}_D^{(q)} \quad \text{only,}$$

and

$H_R(0)$ is block diagonal in the charge sectors $q = +1, -1, 0$.

If these hold, then

$$[Q_*, \Sigma(0, 0)] = 0, \quad \Sigma(0, 0) \cap \mathcal{M}_{\text{actual}} = \{0\}.$$

Therefore:

in the working truncation, Gate 4 is already closed by Route I;

in enlarged dangerous truncations, Gate 4 reduces to a finite Route-II charge-table audit.

CXXII. ACTUAL PASS STAMP FOR GATE 3 AND REDUCTION OF GATE 2 TO THE FINAL LONGITUDINAL FRONTIER

CXXIII. 1. INTEGRATED CARRIER LOGIC

The integrated carrier acceptance condition before the one-defect Gate 5 is

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0), \quad A_{\text{res}} \cap \mathcal{M}_D = \{0\}. \quad (280)$$

In the current line, Gate 4 has already been reduced to a closed working-truncation statement, so the only remaining carrier-side issue is to decide Gates 2 and 3 as actual microscopic pass/fail statements.

CXXIV. 2. GATE 3 IS EFFECTIVELY PASSED ON THE WORKING BRANCH

The exact Gate-3 shell-side pass condition is

$$(c_E, c_O) \neq (0, 0), \quad (C_E, C_O) \neq (0, 0). \quad (281)$$

On the explicit folded-Bragg Class II working branch, this compresses to the practical sufficient test

$$A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0, \quad (282)$$

together with the Bloch calibration condition

$$(C_E, C_O) \neq (0, 0). \quad (283)$$

Moreover, the transverse witness is already explicitly realized in the seed construction:

$$(\alpha, \beta) \neq (0, 0) \text{ is explicitly realizable.} \quad (284)$$

Therefore, at the working-branch level,

$$\text{Gate 3 is no longer the genuine frontier.} \quad (285)$$

Its honest status is:

$$\text{finite explicit overlap/calibration computation left on the working branch,} \quad (286)$$

but the transverse carrier itself is already structurally and explicitly realized.

Hence the strongest honest working verdict is:

$$\text{Gate 3 is effectively passed modulo finite explicit overlap insertion.} \quad (287)$$

CXXV. 3. GATE 2 REMAINS THE ACTUAL FRONTIER

For Gate 2, the exact longitudinal chain is

$$\boxed{\ell_{\parallel A} \longrightarrow c_{\parallel}^{(\text{mic})} \longrightarrow J_{\parallel}^{(3)} \longrightarrow C_c \longrightarrow \lambda_{\parallel}.} \quad (288)$$

The practical Gate-2 verdict is

$$\boxed{\exists A : \ell_{\parallel A} \neq 0, \quad c_{\parallel}^{(\text{mic})} \neq 0, \quad J_{\parallel}^{(3)} \neq 0, \quad C_c \neq 0 \implies \lambda_{\parallel} \neq 0.} \quad (289)$$

However, the honest carrier-audit status table states:

$$\boxed{\text{Longitudinal primitive seed: open; this is the honest remaining frontier.}} \quad (290)$$

The same conclusion appears in the basis-overlap formulation:

$$\boxed{\exists A : \ell_{\parallel A} \neq 0} \quad (291)$$

is the unresolved microscopic yes/no question.

Equivalently, in the canonical enriched-shell extraction language, the longitudinal carrier condition is

$$\boxed{v_{\parallel} \neq 0,} \quad (292)$$

where

$$\boxed{v_{\parallel} = \frac{1}{4} \text{Tr}_{\text{int}} [\partial_{p_{\parallel}} (E_+(p) - E_-(p))]_{p=0}.} \quad (293)$$

Thus Gate 2 is now reduced to a single primitive longitudinal survival question:

$$\boxed{\text{does the actual microscopic refined locked + Class II branch realize } v_{\parallel} \neq 0?} \quad (294)$$

CXXVI. 4. STRONGEST HONEST INTEGRATED VERDICT

Since Gate 3 is effectively passed on the working branch, while Gate 2 remains open, the integrated carrier admissibility status is:

$$\boxed{\text{transverse solved, longitudinal remaining.}} \quad (295)$$

Equivalently,

$$\boxed{\text{carrier admissibility is reduced to the single unresolved condition } \lambda_{\parallel} \neq 0 \text{ (or } v_{\parallel} \neq 0\text{).}} \quad (296)$$

In basis-overlap language this is

$$\boxed{\exists A : a_A \neq 0,} \quad (297)$$

where

$$E_A = a_A B_{\parallel} + b_A B_I + c_A B_J + R_A.$$

So the full carrier problem is no longer a broad multi-gate issue. It is one final longitudinal primitive-channel survival audit.

CXXVII. 5. CONSEQUENCE FOR EXPLICIT BRANCH ADMISSIBILITY

Therefore the current refined locked + Class II explicit branch has not yet received the final actual carrier-admissibility stamp.

But the reason is now unique and sharply isolated:

$$\boxed{\text{the only remaining carrier question is the survival of the primitive longitudinal channel.}} \quad (298)$$

So the explicit branch admissibility problem has been reduced to:

$$\boxed{\text{Gate 3: effectively yes,} \quad \text{Gate 4: closed in the working truncation,} \quad \text{Gate 2: final unresolved microscopic audit.}} \quad (299)$$

CXXVIII. FINAL SUMMARY

The carrier-level acceptance logic is

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0), \quad A_{\text{res}} \cap \mathcal{M}_D = \{0\}.$$

At the current stage:

$$\boxed{\text{Gate 4 is closed in the working truncation,}}$$

$$\boxed{\text{Gate 3 is effectively passed on the explicit folded-Bragg Class II working branch,}}$$

because its exact pass condition

$$(c_E, c_O) \neq (0, 0), \quad (C_E, C_O) \neq (0, 0)$$

is already compressed to the finite practical test

$$A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0, \quad (C_E, C_O) \neq (0, 0),$$

and the transverse witness is explicitly realizable.

By contrast, Gate 2 remains the honest remaining frontier:

$$\exists A : \ell_{\parallel A} \neq 0$$

or equivalently

$$v_{\parallel} \neq 0, \quad v_{\parallel} = \frac{1}{4} \text{Tr}_{\text{int}} [\partial_{p_{\parallel}} (E_+(p) - E_-(p))]_{p=0}.$$

Therefore the integrated carrier status is

$$\boxed{\text{transverse solved, longitudinal remaining.}}$$

Equivalently, the explicit branch admissibility problem is now reduced to the single unresolved longitudinal primitive-channel question

$$\boxed{\lambda_{\parallel} \neq 0 \quad (\text{equivalently } v_{\parallel} \neq 0).}$$

CXXIX. ACTUAL-BASIS OVERLAP CLOSURE OF GATE 2

CXXX. 1. EXACT LEVEL-A / LEVEL-B / LEVEL-C EQUIVALENCE

The broad basis-level longitudinal audit is:

$$\ell_{\parallel A} := E_{A,ij}^3 \hat{n}_i \hat{n}_j, \quad \exists A : \ell_{\parallel A} \neq 0. \quad (300)$$

This is exactly equivalent to the nonvanishing of the primitive longitudinal seed coefficient in the shell-seed decomposition

$$P_{\text{seed}} N_{\text{mic}} = c_{\parallel}^{(\text{mic})} B_{\parallel} + c_I^{(\text{mic})} B_I + c_J^{(\text{mic})} B_J, \quad (301)$$

namely

$$\exists A : \ell_{\parallel A} \neq 0 \iff c_{\parallel}^{(\text{mic})} \neq 0. \quad (302)$$

The projected carrier coefficient is then

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad I_{\theta'^3} \neq 0. \quad (303)$$

Hence

$$c_{\parallel}^{(\text{mic})} \neq 0 \implies \lambda_{\parallel} \neq 0. \quad (304)$$

So Gate 2 is not an infinite-dimensional conceptual problem any more. It is a finite overlap extraction problem. :contentReference[oaicite:7]index=7

CXXXI. 2. BRANCH-SENSITIVE DECISION TREE AND CURRENT SOLVED BRANCH

The exact branch-sensitive longitudinal parameterization is

$$a_3(\xi) = \Lambda_{\parallel}^{(0)} \theta'(\xi) + \frac{\lambda_c}{2} \theta'(\xi)^3 + a_3^{\text{higher}}(\xi). \quad (305)$$

In the current solved lowest-order local realization,

$$\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0. \quad (306)$$

Therefore the current practical longitudinal route is forced to Branch B, not Branch A. :contentReference[oaicite:8]index=8
Equivalently, the primitive longitudinal transport block is

$$V_{(\parallel),\text{tr}}^{\mu}(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^{\mu} \sigma_3, \quad (307)$$

with

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}. \quad (308)$$

So the branch-level question is reduced to the finite extraction

$$\lambda_{c;OO} \neq 0 ? \quad (309)$$

or equivalently to the actual projected overlap criteria below. :contentReference[oaicite:9]index=9

CXXXII. 3. FINAL PROJECTED MATRIX-ELEMENT FORMULATION

The remaining carrier-side longitudinal frontier is reduced to two explicit projected quantities:

$$\boxed{\Delta_{00} = \eta_+^\dagger \Sigma_3 \eta_-}, \quad (310)$$

where $\eta_\pm$ are the normalized low internal eigenvectors of the valley-diagonal 2×2 blocks $E_\pm(0)$, and

$$\boxed{L_* = m_\parallel u_\parallel = \frac{m_\parallel}{2} \text{Tr}(\Sigma_3 \hat{n}^j \mathcal{U}_j(0)) = \frac{m_\parallel}{2} (u_{11} - u_{22})}, \quad (311)$$

with

$$(\mathcal{U}_j(0))_{ab} = \langle -, a | \hat{J}_j(G_*) | +, b \rangle. \quad (312)$$

Thus the current actual-basis longitudinal slot is already reduced to the finite projected matrix extraction of:

$$E_\pm(0), \quad \mathcal{U}_j(0), \quad m_\parallel.$$

:contentReference[oaicite:10]index=10

CXXXIII. 4. PRACTICAL SUFFICIENT CONDITION

Under first-shell dominant intervalley compression, the practical sufficient condition is

$$\boxed{|\Delta_{00} \alpha_X L_* A_{G_*}| > |R_N| \implies \lambda_\parallel \neq 0.} \quad (313)$$

In the ultra-minimal truncation $R_N = 0$, this collapses to

$$\boxed{\Delta_{00} \neq 0, \quad L_* \neq 0, \quad \alpha_X \neq 0, \quad A_{G_*} \neq 0 \implies \lambda_\parallel \neq 0.} \quad (314)$$

So in the current branch, there is no remaining structural ambiguity. The only remaining work is the literal finite insertion of the projected matrix elements.

CXXXIV. 5. EXPLICIT SIGN/VANISHING CONDITIONS INSIDE THE PROJECTED MATRICES

The overlap factor Δ_{00} is explicitly

$$\boxed{\Delta_{00} = \frac{1 - \bar{r}_+ r_-}{\sqrt{(1 + |r_+|^2)(1 + |r_-|^2)}}, \quad r_\pm = -\frac{c_\pm}{a_\pm - \varepsilon_\pm^{(\text{low})}}.} \quad (315)$$

Hence

$$\boxed{\Delta_{00} = 0 \iff \bar{r}_+ r_- = 1.} \quad (316)$$

Likewise,

$$\boxed{L_* = 0 \iff m_\parallel = 0 \quad \text{or} \quad u_{11} = u_{22}.} \quad (317)$$

Therefore, in the ultra-minimal truncation, a sharp finite pass sheet is:

$$\boxed{\bar{r}_+ r_- \neq 1, \quad m_\parallel \neq 0, \quad u_{11} \neq u_{22}, \quad \alpha_X \neq 0, \quad A_{G_*} \neq 0 \implies \lambda_\parallel \neq 0.} \quad (318)$$

CXXXV. 6. CALIBRATION LAYER

Finally, the exact normalization layer is

$$\boxed{g_c = \frac{\lambda_{\parallel}}{C_{\parallel}}.} \quad (319)$$

So for the carrier-level longitudinal survival itself, $\lambda_{\parallel} \neq 0$ is the primitive condition. For the downstream one-defect acceptance audit, one also needs

$$\boxed{C_{\parallel} \neq 0.} \quad (320)$$

Thus the full practical longitudinal pass stamp on the current solved Branch-B working route is:

$$\boxed{|\Delta_{00} \alpha_X L_* A_{G_*}| > |R_N|, \quad C_{\parallel} \neq 0 \implies \lambda_{\parallel} \neq 0 \implies g_c \neq 0.} \quad (321)$$

CXXXVI. 7. HONEST ENDPOINT

The current longitudinal frontier is therefore no longer

$$\exists A : \ell_{\parallel A} \neq 0$$

as an abstract statement.

It is the finite actual-basis extraction of

$$E_{\pm}(0), \quad \mathcal{U}_j(0), \quad m_{\parallel}, \quad A_{G_*}, \quad C_{\parallel}.$$

So the exact honest status is:

$$\boxed{\text{Gate 2 is reduced to a finite projected-matrix audit on the current Branch-B working realization.}} \quad (322)$$

CXXXVII. FINAL SUMMARY

The broad longitudinal overlap statement

$$\exists A : \ell_{\parallel A} \neq 0$$

is already equivalent to the shell-seed condition

$$c_{\parallel}^{(\text{mic})} \neq 0,$$

and this implies

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3} \neq 0 \quad (I_{\theta'^3} \neq 0).$$

In the current solved lowest-order local class,

$$\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0,$$

so the practical longitudinal route is forced to Branch B.

The remaining carrier-side longitudinal frontier is reduced to the finite projected matrix elements

$$\Delta_{00} = \eta_+^{\dagger} \Sigma_3 \eta_-, \quad L_* = \frac{m_{\parallel}}{2} \text{Tr}(\Sigma_3 \hat{n}^j \mathcal{U}_j(0)) = \frac{m_{\parallel}}{2} (u_{11} - u_{22}),$$

computed from

$$E_{\pm}(0), \quad \mathcal{U}_j(0), \quad m_{\parallel}.$$

A practical sufficient condition is

$$|\Delta_{00} \alpha_X L_* A_{G_*}| > |R_N| \implies \lambda_{\parallel} \neq 0.$$

In the ultra-minimal truncation $R_N = 0$, it is enough to verify

$$\Delta_{00} \neq 0, \quad L_* \neq 0, \quad \alpha_X \neq 0, \quad A_{G_*} \neq 0.$$

Equivalently, using

$$\Delta_{00} = 0 \iff \bar{r}_+ r_- = 1, \quad L_* = 0 \iff m_{\parallel} = 0 \text{ or } u_{11} = u_{22},$$

a sharp finite pass sheet is

$$\bar{r}_+ r_- \neq 1, \quad m_{\parallel} \neq 0, \quad u_{11} \neq u_{22}, \quad \alpha_X \neq 0, \quad A_{G_*} \neq 0 \implies \lambda_{\parallel} \neq 0.$$

Finally, for downstream one-defect acceptance one also needs

$$C_{\parallel} \neq 0, \quad g_c = \frac{\lambda_{\parallel}}{C_{\parallel}}.$$

Thus Gate 2 is no longer an abstract open problem. It is a finite projected-matrix audit on the current Branch-B working realization.

CXXXVIII. ULTRA-MINIMAL ACTUAL PASS SHEET FOR GATE 2

CXXXIX. 1. FINAL FINITE REDUCTION OF THE LONGITUDINAL SLOT

The carrier-side longitudinal audit is already reduced to two explicit projected scalars:

$$\boxed{\Delta_{00} = \eta_+^{\dagger} \Sigma_3 \eta_-, \quad L_* = m_{\parallel} u_{\parallel}.} \quad (323)$$

Under first-shell dominant intervalley compression, the practical sufficient condition is

$$\boxed{|\Delta_{00} \alpha_X L_* A_{G_*}| > |R_N| \implies \lambda_{\parallel} \neq 0.} \quad (324)$$

In the ultra-minimal truncation $R_N = 0$, it is enough to verify

$$\boxed{\Delta_{00} \neq 0, \quad L_* \neq 0, \quad \alpha_X \neq 0, \quad A_{G_*} \neq 0.} \quad (325)$$

CXL. 2. EXPLICIT VANISHING LOCI

Write the two valley-diagonal projected blocks as

$$E_{\pm}(0) = \begin{pmatrix} a_{\pm} & c_{\pm} \\ c_{\pm} & b_{\pm} \end{pmatrix}, \quad r_{\pm} = -\frac{c_{\pm}}{a_{\pm} - \varepsilon_{\pm}^{(\text{low})}}.$$

Then

$$\boxed{\Delta_{00} = \frac{1 - \bar{r}_+ r_-}{\sqrt{(1 + |r_+|^2)(1 + |r_-|^2)}},} \quad (326)$$

hence

$$\boxed{\Delta_{00} = 0 \iff \bar{r}_+ r_- = 1.} \quad (327)$$

Next define

$$U_{\parallel} = \hat{n}^j \mathcal{U}_j(0) = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}, \quad u_{\parallel} = \frac{1}{2}(u_{11} - u_{22}).$$

Then

$$\boxed{L_* = \frac{m_{\parallel}}{2}(u_{11} - u_{22}),} \quad (328)$$

so

$$\boxed{L_* = 0 \iff m_{\parallel} = 0 \quad \text{or} \quad u_{11} = u_{22}.} \quad (329)$$

Therefore the ultra-minimal actual pass sheet is

$$\boxed{\bar{r}_+ r_- \neq 1, \quad m_{\parallel} \neq 0, \quad u_{11} \neq u_{22}, \quad \alpha_X \neq 0, \quad A_{G_*} \neq 0 \implies \lambda_{\parallel} \neq 0.} \quad (330)$$

CXLI. 3. EXPLICIT CANONICAL WITNESS BRANCH $T_* = T^1$

For the canonical locked basis

$$z_0 = e_1, \quad u_1 = e_2, \quad u_2 = e_3,$$

choose the active generator

$$\boxed{T_* = T^1.}$$

Then

$$t_{10} = \frac{1}{2}, \quad t_{20} = 0,$$

and the shell-normal reduced J -split becomes

$$\boxed{u_{\parallel}(T^1) = \frac{1}{2} \Re(\kappa_J).} \quad (331)$$

Hence

$$\boxed{L_*(T^1) = \frac{m_{\parallel}}{2} \Re(\kappa_J).} \quad (332)$$

Using

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X L_* I_{\theta'^3}, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3,$$

one gets

$$\boxed{\lambda_{\parallel}(T^1) = \frac{1}{4} \alpha_X m_{\parallel} \Re(\kappa_J) I_{\theta'^3}.} \quad (333)$$

Therefore an explicit practical sufficient witness condition is

$$\boxed{\alpha_X \neq 0, \quad m_{\parallel} \neq 0, \quad \Re(\kappa_J) \neq 0, \quad \theta_0 \kappa \neq 0 \implies \lambda_{\parallel}(T^1) \neq 0.} \quad (334)$$

CXLII. 4. HONEST ENDPOINT

Thus Gate 2 is no longer structurally open. At the current stage it is reduced to the literal finite extraction of

$$E_{\pm}(0), \quad U_{\parallel}, \quad m_{\parallel},$$

or, in the explicit witness branch, to the finite scalar insertion

$$m_{\parallel}, \quad \Re(\kappa_J), \quad \alpha_X, \quad \theta_0 \kappa.$$

So the only remaining work is explicit branch evaluation, not further theorem-building.

CXLIII. FINAL SUMMARY

The longitudinal Gate-2 problem is already reduced to two explicit projected scalars:

$$\Delta_{00} = \eta_+^\dagger \Sigma_3 \eta_-, \quad L_* = m_{\parallel} u_{\parallel}.$$

The practical sufficient condition is

$$|\Delta_{00} \alpha_X L_* A_{G_*}| > |R_N| \implies \lambda_{\parallel} \neq 0.$$

In the ultra-minimal truncation $R_N = 0$, this becomes

$$\Delta_{00} \neq 0, \quad L_* \neq 0, \quad \alpha_X \neq 0, \quad A_{G_*} \neq 0 \implies \lambda_{\parallel} \neq 0.$$

The two exact zero loci are:

$$\Delta_{00} = 0 \iff \bar{r}_+ r_- = 1,$$

and

$$L_* = 0 \iff m_{\parallel} = 0 \quad \text{or} \quad u_{11} = u_{22}.$$

Hence the sharp ultra-minimal actual pass sheet is

$$\boxed{\bar{r}_+ r_- \neq 1, \quad m_{\parallel} \neq 0, \quad u_{11} \neq u_{22}, \quad \alpha_X \neq 0, \quad A_{G_*} \neq 0 \implies \lambda_{\parallel} \neq 0.}$$

For the explicit canonical witness branch $T_* = T^1$,

$$u_{\parallel}(T^1) = \frac{1}{2} \Re(\kappa_J), \quad L_*(T^1) = \frac{m_{\parallel}}{2} \Re(\kappa_J),$$

so

$$\lambda_{\parallel}(T^1) = \frac{1}{4} \alpha_X m_{\parallel} \Re(\kappa_J) I_{\theta'^3}.$$

Therefore

$$\boxed{\alpha_X \neq 0, \quad m_{\parallel} \neq 0, \quad \Re(\kappa_J) \neq 0, \quad \theta_0 \kappa \neq 0 \implies \lambda_{\parallel}(T^1) \neq 0.}$$

So Gate 2 is no longer a conceptual frontier. It is a finite projected-matrix extraction problem, and on the explicit canonical witness branch it already has a practical sufficient pass stamp.

CXLIV. ACTUAL PASS STAMP OF THE CALIBRATION LAYER

CXLV. 1. EXACT CALIBRATION LAYER

The exact reduced coefficient extraction is

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 M^{\parallel}), \quad (335)$$

$$\alpha = \frac{1}{4} \text{Tr}(\sigma_1 M_1 + \sigma_2 M_2), \quad \beta = \frac{1}{4} \text{Tr}(\sigma_2 M_1 - \sigma_1 M_2). \quad (336)$$

The exact Bloch/transport normalization layer is

$$\lambda_{\parallel} = C_{\parallel} g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O. \quad (337)$$

Equivalently,

$$g_c = \frac{\lambda_{\parallel}}{C_{\parallel}}, \quad g_E = \frac{\alpha}{C_E}, \quad g_O = \frac{\beta}{C_O}. \quad (338)$$

Thus the calibration constants are not new dynamical coefficients. They are the overlap-normalization factors converting the reduced shell coefficients

$$(\lambda_{\parallel}, \alpha, \beta)$$

into the transport-normalized one-defect data

$$(g_c, g_E, g_O).$$

CXLVI. 2. CONCRETE ONE-MEDIATOR EXPLICIT BRANCH

In the concrete one-mediator truncation, the microscopic effective cross terms are

$$\mathcal{L}_{\text{eff}} \supset -\frac{\alpha_X \beta_X}{M_X^2} J_i^a K_i^a - \frac{\beta_X \gamma_E}{M_X^2} K_i^a \mathcal{E}_i^a - \frac{\beta_X \gamma_O}{M_X^2} K_i^a \mathcal{O}_i^a. \quad (339)$$

The corresponding normalized overlap constants are defined by

$$C_{\parallel} := A_a(\theta) k_a(\theta) \big|_{\text{normalized}}, \quad (340)$$

and

$$C_E := \text{normalized coefficient of } K_i^a \mathcal{E}_i^a, \quad C_O := \text{normalized coefficient of } K_i^a \mathcal{O}_i^a. \quad (341)$$

Hence the explicit concrete coefficients are

$$\lambda_c^{(1)} = -\frac{\alpha_X \beta_X}{M_X^2} C_{\parallel}, \quad (342)$$

$$\tilde{\Lambda}_E^{(1)} = -\frac{\beta_X \gamma_E}{M_X^2} C_E, \quad \tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2} C_O. \quad (343)$$

Therefore the concrete witness branch already shows that the entire calibration layer is absorbed into the three overlap factors

$$C_{\parallel}, \quad C_E, \quad C_O.$$

CXLVII. 3. CANONICAL EXPLICIT NORMALIZATION

The crucial practical simplification is that in the canonical explicit normalization one may choose

$$\boxed{C_{\parallel} = C_E = C_O = 1.} \quad (344)$$

With this choice,

$$\boxed{\lambda_c^{(1)} = -\frac{\alpha_X \beta_X}{M_X^2}, \quad \tilde{\Lambda}_E^{(1)} = -\frac{\beta_X \gamma_E}{M_X^2}, \quad \tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2}.} \quad (345)$$

At the shell-block level, the explicit canonical Branch-B block becomes

$$\boxed{M^{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} I_3 \sigma_3,} \quad (346)$$

$$\boxed{M^1 = -\frac{\beta_X \gamma_E}{M_X^2} J_1 \sigma_1 - \frac{\beta_X \gamma_O}{M_X^2} J_1 \sigma_2,} \quad (347)$$

$$\boxed{M^2 = +\frac{\beta_X \gamma_O}{M_X^2} J_1 \sigma_1 - \frac{\beta_X \gamma_E}{M_X^2} J_1 \sigma_2.} \quad (348)$$

Therefore

$$\boxed{A_1(p) = -\frac{\alpha_X \beta_X}{2M_X^2} I_3 p_{\parallel} \sigma_3 - \frac{\beta_X \gamma_E}{M_X^2} J_1 (p_1 \sigma_1 + p_2 \sigma_2) - \frac{\beta_X \gamma_O}{M_X^2} J_1 (-p_2 \sigma_1 + p_1 \sigma_2).} \quad (349)$$

So on the explicit canonical witness branch, the calibration layer does not suppress any nonzero transport coefficient:

$$\boxed{C_{\parallel} = C_E = C_O = 1 \implies g_c = \lambda_{\parallel}, \quad g_E = \alpha, \quad g_O = \beta.} \quad (350)$$

CXLVIII. 4. PRACTICAL CARRIER PASS STAMP AFTER CALIBRATION

Recall the current carrier-side status:

1. Gate 4 is already closed in the working truncation;
2. Gate 3 is effectively passed once the explicit branch realizes a nonzero transverse witness and $(C_E, C_O) \neq (0, 0)$;
3. Gate 2 is reduced to the practical witness condition

$$|\Delta_{00} \alpha_X L_* A_{G*}| > |R_N| \implies \lambda_{\parallel} \neq 0.$$

Under the canonical explicit normalization (344), the calibration conditions become automatic:

$$C_{\parallel} \neq 0, \quad (C_E, C_O) \neq (0, 0).$$

Therefore the carrier-level practical pass stamp reduces to the transport-side witness conditions alone.
[Practical carrier pass stamp on the explicit canonical witness branch] Assume:

1. the working truncation is the current refined locked + Class II branch;
2. the shell tangent basis is canonically normalized so that

$$C_{\parallel} = C_E = C_O = 1;$$

3. Gate 4 is closed in the working truncation;
4. the explicit transverse branch realizes a nonzero witness pair

$$(c_E, c_O) \neq (0, 0);$$

5. the longitudinal Branch-B witness satisfies

$$|\Delta_{00}\alpha_X L_* A_{G_*}| > |R_N|.$$

Then

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0),} \quad (351)$$

hence the branch passes the carrier-level practical admissibility test.

Under canonical normalization,

$$C_{\parallel} = C_E = C_O = 1,$$

so the exact normalization layer becomes

$$g_c = \lambda_{\parallel}, \quad g_E = \alpha, \quad g_O = \beta.$$

Therefore no nonzero transport witness is killed by the Bloch overlap layer.

The longitudinal witness implies $\lambda_{\parallel} \neq 0$. The transverse witness implies $(\alpha, \beta) \neq (0, 0)$. Together with the already closed Gate 4, this is exactly the carrier-level admissibility condition.

CXLIX. 5. HONEST SCOPE OF THIS CLOSURE

This does *not* mean that the final complete microscopic TECT action has already been fully and unconditionally calibrated.

What it means is narrower and honest:

$$\boxed{\text{on the strongest explicit canonical witness branch, the normalization layer itself is no longer an obstruction.}} \quad (352)$$

The remaining frontier of the full theory is still the actual coefficient extraction from the final complete TECT operator,

$$L(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta),$$

not the algebraic form of the calibration layer.

CL. FINAL SUMMARY

The exact calibration layer is

$$\lambda_{\parallel} = C_{\parallel} g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O,$$

equivalently

$$g_c = \frac{\lambda_{\parallel}}{C_{\parallel}}, \quad g_E = \frac{\alpha}{C_E}, \quad g_O = \frac{\beta}{C_O}.$$

In the concrete one-mediator explicit branch, the calibration constants are the normalized overlap factors appearing in

$$\lambda_c^{(1)} = -\frac{\alpha_X \beta_X}{M_X^2} C_{\parallel}, \quad \tilde{\Lambda}_E^{(1)} = -\frac{\beta_X \gamma_E}{M_X^2} C_E, \quad \tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2} C_O.$$

In the canonical explicit normalization one may choose

$$C_{\parallel} = C_E = C_O = 1.$$

Therefore, on the explicit canonical witness branch,

$$g_c = \lambda_{\parallel}, \quad g_E = \alpha, \quad g_O = \beta.$$

So the calibration layer no longer kills any nonzero transport witness.

Hence the practical carrier pass stamp becomes:

if

$$|\Delta_{00}\alpha_X L_* A_{G_*}| > |R_N|, \quad (c_E, c_O) \neq (0, 0),$$

and Gate 4 is already closed in the working truncation, then

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0),$$

so the explicit canonical witness branch passes the carrier-level practical admissibility test.

This is an honest working-branch closure. It does not yet replace the final complete microscopic coefficient extraction from the full TECT operator.

CLI. ORTHODOX COMPLETION ROUTE FROM THE MICROSCOPIC TECT DEFECT OPERATOR

CLII. 1. HONEST ENDPOINT OF THE PREVIOUS LINE

The previous witness/certificate line should now be regarded as complete at its honest endpoint.

What it achieved is:

$$\boxed{\text{support theorems, carrier witness structure, and one-defect finite audit are in place.}} \quad (353)$$

What it did *not* yet achieve is the true first-principles microscopic derivation. The genuinely unfinished core remains:

$$\boxed{\text{derive the actual TECT defect fluctuation operator and compute the explicit microscopic overlap coefficients.}} \quad (354)$$

Therefore the correct next step is no longer another auxiliary theorem. It is to return to the main line:

$$\boxed{\text{microscopic TECT defect PDE} \longrightarrow \text{single-defect operator} \longrightarrow \text{single-defect zero mode} \longrightarrow \text{three-defect projector} \longrightarrow \text{sector}} \quad (355)$$

CLIII. 2. MINIMAL MICROSCOPIC MULTIPLY AND WORKING ACTION

We adopt the minimal working microscopic multiplet

$$\mathfrak{B} = (P^A, \Psi, X_i^a). \quad (356)$$

Here:

- $P^A(x)$ is the locked condensate/order-parameter background,
- $\Psi(x)$ is the fluctuation field whose low spectral sector will generate the family space,
- $X_i^a(x)$ is the heavy Class-II mediator field.

The minimal structurally correct microscopic working ansatz is

$$\mathcal{L}_{\text{mic}} = \mathcal{L}_{\Psi}^{\text{ref}}[P, \Psi] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[P, \Psi] + \beta_X X_i^a K_i^a[P, \Psi]. \quad (357)$$

The reference Ψ -sector is taken in first-order form:

$$\mathcal{L}_{\Psi}^{\text{ref}} = \Psi^{\dagger} \left(iK^0(P) \partial_t + iK^i(P) \partial_i - \mathcal{M}(P) \right) \Psi + \dots \quad (358)$$

where $K^0(P)$, $K^i(P)$, and $\mathcal{M}(P)$ are constitutive TECT coefficient fields.

CLIV. 3. DEFECT BACKGROUND AND LINEARIZATION

Choose a static one-defect background

$$\boxed{\bar{P}(x), \quad \bar{X}_i^a(x),} \quad (359)$$

with asymptotic bulk limit

$$\bar{P}(x) \rightarrow P_\infty, \quad \bar{X}_i^a(x) \rightarrow X_{i,\infty}^a \quad (|x| \rightarrow \infty).$$

Now linearize the microscopic equations around the defect:

$$P = \bar{P} + \pi, \quad X = \bar{X} + \chi, \quad \Psi = \bar{\Psi} + \psi.$$

Keeping only the fluctuation operator acting on ψ , one obtains

$$\boxed{iG_{\text{def}}^0(x) \partial_t \psi = \left(-iG_{\text{def}}^i(x) \partial_i + M_{\text{def}}(x) \right) \psi + W_{\text{def}}(x) \chi + \dots} \quad (360)$$

with defect-background coefficients

$$\boxed{G_{\text{def}}^0 = K^0(\bar{P}, \bar{X}), \quad G_{\text{def}}^i = K^i(\bar{P}, \bar{X}), \quad M_{\text{def}} = \mathcal{M}(\bar{P}, \bar{X}, \partial \bar{P}, \partial \bar{X}).} \quad (361)$$

CLV. 4. ELIMINATION OF THE HEAVY MEDIATOR

At leading order in the heavy-mediator approximation, the χ -equation is algebraic:

$$\boxed{M_X^2 \chi_i^a + \alpha_X \delta J_i^a + \beta_X \delta K_i^a = 0 + O(\partial \chi, \chi^2).} \quad (362)$$

Hence

$$\boxed{\chi_i^a = -\frac{1}{M_X^2} (\alpha_X \delta J_i^a + \beta_X \delta K_i^a) + O(M_X^{-4}).} \quad (363)$$

Substituting (363) back into (360) gives a closed effective linear fluctuation equation for ψ :

$$\boxed{i\widehat{G}_{\text{def}}^0(x) \partial_t \psi = \left(-i\widehat{G}_{\text{def}}^i(x) \partial_i + \widehat{M}_{\text{def}}(x) \right) \psi} \quad (364)$$

where the hatted coefficients include the Class-II Schur-complement correction.

This is the first genuinely orthodox microscopic step: the effective single-defect operator is now derived from the microscopic defect background rather than postulated as an abstract matrix ansatz.

CLVI. 5. CANONICAL FIRST-ORDER DEFECT OPERATOR

Assume $\widehat{G}_{\text{def}}^0(x)$ is Hermitian and positive in the asymptotic bulk and remains invertible along the defect profile. Then we canonically normalize the time-kinetic factor:

$$\psi = (\widehat{G}_{\text{def}}^0)^{-1/2} \tilde{\psi}.$$

The equation becomes

$$\boxed{i\partial_t \tilde{\psi} = \mathbb{O}_{1\text{def}}^{(0)} \tilde{\psi},} \quad (365)$$

with

$$\boxed{\mathbb{O}_{1\text{def}}^{(0)} = i\tilde{\Gamma}_{\text{def}}^i (\partial_i + \Omega_i^{\text{def}}) + \Phi_{\text{def}},} \quad (366)$$

where

$$\boxed{\tilde{\Gamma}_{\text{def}}^i = (\hat{G}_{\text{def}}^0)^{-1/2} \hat{G}_{\text{def}}^i (\hat{G}_{\text{def}}^0)^{-1/2},} \quad (367)$$

$$\boxed{\Omega_i^{\text{def}} = \frac{1}{2} (\hat{G}_{\text{def}}^0)^{1/2} \partial_i (\hat{G}_{\text{def}}^0)^{-1/2} + \Omega_{i,\text{mic}}^{\text{def}},} \quad (368)$$

$$\boxed{\Phi_{\text{def}} = (\hat{G}_{\text{def}}^0)^{-1/2} \hat{M}_{\text{def}} (\hat{G}_{\text{def}}^0)^{-1/2}.} \quad (369)$$

Thus the abstract reference-operator template is now promoted to the actual microscopic target of the orthodox TECT program.

CLVII. 6. THE SINGLE-DEFECT ZERO-MODE PROBLEM

The single-defect light-mode equation is

$$\boxed{\mathbb{O}_{1\text{def}}^{(0)} \psi_{\star} = 0.} \quad (370)$$

The theorem schema to be completed inside TECT is:

Fredholm / Callias structure + asymptotic bulk invertibility + grading + $\text{Ind}(\mathbb{O}_{1\text{def}}^{(0)}) = 1 \implies \dim \ker \mathbb{O}_{1\text{def}}^{(0)} = 1$ in the protected	(371)
--	-------

At present this is still conditional in TECT. So the honest next mathematical tasks are:

1. prove self-adjointness / Fredholmness of $\mathbb{O}_{1\text{def}}^{(0)}$,
2. prove asymptotic invertibility of Φ_{def} ,
3. identify the relevant topological class,
4. prove

$$\text{Ind}(\mathbb{O}_{1\text{def}}^{(0)}) = 1.$$

CLVIII. 7. THREE-DEFECT FAMILY PROJECTOR

If the single-defect theorem is established and the isolated defect centered at X_a supports a localized mode

$$\psi_{\star}(x - X_a), \quad a = 1, 2, 3,$$

then for sufficiently separated defects the three-defect operator

$$\mathbb{O}_{3\text{def}}^{(0)}$$

has an isolated low-energy three-state cluster.

The exact family projector is then

$$\boxed{P_L = \frac{1}{2\pi i} \oint_{\Gamma} (z - \mathbb{O}_{3\text{def}}^{(0)})^{-1} dz,} \quad (372)$$

with

$$\text{rank } P_L = 3, \quad \mathcal{H}_L = \text{Ran}(P_L).$$

This is the orthodox derivation of the family space: it is not assumed, but extracted spectrally from the microscopic three-defect operator.

CLIX. 8. SECTORAL OVERLAP OPERATORS

For each sector $f \in \{u, d, e, \nu\}$, the microscopic sectoral family operator is obtained by projection:

$$\boxed{\mathcal{Y}_f = P_L \delta \mathbb{O}_f P_L - P_L \delta \mathbb{O}_f Q_H (\mathbb{O}_H)^{-1} Q_H \delta \mathbb{O}_f P_L + \cdots}, \quad (373)$$

where $Q_H = 1 - P_L$ is the heavy-sector complement.

In a family basis $\{\phi_1, \phi_2, \phi_3\} \subset \mathcal{H}_L$,

$$\boxed{(M_f)_{ab} = \langle \phi_a, \mathcal{Y}_f \phi_b \rangle}. \quad (374)$$

This is the exact orthodox analogue of the overlap-integral construction in string/KK models: the effective family operator is derived from the microscopic zero-mode wavefunctions and their projected couplings.

CLX. 9. RELATION TO THE PREVIOUS WITNESS CHAIN

The previous reduced witness chain

$$P_B \rightarrow V^\mu \rightarrow u^\pm \rightarrow (M^\mu)_{ab} \rightarrow P_* K_i P_* \rightarrow (\lambda_\parallel, \alpha, \beta)$$

is now reinterpreted correctly: it is not the final theory, but the shell-projected low-energy shadow of the orthodox microscopic chain

$$\mathfrak{B} \rightarrow \mathbb{O}_{\text{def}}^{(0)} \rightarrow \psi_* \rightarrow P_L \rightarrow \mathcal{Y}_f.$$

So the two are consistent, but the latter is the fundamental route.

CLXI. 10. FINAL SHARP STATUS

The operator-theoretic and coefficient-extraction skeleton is now fixed. What remains open is not the route itself, but the actual TECT constitutive insertion:

$$Z^0(P), \quad Z^i(P), \quad M_0(P), \quad A_i^a(P), \quad B_{ij}^a(P), \quad C_i^a(P), \quad D_{ij}^a(P),$$

the explicit locked BCC defect background $(\bar{P}, \bar{X})$, the asymptotic bulk invertibility proof, the grading/index analysis, and the actual overlap integrals.

Hence the correct orthodox completion problem is:

$$\boxed{\text{insert the actual TECT constitutive data into the already-fixed microscopic operator skeleton.}} \quad (375)$$

CLXII. FINAL SUMMARY

The previous witness/certificate line should now be regarded as complete at its honest endpoint. The correct orthodox route is:

$$\boxed{\mathfrak{B} = (P^A, \Psi, X_i^a) \longrightarrow \mathbb{O}_{\text{def}}^{(0)} \longrightarrow \ker \mathbb{O}_{\text{def}}^{(0)} \longrightarrow P_L \longrightarrow \mathcal{Y}_f.}$$

The minimal microscopic working action is

$$\mathcal{L}_{\text{mic}} = \mathcal{L}_{\Psi}^{\text{ref}}[P, \Psi] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[P, \Psi] + \beta_X X_i^a K_i^a[P, \Psi].$$

Linearizing around a static defect background $(\bar{P}, \bar{X})$ and eliminating the heavy mediator gives the closed effective defect fluctuation equation

$$i\widehat{G}_{\text{def}}^0 \partial_t \psi = (-i\widehat{G}_{\text{def}}^i \partial_i + \widehat{M}_{\text{def}}) \psi.$$

After canonical normalization, this becomes the first-order defect operator

$$\mathbb{O}_{\text{def}}^{(0)} = i\tilde{\Gamma}_{\text{def}}^i(\partial_i + \Omega_i^{\text{def}}) + \Phi_{\text{def}}.$$

The single-defect theorem to be completed inside TECT is:

Fredholm / Callias structure + asymptotic bulk invertibility + grading + $\text{Ind}(\mathbb{O}_{\text{def}}^{(0)}) = 1 \implies$ one protected light mode.

If this is established, then for three sufficiently separated defects the family projector is

$$P_L = \frac{1}{2\pi i} \oint_{\Gamma} (z - \mathbb{O}_{\text{def}}^{(0)})^{-1} dz, \quad \text{rank } P_L = 3,$$

and the sectoral family operator is obtained by projection:

$$\mathcal{Y}_f = P_L \delta \mathbb{O}_f P_L - P_L \delta \mathbb{O}_f Q_H (\mathbb{O}_H)^{-1} Q_H \delta \mathbb{O}_f P_L + \dots$$

So the genuine unfinished core is no longer the witness algebra. It is the actual TECT constitutive insertion into this already-fixed microscopic operator skeleton.

CLXIII. ORTHODOX STEP 1 ON THE SAME WORKING BRANCH: ACTUAL DEFECT OPERATOR AND RADIAL REDUCTION

CLXIV. 1. REUSE-ONLY PRINCIPLE

We do *not* introduce a new microscopic branch. We keep exactly the same strongest explicit working branch used before.

Hence we reuse, without rederiving, the already fixed microscopic coefficient template

$$\mathcal{L}_{\text{mic}} = \mathcal{L}_P[P] + \frac{i}{2} \Psi^\dagger Z^0(P) \overset{\leftrightarrow}{\partial}_t \Psi + \frac{i}{2} \Psi^\dagger \left(Z^i(P) + X_j^a V_{ji}^a(P) \right) \overset{\leftrightarrow}{\partial}_i \Psi - \Psi^\dagger \left(M_0(P) - X_i^a U_i^a(P) \right) \Psi + \frac{M_X^2}{2} X_i^a X_i^a. \quad (376)$$

We also reuse the same minimal constitutive ansatz

$$Z^0(P) = z_0(\sigma) \mathbf{1}, \quad Z^i(P) = v_\perp(\sigma) \Gamma^i + v_c(\sigma) \hat{n}^i \Gamma_{\hat{n}}, \quad M_0(P) = m(\sigma) \Gamma^\sharp, \quad (377)$$

and the same Class II local-current coefficients

$$U_i^a(P) = u_X(\sigma) E_i^a \Gamma^\sharp, \quad (378)$$

$$V_{ij}^a(P) = E_i^a \left[v_E^{(X)}(\sigma) \delta_{j\ell} + v_O^{(X)}(\sigma) \epsilon_{j\ell m} \hat{n}_m + v_c^{(X)}(\sigma) \hat{n}_j \hat{n}_\ell \right] \Gamma^\ell. \quad (379)$$

All of the above are already fixed by the previous working-branch construction. The new calculation begins only now.

CLXV. 2. RADIAL SINGLE-DEFECT BACKGROUND

Take a static radial one-defect background centered at the origin:

$$r := |x|, \quad \hat{n}_i(x) := \frac{x_i}{r}, \quad \bar{\sigma}(x) = \bar{\sigma}(r).$$

Choose the locked orthonormal frame $E_i^a(x) \in SO(3)$ adapted to the defect,

$$E_i^3(x) = \hat{n}_i(x),$$

with E_i^1, E_i^2 spanning the tangent frame on S^2 .

The most general rotationally covariant locked mediator profile is

$$\bar{X}_i^a(x) = X_\perp(r)(\delta_i^a - \hat{n}_i \hat{n}^a) + X_\parallel(r) \hat{n}_i \hat{n}^a. \quad (380)$$

Its locked scalar contraction is

$$\chi_X(r) = \frac{1}{3} \bar{X}_i^a E_i^a = \frac{1}{3} (2X_\perp(r) + X_\parallel(r)). \quad (381)$$

CLXVI. 3. NEW CALCULATION: EXPLICIT DEFECT COEFFICIENT FIELDS

Substituting the radial background $(\bar{P}, \bar{X})$ into the reused constitutive template yields the actual microscopic defect coefficients.

Define

$$g_E(r) := v_\perp(\bar{\sigma}(r)) + 3\chi_X(r) v_E^{(X)}(\bar{\sigma}(r)), \quad (382)$$

$$g_O(r) := 3\chi_X(r) v_O^{(X)}(\bar{\sigma}(r)), \quad (383)$$

$$g_c(r) := v_c(\bar{\sigma}(r)) + 3\chi_X(r) v_c^{(X)}(\bar{\sigma}(r)), \quad (384)$$

and

$$m_{\text{eff}}(r) := m(\bar{\sigma}(r)) - 3\chi_X(r) u_X(\bar{\sigma}(r)). \quad (385)$$

Then the actual defect coefficients are

$$K_{\text{def}}^0(r) = z_0(\bar{\sigma}(r)) \mathbf{1}, \quad (386)$$

$$K_{\text{def}}^i(r) = g_E(r) \Gamma^i + g_O(r) \epsilon_{i\ell m} \hat{n}_m \Gamma^\ell + g_c(r) \hat{n}_i \Gamma_{\hat{n}}. \quad (387)$$

Introduce the transverse projector

$$\Pi_{ij}^\perp := \delta_{ij} - \hat{n}_i \hat{n}_j, \quad \Gamma_i^\perp := \Pi_{i\ell}^\perp \Gamma^\ell, \quad \Gamma^i = \Gamma_i^\perp + \hat{n}_i \Gamma_{\hat{n}}.$$

Also define

$$J_i(\hat{n}) := \epsilon_{i\ell m} \hat{n}_m \Gamma^\ell.$$

Since $\hat{n}_i J_i(\hat{n}) = 0$, the transport tensor splits exactly into three channels:

$$K_{\text{def}}^i(r) = g_E(r) \Gamma_i^\perp + g_R(r) \hat{n}_i \Gamma_{\hat{n}} + g_O(r) J_i(\hat{n}), \quad g_R(r) := g_E(r) + g_c(r). \quad (388)$$

This is the actual microscopic radial lift of the previously reduced (g_E, g_O, g_c) channels.

CLXVII. 4. NEW CALCULATION: EXPLICIT ZERO-ORDER TERM

We now compute the zeroth-order term

$$M_{\text{def}} = \frac{i}{2} \partial_i K_{\text{def}}^i - m_{\text{eff}}(r) \Gamma^\sharp.$$

Using

$$\partial_i \hat{n}_j = \frac{1}{r} \Pi_{ij}^\perp, \quad \partial_i \hat{n}_i = \frac{2}{r}, \quad \partial_i \Gamma_{\hat{n}} = \frac{1}{r} \Pi_{ij}^\perp \Gamma^j,$$

one finds:

a. *Radial part.*

$$\partial_i(g_R(r)\hat{n}_i\Gamma_{\hat{n}}) = \left(g'_R(r) + \frac{2g_R(r)}{r}\right)\Gamma_{\hat{n}}.$$

b. *Even tangential part.* Since $\hat{n}_i\Gamma_i^\perp = 0$ and

$$\partial_i\Gamma_i^\perp = -\frac{2}{r}\Gamma_{\hat{n}},$$

one obtains

$$\partial_i(g_E(r)\Gamma_i^\perp) = -\frac{2g_E(r)}{r}\Gamma_{\hat{n}}.$$

c. *Odd tangential part.* Because

$$\hat{n}_i J_i(\hat{n}) = 0, \quad \partial_i J_i(\hat{n}) = 0,$$

one gets

$$\partial_i(g_O(r)J_i(\hat{n})) = 0.$$

Therefore

$$\partial_i K_{\text{def}}^i(r) = \left(g'_R(r) + \frac{2g_c(r)}{r}\right)\Gamma_{\hat{n}}. \quad (389)$$

Hence

$$M_{\text{def}}(r) = \frac{i}{2} \left(g'_R(r) + \frac{2g_c(r)}{r}\right)\Gamma_{\hat{n}} - m_{\text{eff}}(r)\Gamma^\sharp. \quad (390)$$

This is a genuinely new step: the actual defect zeroth-order operator is now explicit on the same working branch.

CLXVIII. 5. CANONICAL NORMALIZATION

Let

$$z(r) := z_0(\bar{\sigma}(r)) > 0.$$

Since $K_{\text{def}}^0(r) = z(r)\mathbf{1}$, the canonical normalization is scalar:

$$S(r) = \sqrt{z(r)}\mathbf{1}, \quad \chi = S\psi.$$

Then

$$i\partial_t\chi = \mathbb{H}_{\text{def}}\chi,$$

with

$$\mathbb{H}_{\text{def}} = -i\Gamma_{\text{def}}^i(r)\partial_i + W_{\text{def}}(r, \hat{n}),$$

and because S is scalar,

$$\Gamma_{\text{def}}^i(r) = \frac{1}{z(r)}K_{\text{def}}^i(r). \quad (391)$$

Define the normalized radial coefficients

$$a(r) := \frac{g_R(r)}{z(r)}, \quad b(r) := \frac{g_E(r)}{z(r)}, \quad c(r) := \frac{g_O(r)}{z(r)}, \quad \mu(r) := \frac{m_{\text{eff}}(r)}{z(r)}. \quad (392)$$

A direct computation gives

$$W_{\text{def}}(r, \hat{n}) = -i \omega_r(r) \Gamma_{\hat{n}} + \mu(r) \Gamma^{\sharp}, \quad (393)$$

where

$$\omega_r(r) = \frac{1}{2z(r)} \left(g'_R(r) + \frac{2g_c(r)}{r} - g_R(r) \frac{z'(r)}{z(r)} \right). \quad (394)$$

Thus the canonically normalized defect operator is

$$\mathbb{H}_{\text{def}} = -i a(r) \Gamma_{\hat{n}} \partial_r - \frac{i b(r)}{r} \Gamma_i^{\perp} \nabla_i^{S^2} - \frac{i c(r)}{r} J_i(\hat{n}) \nabla_i^{S^2} - i \omega_r(r) \Gamma_{\hat{n}} + \mu(r) \Gamma^{\sharp}. \quad (395)$$

CLXIX. 6. SPHERICAL DIRAC REDUCTION

Define the spherical Dirac operator

$$\mathcal{D}_{S^2} := -i \Gamma_{\hat{n}} \Gamma_i^{\perp} \nabla_i^{S^2}. \quad (396)$$

Using

$$\Gamma_i^{\perp} \nabla_i^{S^2} = i \Gamma_{\hat{n}} \mathcal{D}_{S^2}, \quad J_i(\hat{n}) \nabla_i^{S^2} = -\mathcal{D}_{S^2},$$

the operator becomes

$$\mathbb{H}_{\text{def}} = -i a(r) \Gamma_{\hat{n}} \partial_r + \frac{b(r)}{r} \Gamma_{\hat{n}} \mathcal{D}_{S^2} + \frac{i c(r)}{r} \mathcal{D}_{S^2} - i \omega_r(r) \Gamma_{\hat{n}} + \mu(r) \Gamma^{\sharp}. \quad (397)$$

Now choose an angular eigenbasis $Y_{\kappa m \varsigma}$ satisfying

$$\mathcal{D}_{S^2} Y_{\kappa m \varsigma} = \kappa Y_{\kappa m \varsigma}, \quad \Gamma^{\sharp} Y_{\kappa m \varsigma} = \varsigma Y_{\kappa m \varsigma}, \quad \varsigma = \pm 1.$$

Because

$$\{\Gamma_{\hat{n}}, \mathcal{D}_{S^2}\} = 0, \quad \{\Gamma_{\hat{n}}, \Gamma^{\sharp}\} = 0,$$

each channel closes on

$$\text{span}\{Y_{\kappa m \varsigma}, \Gamma_{\hat{n}} Y_{\kappa m \varsigma}\}.$$

Thus a stationary mode of energy ω can be written as

$$\chi(r, \Omega) = u(r) Y(\Omega) + v(r) \Gamma_{\hat{n}} Y(\Omega),$$

and the full zero-mode problem reduces to the exact 1D first-order system

$$\omega u = -i a(r) v'(r) - i \omega_r(r) v(r) - \frac{b(r) \kappa}{r} v(r) + i \frac{c(r) \kappa}{r} u(r) + \varsigma \mu(r) u(r), \quad (398)$$

$$\omega v = -i a(r) u'(r) - i \omega_r(r) u(r) + \frac{b(r) \kappa}{r} u(r) - i \frac{c(r) \kappa}{r} v(r) - \varsigma \mu(r) v(r). \quad (399)$$

This is the actual orthodox zero-mode equation on the same working branch.

CLXX. 7. GAUGE-REMOVAL OF THE RADIAL CONNECTION TERM

The ω_r -term is kinematical and can be removed. Define

$$\boxed{\mathcal{I}(r) := \exp \left(\int^r \frac{\omega_r(s)}{a(s)} ds \right)}, \quad (400)$$

and rescale

$$u(r) = \mathcal{I}(r)^{-1} \tilde{u}(r), \quad v(r) = \mathcal{I}(r)^{-1} \tilde{v}(r).$$

Then the system becomes

$$\boxed{\omega \tilde{u} = -i a(r) \tilde{v}'(r) - \frac{b(r)\kappa}{r} \tilde{v}(r) + i \frac{c(r)\kappa}{r} \tilde{u}(r) + \varsigma \mu(r) \tilde{u}(r)}, \quad (401)$$

$$\boxed{\omega \tilde{v} = -i a(r) \tilde{u}'(r) + \frac{b(r)\kappa}{r} \tilde{u}(r) - i \frac{c(r)\kappa}{r} \tilde{v}(r) - \varsigma \mu(r) \tilde{v}(r)}. \quad (402)$$

So the genuinely dynamical coefficients are only

$$a(r), \quad b(r), \quad c(r), \quad \mu(r).$$

CLXXI. 8. PRECISE RELATION TO THE PREVIOUS WORKING-BRANCH AUDIT

The previous reduced witness/audit chain is recovered as the asymptotic or shell-projected shadow of these microscopic radial coefficients.

If the branch is asymptotically the simplest vacuum branch,

$$X_{\perp, \infty} = X_{\parallel, \infty} = 0 \quad \implies \quad \chi_{X, \infty} = 0,$$

then

$$\boxed{g_E(\infty) = v_{\perp}(\sigma_{\infty}), \quad g_O(\infty) = 0, \quad g_c(\infty) = v_c(\sigma_{\infty}), \quad m_{\text{eff}}(\infty) = m(\sigma_{\infty})}. \quad (403)$$

Thus the old finite audit quantities are not discarded. They are the asymptotic / projected reductions of the same actual microscopic defect operator.

CLXXII. 9. HONEST ENDPOINT OF THIS STEP

We have not yet solved the zero-mode problem. But we have now done the first genuinely orthodox calculation on the same working branch:

$$\boxed{(\bar{P}, \bar{X}) \implies K_{\text{def}}^0, K_{\text{def}}^i, M_{\text{def}} \implies \mathbb{H}_{\text{def}} \implies \text{exact 1D radial zero-mode system.}} \quad (404)$$

So the next frontier is no longer coefficient extraction. It is the analysis of the radial ODE (401)–(402).

CLXXIII. FINAL SUMMARY

Using the same working microscopic branch as before, and reusing the already fixed constitutive template, the new calculation is the explicit radial defect reduction.

For a radial locked defect with

$$\bar{X}_i^a = X_{\perp}(r)(\delta_i^a - \hat{n}_i \hat{n}^a) + X_{\parallel}(r) \hat{n}_i \hat{n}^a, \quad \chi_X(r) = \frac{1}{3}(2X_{\perp}(r) + X_{\parallel}(r)),$$

the actual defect coefficients are

$$K_{\text{def}}^0(r) = z_0(\bar{\sigma}(r)) \mathbf{1},$$

$$K_{\text{def}}^i(r) = g_E(r) \Gamma_i^\perp + g_R(r) \hat{n}_i \Gamma_{\hat{n}} + g_O(r) J_i(\hat{n}), \quad g_R(r) = g_E(r) + g_c(r),$$

with

$$g_E = v_\perp + 3\chi_X v_E^{(X)}, \quad g_O = 3\chi_X v_O^{(X)}, \quad g_c = v_c + 3\chi_X v_c^{(X)},$$

and

$$m_{\text{eff}} = m - 3\chi_X u_X.$$

The zeroth-order term is

$$M_{\text{def}}(r) = \frac{i}{2} \left(g'_R(r) + \frac{2g_c(r)}{r} \right) \Gamma_{\hat{n}} - m_{\text{eff}}(r) \Gamma^\sharp.$$

After canonical normalization with

$$a(r) = \frac{g_R(r)}{z_0(r)}, \quad b(r) = \frac{g_E(r)}{z_0(r)}, \quad c(r) = \frac{g_O(r)}{z_0(r)}, \quad \mu(r) = \frac{m_{\text{eff}}(r)}{z_0(r)},$$

the defect operator becomes

$$\mathbb{H}_{\text{def}} = -i a(r) \Gamma_{\hat{n}} \partial_r + \frac{b(r)}{r} \Gamma_{\hat{n}} \mathcal{D}_{S^2} + \frac{i c(r)}{r} \mathcal{D}_{S^2} - i \omega_r(r) \Gamma_{\hat{n}} + \mu(r) \Gamma^\sharp.$$

Each angular channel (κ, m, ς) reduces to a closed 1D first-order system for two radial functions. After removing the radial connection term, the genuinely dynamical data are only

$$a(r), \quad b(r), \quad c(r), \quad \mu(r).$$

Therefore the orthodox zero-mode problem on the same working branch is now fully reduced to an explicit radial ODE problem.

CLXXIV. ROADMAP FOR THE ORTHODOX TECT FLAVOR PROGRAM: K1 → K4

CLXXV. CURRENT STEP

The current logical position is

$$\boxed{C4 \supset K1A1 : \text{single-defect seed localization / one-defect acceptance.}}$$

The global internal C4 order is

$$\boxed{K1 : \text{common left-handed family subspace,} \quad K2 : \text{canonical kinetic normalization,}}$$

$$\boxed{K3 : \text{nondegenerate splittings,} \quad K4 : \text{hierarchy seed,} \quad K5 : \text{irreducible CP seed.}}$$

The logical order is

$$\boxed{K1 \rightarrow K2 \rightarrow (K3, K4) \rightarrow K5.}$$

From now on, we commit to the orthodox derivation order

$$\boxed{K1 \rightarrow K2 \rightarrow K3 \rightarrow K4}$$

without detouring into K5 or final CKM fitting before the derivation is closed.

CLXXVI. K1. COMMON LEFT-HANDED FAMILY SUBSPACE

A. Target statement

K1 asks for a common three-dimensional left-handed family space

$$\mathcal{H}_{\text{fam}} \cong \mathbb{C}^3$$

on which all relevant sectoral family operators act.

This is the first indispensable precondition. Without a common family space, CKM is not even well-defined.

B. Internal decomposition

The current program already decomposes K1 as

$$K1A \rightarrow K1B \rightarrow K1C \rightarrow K1D,$$

with:

$K1A$: common carrier sector exists, $K1B$: a three-dimensional family subspace is isolated,

$K1C$: sector perturbations preserve the family space effectively, $K1D$: physical left-handed interpretation.

C. Orthodox derivation chain

The orthodox chain is:

$$\text{microscopic TECT defect PDE} \rightarrow \mathbb{O}_{1\text{def}}^{(0)} \rightarrow \ker \mathbb{O}_{1\text{def}}^{(0)} \rightarrow \mathbb{O}_{3\text{def}}^{(0)} \rightarrow P_L \rightarrow \mathcal{H}_{\text{fam}}.$$

More explicitly:

a. K1A. Single-defect light mode. One must derive the actual single-defect operator

$$\mathbb{O}_{1\text{def}}^{(0)} = i\Gamma^i(\partial_i + \Omega_i^{\text{def}}) + \Phi_{\text{def}}$$

from the microscopic defect background and prove:

$$\dim \ker \mathbb{O}_{1\text{def}}^{(0)} = 1$$

in the protected light sector.

This step is the most microscopic part of K1. It requires:

1. self-adjointness / Fredholmness,
2. asymptotic invertibility of Φ_{def} ,
3. the relevant topological charge,
4. index value = 1.

b. K1B. Three-defect light-family cluster. Assuming one exponentially localized light mode per defect, for three sufficiently separated defects the three-defect operator

$$\mathbb{O}_{3\text{def}}^{(0)}$$

must have an isolated rank-three low spectral cluster. Then define the exact light projector

$$P_L = \frac{1}{2\pi i} \oint_{\Gamma} (z - \mathbb{O}_{3\text{def}}^{(0)})^{-1} dz.$$

The family space is

$$\mathcal{H}_{\text{fam}} := \text{Ran}(P_L), \quad \dim \mathcal{H}_{\text{fam}} = 3.$$

c. K1C. Stability under sector perturbations. For each physical sector f , the perturbation

$$\delta\mathbb{O}_f$$

must preserve the family sector effectively, meaning the low-heavy splitting stays controlled and the Schur-complement reduction is well-defined:

$$\mathcal{Y}_f = P_L \delta\mathbb{O}_f P_L - P_L \delta\mathbb{O}_f Q_H (\mathbb{O}_H)^{-1} Q_H \delta\mathbb{O}_f P_L + \dots$$

d. K1D. Left-handed interpretation. Finally one must identify the physical chirality/representation assignment of $\mathcal{H}_{\text{fam}}$, i.e. prove that the common family space is the left-handed physical one rather than an arbitrary low subspace.

D. K1 completion criterion

K1 is complete iff all of the following hold:

$$\dim \ker \mathbb{O}_{\text{def}}^{(0)} = 1, \quad \dim \text{Ran}(P_L) = 3,$$

$$\mathcal{Y}_f : \mathcal{H}_{\text{fam}} \rightarrow \mathcal{H}_{\text{fam}} \text{ is well-defined for the relevant sectors,}$$

$$\mathcal{H}_{\text{fam}} \text{ has the correct physical left-handed interpretation.}$$

CLXXVII. K2. CANONICAL KINETIC NORMALIZATION

A. Target statement

K2 requires that after projection to the family space, the kinetic form becomes the identity:

$$Z_{AB} \rightarrow \delta_{AB}.$$

This is necessary because diagonalizing a Hamiltonian before canonical normalization is not physically meaningful.

B. Orthodox derivation chain

Once K1 gives $\mathcal{H}_{\text{fam}}$, choose any basis $\{\phi_A\}_{A=1}^3 \subset \mathcal{H}_{\text{fam}}$. Define the projected kinetic Gram matrix

$$(Z_f)_{AB} = \langle \phi_A, \mathcal{K}_f \phi_B \rangle,$$

where $\mathcal{K}_f$ is the sectoral time-kinetic form inherited from the microscopic operator.

Then prove:

1. Hermiticity:

$$Z_f^\dagger = Z_f,$$

2. positivity:

$$Z_f > 0,$$

3. canonicalization by

$$\phi_A \mapsto \hat{\phi}_A = \sum_B (Z_f^{-1/2})_{BA} \phi_B.$$

The normalized family Hamiltonian is then

$$H_f^{\text{can}} = Z_f^{-1/2} \mathcal{Y}_f Z_f^{-1/2}.$$

C. K2 completion criterion

K2 is complete iff:

$$\boxed{Z_f \text{ is positive-definite on } \mathcal{H}_{\text{fam}},}$$

and thus

$$\boxed{H_f^{\text{can}} \text{ is the correctly normalized Hermitian family operator.}}$$

CLXXVIII. K3. NONDEGENERATE SPLITTINGS

A. Target statement

K3 requires that the projected family Hamiltonians have genuine nondegenerate splittings:

$$\boxed{\Delta_{21} \neq 0, \quad \Delta_{32} \neq 0, \quad \Delta_{31} \neq 0.}$$

Without this, the mixing problem is degenerate and CKM is not physically sharp.

B. Orthodox derivation chain

For each sector f , compute the canonically normalized Hermitian family operator

$$H_f^{\text{can}}.$$

Let its eigenvalues be

$$\lambda_1^{(f)}, \lambda_2^{(f)}, \lambda_3^{(f)}.$$

Then define

$$\Delta_{ij}^{(f)} = \lambda_i^{(f)} - \lambda_j^{(f)}.$$

The task is to prove that the sectoral microscopic overlap operator generates genuine diagonal splittings. In practice this comes from:

1. direct low-sector matrix elements V_{LL} ,
2. heavy-mediated Schur corrections

$$-V_{LH}(H^{(0)} + V_{HH})^{-1}V_{HL},$$

3. possible sector-sensitive primitive kernels.

C. K3 completion criterion

K3 is complete iff for the relevant sectors

$$\boxed{\text{spec}(H_f^{\text{can}}) \text{ is simple.}}$$

Equivalently,

$$\boxed{\Delta_{21}^{(f)} \neq 0, \quad \Delta_{32}^{(f)} \neq 0, \quad \Delta_{31}^{(f)} \neq 0.}$$

CLXXIX. K4. HIERARCHY SEED

A. Target statement

K4 requires that the off-diagonal family couplings carry a structural hierarchy seed:

$$\boxed{|h_{12}|, |h_{23}| \gg |h_{13}|}$$

or, in the sharper desired regime,

$$\boxed{|h_{12}| > |h_{23}| > |h_{13}|.}$$

This is the microscopic seed for CKM hierarchy.

B. Orthodox derivation chain

Starting from the same H_f^{can} , write its off-diagonal entries

$$h_{AB} = (H_f^{\text{can}})_{AB}.$$

In the orthodox route these entries are not guessed. They must be computed from:

1. the geometric separation of the three defects,
2. the tail decay of the localized single-defect zero modes,
3. the projected overlap kernels,
4. heavy-mediated Schur-complement corrections.

So the hierarchy theorem should have the form:

$$\text{defect geometry} + \text{exponential localization} + \text{allowed operator channels} \implies |h_{12}| > |h_{23}| > |h_{13}|$$

in the relevant parameter regime.

C. K4 completion criterion

K4 is complete iff one proves a structural off-diagonal ordering region:

$$\boxed{\exists \mathcal{R}_{\text{hier}} \text{ in parameter space such that } |h_{12}| > |h_{23}| > |h_{13}|.}$$

CLXXX. WHAT IS ALREADY KNOWN VS WHAT IS STILL OPEN

A. Already structurally organized

The C4 program is already organized in the right logical order:

$$K1 \rightarrow K2 \rightarrow (K3, K4) \rightarrow K5.$$

Moreover the current frontier is still inside K1A1, namely single-defect seed localization / one-defect acceptance.

B. Still genuinely open

The still genuinely open microscopic load-bearing tasks are:

1. derivation of the actual TECT single-defect operator,
2. proof that one defect carries one protected light mode,
3. construction of the exact three-defect projector,
4. explicit evaluation of the sectoral overlap operators,
5. canonical kinetic reduction on the family space,
6. proof of nondegenerate splittings,
7. proof of a structural hierarchy region.

CLXXXI. STRICT EXECUTION POLICY FROM NOW ON

From now on, the execution policy is:

do not jump to CKM fitting or K5 before K1–K4 are derived.

The operational sequence will be:

$K1A \rightarrow K1B \rightarrow K1C \rightarrow K1D \rightarrow K2 \rightarrow K3 \rightarrow K4.$

The immediate next mathematical step is:

$K1A$: derive/verify the single-defect operator and the one-light-mode theorem.

CLXXXII. FINAL SUMMARY

The C4 flavor program must be executed in the strict order

$K1 \rightarrow K2 \rightarrow K3 \rightarrow K4$

before any return to CKM fitting or K5.

The meanings are:

$K1$: common left-handed family subspace,

built through

$K1A$: single-defect light mode, $K1B$: isolated rank-three family projector, $K1C$: sector-stable family subspace, $K1D$:

$K2$: canonical kinetic normalization, $Z_{AB} \rightarrow \delta_{AB}.$

$K3$: nondegenerate splittings, $\Delta_{21}, \Delta_{32}, \Delta_{31} \neq 0.$

$K4$: hierarchy seed, $|h_{12}| > |h_{23}| > |h_{13}|$ in a structural parameter region.

The immediate next step is still

$K1A$: single-defect operator $\rightarrow$ one protected light mode.

So the strict operational order from now on is

$K1A \rightarrow K1B \rightarrow K1C \rightarrow K1D \rightarrow K2 \rightarrow K3 \rightarrow K4.$

CLXXXIII. K1A2: PROTECTED-CHANNEL CLASSIFICATION FROM THE RADIAL CALLIAS CORE

CLXXXIV. 1. STARTING POINT ALREADY ESTABLISHED

On the same working branch, the single-defect problem has already been reduced to a radial first-order system with four real coefficient functions

$$a(r), \quad b(r), \quad c(r), \quad \mu(r),$$

and the large- r normalizable branch is governed by

$$\exp\left(-\frac{|\mu_\infty|}{a_\infty}r\right).$$

So the remaining K1A question is no longer the derivation of the radial operator itself. It is the channel classification problem:

which (κ, ς) channel carries a normalizable protected zero generator?

CLXXXV. 2. RADIAL ZERO-MODE EQUATIONS AT $\omega = 0$

After gauge removal of the radial connection term, each fixed (κ, m, ς) channel is governed by

$$0 = -i a(r) \tilde{v}'(r) - \frac{\kappa b(r)}{r} \tilde{v}(r) + i \frac{\kappa c(r)}{r} \tilde{u}(r) + \varsigma \mu(r) \tilde{u}(r), \quad (405)$$

$$0 = -i a(r) \tilde{u}'(r) + \frac{\kappa b(r)}{r} \tilde{u}(r) - i \frac{\kappa c(r)}{r} \tilde{v}(r) - \varsigma \mu(r) \tilde{v}(r). \quad (406)$$

The notes already identify the graded Callias core by isolating the real superpotential

$$W_\kappa^{\text{core}}(r) = \mu(r) + \frac{\kappa b(r)}{r}. \quad (407)$$

CLXXXVI. 3. GRADED CHIRAL BLOCK FORM

Introduce a grading operator $\mathcal{C}$ satisfying

$$\{\mathcal{C}, \Gamma_{\hat{n}}\} = 0, \quad \{\mathcal{C}, \Gamma^\sharp\} = 0, \quad [\mathcal{C}, \mathcal{D}_{S^2}] = 0.$$

Then the radial operator splits into the genuine graded Callias core plus the odd tangential perturbation:

$$\mathbb{H}_\kappa = \mathbb{H}_{C,\kappa} + \mathbb{V}_{O,\kappa},$$

with

$$\mathbb{V}_{O,\kappa} = i \frac{\kappa c(r)}{r}$$

in the reduced channel.

The Callias core has the chiral block form

$$\mathbb{H}_{C,\kappa} = \begin{pmatrix} 0 & Q_{\kappa,-}^{\text{core}} \\ Q_{\kappa,+}^{\text{core}} & 0 \end{pmatrix}, \quad Q_{\kappa,\pm}^{\text{core}} = \pm a(r) \frac{d}{dr} + W_\kappa^{\text{core}}(r). \quad (408)$$

This is exactly the form already identified in the notes.

CLXXXVII. 4. THE $c(r)$ -TERM DOES NOT AFFECT NORMALIZABILITY

The odd tangential term produces only the purely imaginary shift

$$i \frac{\kappa c(r)}{r}$$

in the channel superpotential. Hence, for a scalar phase

$$\Theta_\kappa(r) := \int^r \frac{\kappa c(s)}{a(s)s} ds,$$

the transformation

$$\psi_\kappa(r) \mapsto e^{-i\Theta_\kappa(r)} \psi_\kappa(r)$$

removes the imaginary part from the first-order generators.

Therefore the L^2 -normalizability of channel generators is controlled entirely by the real superpotential

$$W_\kappa^{\text{core}}(r) = \mu(r) + \frac{\kappa b(r)}{r}.$$

So from this point onward, the zero-mode classification reduces to a real one-dimensional supersymmetric/Callias problem.

CLXXXVIII. 5. EXPLICIT CHANNEL GENERATORS

The zero-generator equations are

$$Q_{\kappa,+}^{\text{core}} \psi_{\kappa,+} = 0, \quad Q_{\kappa,-}^{\text{core}} \psi_{\kappa,-} = 0,$$

that is,

$$a(r) \psi'_{\kappa,+}(r) + W_\kappa^{\text{core}}(r) \psi_{\kappa,+}(r) = 0,$$

$$-a(r) \psi'_{\kappa,-}(r) + W_\kappa^{\text{core}}(r) \psi_{\kappa,-}(r) = 0.$$

Assuming $a(r) > 0$, the formal solutions are

$$\boxed{\psi_{\kappa,+}(r) = C_+ \exp\left(-\int^r \frac{W_\kappa^{\text{core}}(s)}{a(s)} ds\right)}, \quad (409)$$

$$\boxed{\psi_{\kappa,-}(r) = C_- \exp\left(+\int^r \frac{W_\kappa^{\text{core}}(s)}{a(s)} ds\right)}. \quad (410)$$

Hence the channel index is

$$\boxed{\text{ind}_\kappa = \dim \ker Q_{\kappa,+}^{\text{core}} - \dim \ker Q_{\kappa,-}^{\text{core}}}, \quad (411)$$

and it is nonzero iff exactly one of $\psi_{\kappa,\pm}$ is L^2 -normalizable.

CLXXXIX. 6. ASYMPTOTIC CONDITION AT LARGE r

The notes already established the large- r decay law

$$\tilde{f}(r), \tilde{g}(r) \sim \exp\left(-\frac{|\mu_\infty|}{a_\infty} r\right).$$

This means that asymptotically

$$W_\kappa^{\text{core}}(r) \rightarrow \mu_\infty \quad (r \rightarrow \infty)$$

because the $\kappa b(r)/r$ term vanishes like $1/r$.

Therefore:

1. if $\mu_\infty/a_\infty > 0$, then $\psi_{\kappa,+}$ is the decaying asymptotic branch and $\psi_{\kappa,-}$ grows;
2. if $\mu_\infty/a_\infty < 0$, the roles are reversed.

So the asymptotic chirality of the decaying branch is determined by the sign of μ_∞/a_∞ .

CXC. 7. NEAR-CORE REGULARITY CONDITION

Near $r = 0$, the singular part of the real superpotential is

$$W_\kappa^{\text{core}}(r) \sim \mu(0) + \frac{\kappa b(0)}{r}.$$

Hence the formal chiral generators behave like

$$\psi_{\kappa,\pm}(r) \sim r^{\mp \kappa b(0)/a(0)}$$

up to regular multiplicative factors.

Therefore regularity at the core selects one or both local branches depending on the sign and size of

$$\frac{\kappa b(0)}{a(0)}.$$

Thus the full existence problem is a standard matching problem:

$$\text{regular near-core branch} \quad \longleftrightarrow \quad \text{decaying large-}r \text{ branch.}$$

CXCI. 8. PRACTICAL SUFFICIENT CRITERION FOR A PROTECTED CHANNEL

We now state a practical sufficient criterion that is new but follows directly from the reduced real-generator structure. [Practical sufficient protected-channel criterion] Assume:

1. $a(r)$ is continuous and strictly positive on $(0, \infty)$;
2. $\mu_\infty > 0$, so the asymptotically decaying chiral branch is $\psi_{\kappa,+}$;
3. for some channel $\kappa = \kappa_*$, the real superpotential satisfies

$$W_{\kappa_*}^{\text{core}}(r) < 0 \quad \text{for } 0 < r < r_c, \quad W_{\kappa_*}^{\text{core}}(r) > 0 \quad \text{for } r > R,$$

for some $0 < r_c < R < \infty$;

4. for every other channel $\kappa \neq \kappa_*$, the function $W_\kappa^{\text{core}}(r)$ does not produce an L^2 -normalizable chiral generator (for example, it has no sign-changing profile compatible with both core regularity and asymptotic decay).

Then:

$$\dim \ker Q_{\kappa_*,+}^{\text{core}} \geq 1, \quad \dim \ker Q_{\kappa_*, -}^{\text{core}} = 0,$$

so

$$\text{ind}_{\kappa_*} > 0.$$

If moreover no other channel carries nonzero index, then exactly one lowest channel supports one protected light mode.

Because $a(r) > 0$, the chiral generators are controlled by (409)–(410). The positive asymptotic limit $W_{\kappa}^{\text{core}} \rightarrow \mu_{\infty} > 0$ forces $\psi_{\kappa,+}$ to decay and $\psi_{\kappa,-}$ to grow at infinity. The assumed negative core region means the exponent of $\psi_{\kappa,+}$ acquires the correct sign to remain regular/mild near the origin instead of being excluded immediately. This produces an admissible L^2 generator in the $+$ chiral block.

By contrast, $\psi_{\kappa,-}$ necessarily grows at infinity and is not normalizable. Hence $\text{ind}_{\kappa_*} > 0$.

If all other channels have zero index, then exactly one protected channel survives.

CXCII. 9. EXACT LOWEST-CHANNEL REFORMULATION

The notes themselves state the final concrete problem as:

determine which κ -channel has exactly one L^2 -normalizable chiral generator.

So the single-defect one-mode theorem on the actual TECT branch is equivalent to the following channel classification statement:

$\exists! \kappa_* \text{ of lowest } |\kappa| \text{ such that } \text{ind}_{\kappa_*} \neq 0, \quad \text{ind}_{\kappa} = 0 \text{ for all } \kappa \neq \kappa_*.$

(412)

This is now the sharpest K1A2 target.

CXCIII. 10. HONEST REMAINING TASKS

What remains TECT-specific is exactly what the notes already identify:

1. the actual constitutive profiles $a(r), b(r), c(r), \mu(r)$;
2. the explicit grading operator $\mathcal{C}$;
3. the angular spectrum and multiplicities;
4. the proof that exactly one lowest channel carries one L^2 -normalizable chiral generator.

Therefore K1A2 is no longer a broad PDE problem. It is a concrete protected-channel classification problem for the real superpotential

$$W_{\kappa}^{\text{core}}(r) = \mu(r) + \frac{\kappa b(r)}{r}.$$

CXCIV. FINAL SUMMARY

The radial single-defect zero-mode problem on the same working branch is controlled by the graded Callias core

$$\mathbb{H}_{C,\kappa} = \begin{pmatrix} 0 & Q_{\kappa,-}^{\text{core}} \\ Q_{\kappa,+}^{\text{core}} & 0 \end{pmatrix}, \quad Q_{\kappa,\pm}^{\text{core}} = \pm a(r) \frac{d}{dr} + W_{\kappa}^{\text{core}}(r),$$

with real superpotential

$$W_{\kappa}^{\text{core}}(r) = \mu(r) + \frac{\kappa b(r)}{r}.$$

The g_O -term contributes only the imaginary shift

$$i \frac{\kappa c(r)}{r},$$

so it changes only the phase of the channel generator and not its L^2 -normalizability.

Hence the protected-channel classification is controlled entirely by W_κ^{core} .

The formal chiral generators are

$$\psi_{\kappa,+}(r) = C_+ \exp\left(-\int^r \frac{W_\kappa^{\text{core}}(s)}{a(s)} ds\right),$$

$$\psi_{\kappa,-}(r) = C_- \exp\left(+\int^r \frac{W_\kappa^{\text{core}}(s)}{a(s)} ds\right).$$

Thus the channel index is

$$\text{ind}_\kappa = \dim \ker Q_{\kappa,+}^{\text{core}} - \dim \ker Q_{\kappa,-}^{\text{core}},$$

and it is nonzero iff exactly one chiral generator is L^2 -normalizable.

At large r , the asymptotic decay is governed by

$$\exp\left(-\frac{|\mu_\infty|}{a_\infty} r\right),$$

so the asymptotically decaying chirality is fixed by the sign of μ_∞/a_∞ .

Therefore the single-defect one-mode problem is now equivalent to the sharp channel-classification problem:

$$\boxed{\exists! \kappa_* \text{ of lowest } |\kappa| \text{ such that } \text{ind}_{\kappa_*} \neq 0, \quad \text{ind}_\kappa = 0 \text{ for all } \kappa \neq \kappa_*}.$$

So K1A2 has been reduced to the final concrete form: determine the actual TECT profiles $a(r), b(r), c(r), \mu(r)$ and classify which lowest κ -channel has exactly one L^2 -normalizable chiral generator.

CXCV. K1A2: LOWEST- $|\kappa|$ CHANNEL THEOREM AND THE ONE-PROTECTED-MODE CRITERION

CXCVI. 1. REUSED RADIAL CALLIAS CORE

We reuse the already established reduced radial Callias core. For each projected angular channel κ , the graded first-order generators are

$$\boxed{Q_{\kappa,\pm}^{\text{core}} = \pm a(r) \frac{d}{dr} + W_\kappa^{\text{core}}(r), \quad W_\kappa^{\text{core}}(r) = \mu(r) + \frac{\kappa b(r)}{r}.} \quad (413)$$

The odd tangential coefficient $c(r)$ contributes only the phase factor

$$\exp\left(-i\kappa \int^r \frac{c(s)}{a(s)s} ds\right), \quad (414)$$

so it does not affect L^2 -normalizability and therefore does not enter the protected-channel counting.

Hence the full counting problem reduces to the real superpotential W_κ^{core} .

CXCVII. 2. FORMAL CHANNEL GENERATORS

The zero-generator equations are

$$Q_{\kappa,+}^{\text{core}} u_{\kappa,+} = 0, \quad Q_{\kappa,-}^{\text{core}} u_{\kappa,-} = 0.$$

Their formal solutions are

$$u_{\kappa,+}(r) = \exp\left(-\int^r \frac{W_{\kappa}^{\text{core}}(s)}{a(s)} ds\right), \quad u_{\kappa,-}(r) = \exp\left(+\int^r \frac{W_{\kappa}^{\text{core}}(s)}{a(s)} ds\right). \quad (415)$$

The channel index is

$$\text{ind}_{\kappa} = \dim \ker Q_{\kappa,+}^{\text{core}} - \dim \ker Q_{\kappa,-}^{\text{core}}. \quad (416)$$

So $\text{ind}_{\kappa} \neq 0$ iff exactly one of $u_{\kappa,\pm}$ is L^2 -normalizable.

CXCVIII. 3. CORE AND BULK ASYMPTOTICS

Near the core,

$$a(r) = a_0 + O(r^2), \quad b(r) = b_0 + O(r^2), \quad \mu(r) = \mu_0 + O(r^2), \quad (417)$$

hence

$$W_{\kappa}^{\text{core}}(r) = \mu_0 + \frac{\kappa b_0}{r} + O(r). \quad (418)$$

At large radius,

$$a(r) = a_{\infty} + o(1), \quad b(r) = b_{\infty} + o(1), \quad \mu(r) = \mu_{\infty} + o(1), \quad (419)$$

so

$$W_{\kappa}^{\text{core}}(r) = \mu_{\infty} + \frac{\kappa b_{\infty}}{r} + o(1) \rightarrow \mu_{\infty}. \quad (420)$$

Thus if $\mu_{\infty} > 0$, only $u_{\kappa,+}$ can decay at infinity.

CXCIX. 4. NEAR-CORE REGULARITY EXPONENT

Using (418) and $a(r) \sim a_0$, one gets

$$\frac{W_{\kappa}^{\text{core}}(r)}{a(r)} = \frac{\mu_0}{a_0} + \frac{\kappa b_0}{a_0} \frac{1}{r} + O(r).$$

Therefore

$$u_{\kappa,+}(r) \sim r^{-\kappa b_0/a_0}, \quad u_{\kappa,-}(r) \sim r^{+\kappa b_0/a_0}.$$

In the radial L^2 -measure $r^2 dr$, the condition

$$\int_0^\infty r^2 |u(r)|^2 dr < \infty$$

implies:

$$u_{\kappa,+} \in L_{\text{loc}}^2(0) \iff -\frac{\kappa b_0}{a_0} > -\frac{3}{2} \iff \frac{\kappa b_0}{a_0} < \frac{3}{2}, \quad (421)$$

and more sharply, the sufficient margin used in the notes is the stronger condition

$$W_{\kappa}^{\text{core}}(r) \leq -\left(\frac{1}{2} + \varepsilon\right) \frac{a(r)}{r} \quad \text{near } r = 0, \quad (422)$$

which guarantees the decaying asymptotic branch is also core-regular.

Likewise, if

$$W_{\kappa}^{\text{core}}(r) \geq \left(\frac{1}{2} + \varepsilon\right) \frac{a(r)}{r} \quad \text{near } r = 0, \quad (423)$$

then the decaying asymptotic chiral branch fails the core L^2 -regularity condition.

CC. 5. LOWEST SIGNED CHANNEL THEOREM

We now specialize to the most relevant case already isolated in the notes.

Assume:

$$\boxed{\mu_\infty > 0, \quad a_0 > 0, \quad a_\infty > 0, \quad b_0 > 0.} \quad (424)$$

Assume further that after remote-gap projection there exists a unique signed lowest candidate

$$\boxed{\kappa_* < 0} \quad (425)$$

of multiplicity one, while every competing projected channel satisfies

$$\boxed{\kappa \geq \kappa_{\text{comp}} > 0.} \quad (426)$$

Finally assume the core margins

$$\boxed{\frac{|\kappa_*|b_0}{a_0} \geq \frac{1}{2} + \varepsilon_*,} \quad (427)$$

and for every competitor $\kappa \neq \kappa_*$,

$$\boxed{\frac{\kappa b_0}{a_0} \geq \frac{1}{2} + \varepsilon_\kappa} \quad (428)$$

for some $\varepsilon_*, \varepsilon_\kappa > 0$.

[Lowest-channel one-protected-mode sufficient criterion] Under (424)–(428), the projected lowest channel κ_* carries exactly one L^2 -normalizable chiral generator, while all competing channels carry none. Hence

$$\boxed{\text{Ind}(\mathbb{H}_C) = 1,} \quad (429)$$

and the defect carries exactly one protected light mode.

Since $\mu_\infty > 0$, only $u_{\kappa,+}$ can decay at infinity. So channel selection is reduced entirely to near-core L^2 -regularity of $u_{\kappa,+}$.

For the unique signed lowest channel $\kappa_* < 0$, the near-core asymptotic is

$$W_{\kappa_*}^{\text{core}}(r) \sim \mu_0 - \frac{|\kappa_*|b_0}{r}.$$

By (427), the singular term dominates with negative sign and satisfies the sufficient negative core margin (422). Thus $u_{\kappa_*,+}$ is L^2 -regular at the core and decays at infinity, so

$$\dim \ker Q_{\kappa_*,+}^{\text{core}} = 1, \quad \dim \ker Q_{\kappa_*, -}^{\text{core}} = 0.$$

For every competing channel $\kappa > 0$, one has

$$W_\kappa^{\text{core}}(r) \sim \mu_0 + \frac{\kappa b_0}{r},$$

and by (428) the positive singular term enforces the sufficient exclusion margin (423). Hence the decaying asymptotic branch $u_{\kappa,+}$ fails the core L^2 -regularity condition. Therefore

$$\dim \ker Q_{\kappa,+}^{\text{core}} = \dim \ker Q_{\kappa,-}^{\text{core}} = 0 \quad (\kappa \neq \kappa_*).$$

Summing the channel indices gives

$$\text{Ind}(\mathbb{H}_C) = 1.$$

CCI. 6. TRANSLATION INTO PREVIOUSLY EXTRACTED LOW-CORE COEFFICIENTS

The notes already identify the defect-core coefficients with the finite acceptance data:

$$\boxed{a_0 = \frac{\xi_{EGE} + \xi_c g_c}{z_{0,0}}, \quad b_0 = \frac{\xi_{EGE}}{z_{0,0}}, \quad c_0 = \frac{\xi_{OGO}}{z_{0,0}},} \quad (430)$$

and

$$\boxed{\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}.} \quad (431)$$

Substituting (430) into the protected-channel margin (427) yields

$$\boxed{\frac{|\kappa_*| \xi_{EGE}}{\xi_{EGE} + \xi_c g_c} \geq \frac{1}{2} + \varepsilon_*} \quad (432)$$

A sharp sufficient form is

$$\boxed{2|\kappa_*| \xi_{EGE} > \xi_{EGE} + \xi_c g_c} \quad (433)$$

This is exactly the bridge already recorded in the notes between the Bloch/Class-II transport extraction and the radial one-mode criterion.

In the lowest-channel case $|\kappa_*| = 1$, (433) reduces to

$$\boxed{\xi_{EGE} > \xi_c g_c} \quad (434)$$

So the radial protected-channel theorem is perfectly consistent with the previously obtained one-defect finite audit.

CCII. 7. HONEST STATUS AFTER THIS STEP

After this step, the conceptual content of K1A2 is closed:

$$\boxed{\text{remote-gap isolation} \implies \text{lowest projected channel} \implies \text{real superpotential classification} \implies \text{one protected light mode criterion.}} \quad (435)$$

What remains TECT-specific is no longer the logic. It is the explicit verification of:

$$\boxed{\kappa_*, \quad a_0, \quad b_0, \quad \mu_\infty, \quad \text{and the competing-channel inequalities}} \quad (436)$$

for the actual intended branch.

CCIII. FINAL SUMMARY

For each projected channel κ , the protected-mode counting is controlled by the real superpotential

$$W_\kappa^{\text{core}}(r) = \mu(r) + \frac{\kappa b(r)}{r},$$

with formal chiral generators

$$u_{\kappa,+}(r) = \exp\left(-\int^r \frac{W_\kappa^{\text{core}}(s)}{a(s)} ds\right), \quad u_{\kappa,-}(r) = \exp\left(+\int^r \frac{W_\kappa^{\text{core}}(s)}{a(s)} ds\right).$$

If $\mu_\infty > 0$, only $u_{\kappa,+}$ can decay at infinity. Hence channel selection reduces entirely to near-core L^2 -regularity of $u_{\kappa,+}$.

Near the core,

$$W_\kappa^{\text{core}}(r) = \mu_0 + \frac{\kappa b_0}{r} + O(r),$$

so the sign of the singular $1/r$ term is governed by κb_0 .

Therefore, if

$$\mu_\infty > 0, \quad a_0 > 0, \quad b_0 > 0,$$

and after remote-gap projection there exists a unique signed lowest channel

$$\kappa_* < 0$$

with

$$\frac{|\kappa_*| b_0}{a_0} \geq \frac{1}{2} + \varepsilon_*,$$

while every competing channel $\kappa \neq \kappa_*$ satisfies

$$\frac{\kappa b_0}{a_0} \geq \frac{1}{2} + \varepsilon_\kappa,$$

then

$$\text{Ind}(\mathbb{H}_C) = 1,$$

and the defect carries exactly one protected light mode.

Using the previously extracted finite acceptance coefficients

$$a_0 = \frac{\xi_E g_E + \xi_c g_c}{z_{0,0}}, \quad b_0 = \frac{\xi_E g_E}{z_{0,0}},$$

the decisive protected-channel inequality becomes

$$2|\kappa_*| \xi_E g_E > \xi_E g_E + \xi_c g_c.$$

In the lowest-channel case $|\kappa_*| = 1$, this reduces to

$$\boxed{\xi_E g_E > \xi_c g_c.}$$

Thus the radial protected-channel theorem and the previously derived one-defect finite audit are exactly consistent.

CCIV. K1A3: ACTUAL EXTRACTION OF κ_* AND THE COMPETING-CHANNEL STRUCTURE

CCV. 1. GOAL

At the previous step, the one-protected-mode theorem was reduced to the hypotheses:

$$\mu_\infty > 0, \quad a_0 > 0, \quad b_0 > 0,$$

together with the existence of a unique signed lowest projected channel

$$\kappa_* < 0$$

and the exclusion of all competing projected channels.

The present K1A3 task is to extract this κ_* and the competing-channel structure from the actual angular decomposition of the single-defect operator.

CCVI. 2. ANGULAR REDUCTION AND RAW S^2 SPECTRUM

After radial reduction, the angular dependence is governed by the spherical Dirac operator

$$\mathcal{D}_{S^2}.$$

Its spectrum is

$$\boxed{\mathcal{D}_{S^2} Y_{\kappa m \varsigma} = \kappa Y_{\kappa m \varsigma}, \quad \kappa \in \{\pm 1, \pm 2, \pm 3, \dots\}.} \quad (437)$$

Therefore the raw lowest angular magnitude is always

$$\boxed{|\kappa|_{\min} = 1.} \quad (438)$$

So, if after remote-gap isolation and low-slot projection there exists a unique signed lowest surviving channel, then it must be one of

$$\kappa_* = -1 \quad \text{or} \quad \kappa_* = +1.$$

This is already enough to reduce the abstract “unique signed lowest channel” hypothesis to a two-sign choice.

CCVII. 3. SIGN SELECTION FROM THE CORE SUPERPOTENTIAL

The protected-channel counting is controlled by

$$\boxed{W_{\kappa}^{\text{core}}(r) = \mu(r) + \frac{\kappa b(r)}{r}.} \quad (439)$$

Near the core,

$$W_{\kappa}^{\text{core}}(r) = \mu_0 + \frac{\kappa b_0}{r} + O(r).$$

So the sign of the singular $1/r$ term is the sign of κb_0 .

Assume, as in the working theorem, that

$$b_0 > 0.$$

Then:

- for $\kappa < 0$, the singular term is negative;
- for $\kappa > 0$, the singular term is positive.

But under $\mu_{\infty} > 0$, the asymptotically decaying branch is the $+$ -chiral branch. So the only way to get a protected channel is that the same $u_{\kappa,+}$ branch be:

1. decaying at infinity;
2. and L^2 -regular at the core.

The negative singular core sign is exactly what makes this possible in the sufficient theorem. Therefore, once $b_0 > 0$, the protected lowest signed channel must be negative:

$$\boxed{\kappa_* < 0.} \quad (440)$$

Since the lowest possible $|\kappa|$ is 1, it follows that

$$\boxed{\kappa_* = -1.} \quad (441)$$

This is the actual sign extraction hidden inside the earlier profile theorem.

CCVIII. 4. COMPETING-CHANNEL STRUCTURE

Because the raw angular spectrum is

$$\kappa = \pm 1, \pm 2, \pm 3, \dots,$$

and the unique lowest signed protected channel is $\kappa_* = -1$, every competing projected channel must satisfy one of the following:

1. it is removed by the remote-gap theorem;
2. or it survives but has positive sign, hence

$$\kappa \geq 1.$$

Thus the projected competing structure becomes

$$\boxed{\kappa_{\text{comp}} \geq 1.} \quad (442)$$

This is exactly the structural content needed in the previous one-mode theorem.

CCIX. 5. EXPLICIT CANONICAL BRANCH INTERPRETATION

For the explicit canonical branch

$$T_* = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the practical lowest-channel statement already recorded in the notes is

$$\boxed{\kappa_* = -1, \quad \kappa_{\text{comp}} \geq 1.} \quad (443)$$

So K1A3 is not inventing a new condition. It is identifying the precise operator-theoretic meaning of the branch statement already used operationally.

CCX. 6. REDUCTION OF THE DECISIVE INEQUALITY

The previous transport bridge was

$$\boxed{2|\kappa_*| \xi_E g_E > \xi_E g_E + \xi_c g_c.} \quad (444)$$

Now insert the actual lowest-channel extraction

$$|\kappa_*| = 1.$$

Then (444) becomes

$$\boxed{\xi_E g_E > \xi_c g_c.} \quad (445)$$

Likewise, for every competing projected channel $\kappa \geq 1$, the exclusion condition is

$$\boxed{2\kappa \xi_E g_E > \xi_E g_E + \xi_c g_c.} \quad (446)$$

So after K1A3, the entire lowest-channel classification is encoded in the explicit core coefficients and this finite scalar bridge.

CCXI. 7. ACTUAL CONTENT OF K1A3

Therefore K1A3 consists of exactly three operator-theoretic facts:

1. the projected angular spectrum is inherited from the spherical Dirac spectrum, so

$$|\kappa|_{\min} = 1;$$

2. the protected-channel sign is fixed by the sign of κb_0 , and for $b_0 > 0$ this selects $\kappa_* = -1$;
3. every surviving competitor either is removed by remote-gap isolation or satisfies $\kappa \geq 1$.

Thus the abstract hypothesis “unique signed lowest channel” has now been sharpened to the explicit working statement

$$\boxed{\kappa_* = -1, \quad \kappa_{\text{comp}} \geq 1.} \tag{447}$$

CCXII. 8. HONEST ENDPOINT

What remains after K1A3 is no longer the identity of κ_* . It is the explicit verification that the projected operator on the intended branch indeed realizes:

$$\mu_\infty > 0, \quad a_0 > 0, \quad b_0 > 0, \quad \kappa_* = -1, \quad \kappa_{\text{comp}} \geq 1,$$

together with the remote-gap exclusion of all higher sectors.

So K1A3 closes the channel-label part of the problem. The remaining work is the actual branchwise inequality verification.

CCXIII. FINAL SUMMARY

The projected angular channel labels come from the spherical Dirac spectrum

$$\kappa \in \{\pm 1, \pm 2, \pm 3, \dots\},$$

so the raw lowest angular magnitude is always

$$|\kappa|_{\min} = 1.$$

Therefore, if after remote-gap isolation and low-slot projection there exists a unique signed lowest surviving channel, it must be either $\kappa_* = -1$ or $\kappa_* = +1$.

The protected-channel sign is then determined by the core superpotential

$$W_\kappa^{\text{core}}(r) = \mu(r) + \frac{\kappa b(r)}{r}.$$

When $b_0 > 0$, the singular core term has negative sign precisely for $\kappa < 0$. Under $\mu_\infty > 0$, this is the sign needed for the asymptotically decaying chiral generator to remain L^2 -regular at the core.

Hence the protected lowest signed channel is

$$\boxed{\kappa_* = -1.}$$

All competing projected channels are either removed by remote-gap isolation or satisfy

$$\boxed{\kappa_{\text{comp}} \geq 1.}$$

This is exactly the practical lowest-channel structure already used in the explicit canonical branch.

Consequently, the decisive transport bridge

$$2|\kappa_*| \xi_E g_E > \xi_E g_E + \xi_c g_c$$

reduces to

$$\boxed{\xi_E g_E > \xi_c g_c}$$

in the lowest-channel case.

Thus K1A3 closes the channel-label part of the one-mode problem: the remaining work is only the explicit branchwise verification of the core/bulk inequalities and remote-gap exclusion.

CCXIV. K1A4: PRACTICAL ONE-DEFECT ONE-MODE PASS THEOREM ON THE EXPLICIT CANONICAL WORKING BRANCH

CCXV. 1. REUSED INPUTS FROM K1A2 AND K1A3

From the previous K1A steps, we reuse:

$$\boxed{\kappa_* = -1, \quad \kappa_{\text{comp}} \geq 1,} \quad (448)$$

and the radial core coefficients

$$\boxed{a_0 = \frac{\xi_E g_E + \xi_c g_c}{z_{0,0}}, \quad b_0 = \frac{\xi_E g_E}{z_{0,0}}, \quad c_0 = \frac{\xi_O g_O}{z_{0,0}}.} \quad (449)$$

The bulk mass parameter is

$$\boxed{\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}.} \quad (450)$$

The general practical one-defect / one-mode criterion is:

$$\boxed{\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}, \quad \kappa_* < 0 \text{ unique}, \quad \mu_\infty > 0,} \quad (451)$$

and

$$\boxed{\xi_E g_E + \xi_c g_c > 0, \quad \xi_E g_E > 0, \quad 2|\kappa_*| \xi_E g_E > \xi_E g_E + \xi_c g_c,} \quad (452)$$

with every competitor satisfying

$$\boxed{2\kappa \xi_E g_E > \xi_E g_E + \xi_c g_c.} \quad (453)$$

CCXVI. 2. EXPLICIT CANONICAL BRANCH INSERTION

Now specialize to the explicit canonical branch

$$\boxed{T_* = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0.}$$

Then the Bloch/Class-II coefficients are already explicit:

$$\boxed{g_c = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3,} \quad (454)$$

$$\boxed{g_E = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad g_O = 0.} \quad (455)$$

Hence

$$\boxed{c_0 = 0, \quad b_0 = \frac{\xi_E}{z_{0,0}} \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa,} \quad (456)$$

and defining

$$\boxed{R_c := \frac{\xi_c g_c}{\xi_E g_E} = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2,} \quad (457)$$

one gets

$$\boxed{a_0 = b_0(1 + R_c).} \quad (458)$$

:contentReference[oaicite:10]index=10

CCXVII. 3. COLLAPSE OF THE LOWEST-CHANNEL INEQUALITIES

Since $\kappa_* = -1$, one has $|\kappa_*| = 1$. Therefore the decisive bridge inequality

$$2|\kappa_*| \xi_{EGE} > \xi_{EGE} + \xi_c g_c$$

reduces to

$$\boxed{\xi_{EGE} > \xi_c g_c.} \quad (459)$$

In terms of R_c , the three low-core inequalities become:

$$\xi_{EGE} > 0 \iff b_0 > 0,$$

$$\xi_{EGE} + \xi_c g_c > 0 \iff a_0 > 0 \iff 1 + R_c > 0,$$

and

$$\xi_{EGE} > \xi_c g_c \iff R_c < 1.$$

Hence the entire low-core part is equivalent to

$$\boxed{-1 < R_c < 1, \quad \alpha > 0.} \quad (460)$$

Thus the practical lowest-channel one-mode criterion becomes

$$\boxed{-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho}.} \quad (461)$$

This is exactly the explicit canonical branch reduction already recorded in the notes.

CCXVIII. 4. PRIMITIVE WITNESS INSERTION FOR THE LONGITUDINAL CHANNEL

The same explicit canonical witness branch also gives the more microscopic longitudinal coefficient:

$$\boxed{\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \Gamma_{JJ} m_{\parallel} u_{\parallel}(T^1) \theta_0^3 \kappa^3,} \quad (462)$$

while the transverse coefficients are

$$\boxed{\alpha = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa, \quad \beta = 0.} \quad (463)$$

Therefore a practical sufficient witness condition is

$$\boxed{\Gamma_{JJ} \neq 0, \quad \beta_X \neq 0, \quad m_{\parallel} \neq 0, \quad u_{\parallel}(T^1) \neq 0, \quad \theta_0 \kappa \neq 0.} \quad (464)$$

Under (464),

$$\lambda_{\parallel} \neq 0, \quad \alpha \neq 0, \quad \beta = 0,$$

so the branch is a nondegenerate carrier witness branch already at the explicit canonical level. :contentReference[oaicite:12]index=12

CCXIX. 5. PRACTICAL K1A4 THEOREM

Combining the low-core reduction (461) with the explicit witness insertion (464), we obtain:

[Practical one-defect one-mode pass theorem on the explicit canonical working branch] Assume the same explicit canonical working branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

and the practical lowest-channel structure

$$\kappa_\star = -1, \quad \kappa_{\text{comp}} \geq 1.$$

If, in addition,

$$\boxed{\Gamma_{JJ} \neq 0, \quad \beta_X \neq 0, \quad m_\parallel \neq 0, \quad u_\parallel(T^1) \neq 0, \quad \theta_0 \kappa \neq 0,} \quad (465)$$

and

$$\boxed{-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho},} \quad (466)$$

then

$$\boxed{\text{Ind}(\mathbb{H}_C) = 1,} \quad (467)$$

and therefore

$$\boxed{\text{one defect} \leftrightarrow \text{one protected light mode.}} \quad (468)$$

Condition (465) implies

$$\lambda_\parallel \neq 0, \quad \alpha \neq 0, \quad \beta = 0,$$

so the explicit canonical branch realizes both a longitudinal witness and a transverse witness.

Condition (466) is exactly the reduced practical lowest-channel one-mode criterion on that branch.

By the previously established working one-defect/one-mode theorem, these imply

$$\text{Ind}(\mathbb{H}_C) = 1,$$

hence one defect carries exactly one protected light mode.

CCXX. 6. HONEST SCOPE

This is the strongest honest K1A4 result on the present explicit canonical working branch.

It is already enough for a practical theorem-level completion of the single-defect one-mode statement on that branch.

However, it is still not the fully unconditional final microscopic TECT theorem, because the notes explicitly keep open the actual branch evaluation of:

$$\xi_E, \quad \xi_C, \quad \mu_\infty, \quad \Delta_{\text{bench},\rho}^{\text{full}} - \eta_{\text{rem},\rho},$$

as well as the final overlap calibrations in the complete theory.

CCXXI. FINAL SUMMARY

At K1A4, the one-defect one-mode problem is no longer structural. On the explicit canonical working branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the coefficients are already explicit:

$$g_c = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3, \quad g_E = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad g_O = 0.$$

With

$$a_0 = \frac{\xi_E g_E + \xi_c g_c}{z_{0,0}}, \quad b_0 = \frac{\xi_E g_E}{z_{0,0}}, \quad R_c = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2,$$

and in the practical lowest-channel case

$$\kappa_* = -1, \quad \kappa_{\text{comp}} \geq 1,$$

the low-core one-mode inequalities reduce exactly to

$$\boxed{-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho}.$$

The primitive witness layer is also explicit:

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \Gamma_{JJ} m_{\parallel} u_{\parallel}(T^1) \theta_0^3 \kappa^3,$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa, \quad \beta = 0.$$

So a practical sufficient witness condition is

$$\Gamma_{JJ} \neq 0, \quad \beta_X \neq 0, \quad m_{\parallel} \neq 0, \quad u_{\parallel}(T^1) \neq 0, \quad \theta_0 \kappa \neq 0.$$

Therefore, under these finite scalar checks,

$$\boxed{\text{Ind}(\mathbb{H}_C) = 1,}$$

and hence

$$\boxed{\text{one defect} \leftrightarrow \text{one protected light mode.}}$$

This closes K1A4 at practical theorem level on the explicit canonical working branch, while the fully unconditional final microscopic theorem still depends on the final finite audits of the intended branch data.

CCXXII. K1B: SPECTRAL ISOLATION OF THE THREE-DEFECT FAMILY SUBSPACE

CCXXIII. 1. INPUT FROM K1A

Assume K1A has already been established at practical theorem level on the chosen working branch. That is, a single isolated defect supports exactly one exponentially localized protected light mode

$$\boxed{\mathbb{O}_{\text{1def}}^{(0)} \psi_{\star} = \lambda_{\star} \psi_{\star}, \quad \|\psi_{\star}(x)\| \lesssim e^{-\mu|x|}, \quad \dim \ker_{\text{low}}(\mathbb{O}_{\text{1def}}^{(0)} - \lambda_{\star}) = 1.} \quad (469)$$

The rank-one single-defect case is exactly the natural family-space input: if one defect carried $q > 1$ protected low modes, then three defects would generate $3q$ light states rather than 3. :contentReference[oaicite:5]index=5

CCXXIV. 2. THREE TRANSLATED DEFECT MODES

Let $X_1, X_2, X_3 \in \mathbb{R}^3$ be three sufficiently separated defect centers, and let the three-defect background be a weakly interacting superposition of three isolated defects.

Define the translated approximate localized modes

$$\psi_a(x) := \psi_\star(x - X_a), \quad a = 1, 2, 3. \quad (470)$$

If a microscopic connection is present, one should use parallel-transported translates

$$\psi_a(x) = U_a(x)\psi_\star(x - X_a),$$

but this does not change the algebra below. :contentReference[oaicite:6]index=6

Let

$$R_{ab} := |X_a - X_b|, \quad R_{\min} := \min_{a \neq b} R_{ab}.$$

Because $\psi_\star$ is exponentially localized,

$$\langle \psi_a, \psi_b \rangle = \delta_{ab} + O(e^{-\mu R_{ab}}). \quad (471)$$

Hence the Gram matrix

$$G_{ab} := \langle \psi_a, \psi_b \rangle \quad (472)$$

satisfies

$$G = I + O(e^{-\mu R_{\min}}). \quad (473)$$

Therefore, for sufficiently large separation,

$$G \text{ is invertible.} \quad (474)$$

So the approximate low subspace

$$\mathcal{S}_{\text{app}} := \text{span}\{\psi_1, \psi_2, \psi_3\} \quad (475)$$

is genuinely three-dimensional. :contentReference[oaicite:7]index=7

CCXXV. 3. PROJECTED APPROXIMATE LIGHT MATRIX

Let

$$\mathbb{O}_{3\text{def}}^{(0)}$$

be the microscopic reference operator on the three-defect background.

Projecting onto the approximate low subspace gives

$$M_{ab}^{\text{app}} := \langle \psi_a, \mathbb{O}_{3\text{def}}^{(0)} \psi_b \rangle. \quad (476)$$

Because each ψ_a is localized near the a -th defect and solves the isolated one-defect problem up to exponentially small interaction corrections, one has

$$M_{ab}^{\text{app}} = \lambda_\star G_{ab} + T_{ab}, \quad T_{ab} = O(e^{-\mu R_{ab}}). \quad (477)$$

Thus the low three-defect cluster is controlled by a finite 3×3 tight-binding-type matrix with exponentially small off-diagonal couplings.

CCXXVI. 4. ORTHONORMALIZED LOCALIZED BASIS

Since G is invertible, define the orthonormalized localized basis

$$\phi_a := \sum_{b=1}^3 \psi_b (G^{-1/2})_{ba}. \quad (478)$$

Then

$$\langle \phi_a, \phi_b \rangle = \delta_{ab}. \quad (479)$$

The projected approximate light matrix becomes

$$L^{\text{app}} := G^{-1/2} M^{\text{app}} G^{-1/2}, \quad L_{ab}^{\text{app}} = \langle \phi_a, \mathbb{O}_{3\text{def}}^{(0)} \phi_b \rangle. \quad (480)$$

So already at the approximate level, the three-defect low sector is a finite Hermitian 3×3 matrix problem. This is exactly the K1B precursor to later family Hamiltonians. :contentReference[oaicite:9]index=9

CCXXVII. 5. HEAVY-GAP HYPOTHESIS AND EXACT LOW PROJECTOR

Now let the rest of the spectrum of $\mathbb{O}_{3\text{def}}^{(0)}$ stay at least a distance $\Delta_H > 0$ away from $\lambda_\star$ in the isolated-defect limit. Assume the interaction/tunneling scale obeys

$$\|T\| + \|G - I\| |\lambda_\star| < \frac{1}{2} \Delta_H. \quad (481)$$

Then the three approximate low eigenvalues obtained from L^{app} remain separated from the heavy spectrum. Therefore there exists a contour Γ enclosing exactly those three low eigenvalues and no other spectrum.

Define the exact Riesz projector

$$P_L = \frac{1}{2\pi i} \oint_{\Gamma} (z - \mathbb{O}_{3\text{def}}^{(0)})^{-1} dz. \quad (482)$$

Then the exact light family space is

$$\mathcal{H}_L := \text{Ran}(P_L). \quad (483)$$

Since Γ encloses exactly three isolated eigenvalues, one gets

$$\dim \mathcal{H}_L = 3. \quad (484)$$

This is precisely the K1B statement that a three-dimensional family subspace is isolated. :contentReference[oaicite:10]index=10

CCXXVIII. 6. PRACTICAL THEOREM-LEVEL K1B STATEMENT

We can now state the practical K1B theorem on the current route.

[Practical K1B triplet-isolation theorem] Assume:

1. K1A holds at practical theorem level, i.e. one isolated defect carries exactly one exponentially localized protected light mode $\psi_\star$;
2. three defect centers X_1, X_2, X_3 are sufficiently well separated;
3. the three-defect background is a weakly interacting superposition of the three isolated defects;
4. the heavy spectrum remains separated by a gap Δ_H satisfying (481).

Then:

$$\boxed{\dim \mathcal{S}_{\text{app}} = 3, \quad \text{rank } P_L = 3, \quad \mathcal{H}_L = \text{Ran}(P_L)} \quad (485)$$

and hence a three-dimensional common light family subspace is isolated.

By exponential localization, the overlaps obey (471), so the Gram matrix satisfies (473) and is invertible for sufficiently large separation. Therefore $\dim \mathcal{S}_{\text{app}} = 3$.

Projecting $\mathbb{O}_{3\text{def}}^{(0)}$ onto $\mathcal{S}_{\text{app}}$ gives the finite matrix M^{app} with exponentially small tunneling correction T . Under the gap condition (481), the three approximate low eigenvalues remain isolated from the heavy spectrum. Hence a contour Γ exists that encloses exactly those three eigenvalues, and the corresponding Riesz projector P_L has rank three.

Therefore the exact isolated family space $\mathcal{H}_L$ is three-dimensional.

CCXXIX. 7. HONEST STATUS

This closes K1B at practical theorem level, but not yet as a first-principles TECT theorem.

The reason is exactly the same as before: the single-defect one-mode statement itself is still conditional on the actual TECT defect operator satisfying the Callias-type hypotheses A1–A4, including asymptotic gap and index one. Only after that is verified does the implication

$$\text{one defect} \leftrightarrow \text{one light mode} \implies \text{rank } P_L = 3$$

become a first-principles theorem inside TECT.

So the correct K1B wording is:

$$\boxed{\text{K1B is closed conditionally at practical theorem level,}} \quad (486)$$

namely conditional on K1A and on a sufficiently large separation / heavy-gap regime.

CCXXX. FINAL SUMMARY

K1B asks for an isolated three-dimensional family subspace.

Assuming K1A has already supplied one exponentially localized protected light mode per defect,

$$\mathbb{O}_{1\text{def}}^{(0)} \psi_\star = \lambda_\star \psi_\star, \quad \|\psi_\star(x)\| \lesssim e^{-\mu|x|},$$

define the translated modes

$$\psi_a(x) = \psi_\star(x - X_a), \quad a = 1, 2, 3.$$

For sufficiently well-separated defect centers,

$$\langle \psi_a, \psi_b \rangle = \delta_{ab} + O(e^{-\mu R_{ab}}),$$

so the Gram matrix

$$G_{ab} = \langle \psi_a, \psi_b \rangle$$

satisfies

$$G = I + O(e^{-\mu R_{\min}})$$

and is invertible. Therefore the approximate low subspace

$$\mathcal{S}_{\text{app}} = \text{span}\{\psi_1, \psi_2, \psi_3\}$$

is three-dimensional.

Projecting the three-defect operator gives

$$M_{ab}^{\text{app}} = \langle \psi_a, \mathbb{O}_{3\text{def}}^{(0)} \psi_b \rangle = \lambda_* G_{ab} + T_{ab}, \quad T_{ab} = O(e^{-\mu R_{ab}}).$$

After orthonormalization, the approximate light matrix is

$$L^{\text{app}} = G^{-1/2} M^{\text{app}} G^{-1/2}.$$

If the heavy spectrum stays separated by a gap Δ_H larger than the tunneling/Gram-correction scale, then the exact Riesz projector

$$P_L = \frac{1}{2\pi i} \oint_{\Gamma} (z - \mathbb{O}_{3\text{def}}^{(0)})^{-1} dz$$

is well-defined and satisfies

$$\boxed{\text{rank } P_L = 3.}$$

Therefore

$$\boxed{\mathcal{H}_L = \text{Ran}(P_L)}$$

is the isolated three-dimensional common family subspace.

So K1B is closed at practical theorem level, conditional on K1A and on sufficiently large defect separation with a surviving heavy gap.

CCXXXI. K1C: EFFECTIVE PRESERVATION OF THE ISOLATED FAMILY SPACE UNDER SECTOR PERTURBATIONS

CCXXXII. 1. INPUT FROM K1B

Assume K1B has already been established at practical theorem level. Thus there exists an isolated three-dimensional low spectral subspace

$$\boxed{\mathcal{H}_L = \text{Ran}(P_L), \quad \dim \mathcal{H}_L = 3,} \tag{487}$$

where

$$\boxed{P_L = \frac{1}{2\pi i} \oint_{\Gamma} (z - \mathbb{O}_{3\text{def}}^{(0)})^{-1} dz} \tag{488}$$

projects onto the isolated low cluster of the reference three-defect operator

$$\mathbb{O}_{3\text{def}}^{(0)}.$$

Let

$$Q_H := 1 - P_L$$

be the heavy-sector projector. Assume the reference low energy is centered near E_* , and the heavy spectrum obeys

$$\boxed{\text{dist}(E_*, \text{spec}(Q_H \mathbb{O}_{3\text{def}}^{(0)} Q_H)) \geq \Delta_H > 0.} \tag{489}$$

CCXXXIII. 2. SECTOR PERTURBATION AND EXACT EFFECTIVE OPERATOR

For each fermion sector f , define the microscopic perturbed operator

$$\mathbb{O}_f = \mathbb{O}_{3\text{def}}^{(0)} + \delta\mathbb{O}_f. \quad (490)$$

Write it in the $P_L \oplus Q_H$ block form:

$$\mathbb{O}_f = \begin{pmatrix} L_f & K_f \\ K_f^\dagger & H_f \end{pmatrix},$$

where

$$L_f = P_L \mathbb{O}_f P_L, \quad K_f = P_L \mathbb{O}_f Q_H, \quad H_f = Q_H \mathbb{O}_f Q_H.$$

Then the exact Feshbach/Schur-complement operator on $\mathcal{H}_L$ is

$$\mathcal{Y}_f(E) = L_f + K_f (E - H_f)^{-1} K_f^\dagger. \quad (491)$$

Equivalently, subtracting the universal reference block and expanding around the unperturbed energy E_* , the exact reduced family operator has the standard form

$$\mathcal{Y}_f = P_L \delta\mathbb{O}_f P_L - P_L \delta\mathbb{O}_f Q_H (Q_H \mathbb{O}_{3\text{def}}^{(0)} Q_H - E_*)^{-1} Q_H \delta\mathbb{O}_f P_L + O(\delta\mathbb{O}_f^3). \quad (492)$$

This is exactly the finite family operator already identified in the notes.

CCXXXIV. 3. MEANING OF “EFFECTIVE PRESERVATION”

K1C does not require the family subspace to remain literally invariant under the full perturbed operator in the strong sense

$$\delta\mathbb{O}_f \mathcal{H}_L \subset \mathcal{H}_L.$$

That would be too strong.

Instead, the correct notion is:

$$\boxed{\text{the low sector remains spectrally isolated and the full dynamics near } E_* \text{ is exactly reducible to a closed operator on } \mathcal{H}_L.} \quad (493)$$

Thus “effective preservation” means:

1. the heavy resolvent exists near the low energy,
2. the low-heavy mixing is perturbative,
3. the exact low dynamics is captured by the finite operator $\mathcal{Y}_f(E)$ on $\mathcal{H}_L$.

CCXXXV. 4. PRACTICAL SUFFICIENT CONDITION

A sufficient practical condition is the finite heavy-gap / mixing inequality

$$\boxed{\|\delta\mathbb{O}_f\| < \frac{\Delta_H}{2}, \quad \|K_f\| < \frac{\Delta_H}{2}.} \quad (494)$$

Then for every E with $|E - E_*| < \Delta_H/4$, one has

$$\boxed{\|(E - H_f)^{-1}\| \leq \frac{2}{\Delta_H}.} \quad (495)$$

Consequently, the heavy-mediated correction is bounded by

$$\|K_f(E - H_f)^{-1}K_f^\dagger\| \leq \frac{2}{\Delta_H} \|K_f\|^2. \quad (496)$$

So the sector perturbation does not destroy the family space: it only renormalizes the finite 3×3 family operator by a controlled heavy-mediated correction.

CCXXXVI. 5. HERMITICITY OF THE REDUCED OPERATOR

Assume $\mathbb{O}_f$ is self-adjoint. Then for real E outside $\text{spec}(H_f)$,

$$(E - H_f)^{-1}$$

is self-adjoint, hence

$$\mathcal{Y}_f(E)^\dagger = \mathcal{Y}_f(E). \quad (497)$$

Therefore the reduced family problem is a finite Hermitian matrix problem on $\mathcal{H}_L$, exactly as required for later K2, K3, and K4.

CCXXXVII. 6. PRACTICAL K1C THEOREM

We now formulate the practical theorem-level closure of K1C.

[Practical K1C theorem] Assume:

1. K1B holds, i.e. the reference three-defect operator has an isolated rank-three low spectral subspace $\mathcal{H}_L = \text{Ran}(P_L)$;
2. the heavy spectrum is separated by a gap $\Delta_H > 0$ as in (489);
3. each sector perturbation $\delta\mathbb{O}_f$ is sufficiently small compared with Δ_H , and the low-heavy mixing block K_f is perturbative.

Then for each sector f :

$$\mathcal{H}_L \text{ is preserved effectively,} \quad (498)$$

in the precise sense that the exact low-energy dynamics near E_* is captured by the finite Hermitian operator

$$\mathcal{Y}_f(E) = L_f + K_f(E - H_f)^{-1}K_f^\dagger \quad (499)$$

or, to second order,

$$\mathcal{Y}_f = P_L \delta\mathbb{O}_f P_L - P_L \delta\mathbb{O}_f Q_H (Q_H \mathbb{O}_{3\text{def}}^{(0)} Q_H - E_*)^{-1} Q_H \delta\mathbb{O}_f P_L + O(\delta\mathbb{O}_f^3). \quad (500)$$

Hence K1C is closed at practical theorem level.

Because of the heavy gap and perturbative low-heavy mixing, the heavy resolvent exists and obeys the norm bound (495). Therefore the Schur complement (491) is well-defined and bounded on the finite-dimensional low space $\mathcal{H}_L$. Self-adjointness of $\mathbb{O}_f$ implies Hermiticity of $\mathcal{Y}_f(E)$. Thus the low sector is preserved effectively and the full sectoral problem reduces to a finite Hermitian operator on the same common family space.

CCXXXVIII. 7. HONEST STATUS

This closes K1C at practical theorem level, but conditionally on K1B and on the survival of the heavy gap under sector perturbations.

So the exact honest conclusion is:

$$\boxed{K1C \text{ is no longer conceptually open.}} \quad (501)$$

What remains is explicit microscopic evaluation of

$$P_L \delta \mathbb{O}_f P_L, \quad P_L \delta \mathbb{O}_f Q_H, \quad Q_H \mathbb{O}_{3\text{def}}^{(0)} Q_H,$$

for the actual TECT defect PDE.

CCXXXIX. FINAL SUMMARY

K1C asks whether sector perturbations preserve the common family space effectively.

Assuming K1B has already isolated a rank-three low spectral subspace

$$\mathcal{H}_L = \text{Ran}(P_L), \quad \dim \mathcal{H}_L = 3,$$

with heavy complement $Q_H = 1 - P_L$ separated by a gap Δ_H , the sector-dependent microscopic operator

$$\mathbb{O}_f = \mathbb{O}_{3\text{def}}^{(0)} + \delta \mathbb{O}_f$$

has the exact low-energy effective operator

$$\mathcal{Y}_f(E) = L_f + K_f(E - H_f)^{-1} K_f^\dagger$$

on $\mathcal{H}_L$.

Expanding around the unperturbed low energy E_* , this becomes

$$\boxed{\mathcal{Y}_f = P_L \delta \mathbb{O}_f P_L - P_L \delta \mathbb{O}_f Q_H (Q_H \mathbb{O}_{3\text{def}}^{(0)} Q_H - E_*)^{-1} Q_H \delta \mathbb{O}_f P_L + O(\delta \mathbb{O}_f^3).}$$

Thus effective preservation means not literal invariance of $\mathcal{H}_L$, but the existence of a closed finite Hermitian low-energy operator on the same three-dimensional family space.

If the heavy gap survives and the low-heavy mixing is perturbative, then $\mathcal{Y}_f$ is well-defined and Hermitian. Therefore K1C is closed at practical theorem level.

So once K1B holds, the flavor problem is rigorously reduced to a finite Hermitian matrix problem on the common family space.

CCXL. K1D: PHYSICAL LEFT-HANDED INTERPRETATION OF THE COMMON FAMILY SPACE

CCXLI. 1. INPUT FROM K1A–K1C

Assume K1A–K1C have already been established at practical theorem level.

Then:

$$\boxed{\mathcal{H}_L = \text{Ran}(P_L), \quad \dim \mathcal{H}_L = 3,} \quad (502)$$

is the isolated common family subspace, and for each sector f the low-energy dynamics is captured by a finite Hermitian operator on $\mathcal{H}_L$.

However, this does not yet imply that $\mathcal{H}_L$ is the *physical left-handed* family space. K1D asks for precisely this additional interpretation.

CCXLII. 2. MIRROR LIFTING AS THE FIRST LEFT-HANDEDNESS REQUIREMENT

Let the defect zero-mode space decompose as

$$\boxed{Z_L = Z_{\text{top}} \oplus Z_{\text{mir}}^L, \quad Z_R = Z_{\text{mir}}^R,} \quad (503)$$

with

$$\dim Z_{\text{top}} = I, \quad \dim Z_{\text{mir}}^L = \dim Z_R =: r.$$

Let the effective chiral-odd mirror-lifting operator on the zero-mode sector be

$$\boxed{M = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & \mu^\dagger \\ 0 & \mu & 0 \end{pmatrix},} \quad (504)$$

relative to the splitting $Z_L = Z_{\text{top}} \oplus Z_{\text{mir}}^L$, Z_R .

If

$$\boxed{\mu : Z_{\text{mir}}^L \rightarrow Z_R \text{ is invertible,}} \quad (505)$$

then all vectorlike mirror pairs are lifted to nonzero energy and only the index-protected chiral sector remains massless:

$$\boxed{\text{massless sector} = Z_{\text{top}} \subset Z_L, \quad Z_R \text{ carries no massless mirror zero modes.}} \quad (506)$$

This is already an exact finite-dimensional theorem.

So after mirror lifting, the right-handed obstruction is reduced to zero at the massless level.

CCXLIII. 3. WEAK-BUNDLE INCLUSION TEST

Now let $\mathcal{E}_W$ be the candidate rank-two weak bundle, with local orthonormal frame

$$\Psi_1, \Psi_2.$$

Let

$$P_L : \mathcal{E}_0 \rightarrow Z_{\text{top}}, \quad P_R : \mathcal{E}_0 \rightarrow Z_R$$

denote the orthogonal projectors onto the protected left-handed sector and the right-handed sector.

Define the two Gram matrices

$$\boxed{G_{AB}^{(L)} = \langle \Psi_A, P_L \Psi_B \rangle, \quad G_{AB}^{(R)} = \langle \Psi_A, P_R \Psi_B \rangle, \quad A, B = 1, 2.} \quad (507)$$

Then the exact weak-chirality criterion is

$$\boxed{G^{(R)} = 0, \quad \det G^{(L)} \neq 0.} \quad (508)$$

Indeed:

$$G^{(R)} = 0 \iff P_R \Psi_A = 0 \text{ for } A = 1, 2,$$

so the weak bundle has no right-handed component, while

$$\det G^{(L)} \neq 0 \iff P_L \Psi_1, P_L \Psi_2 \text{ are linearly independent,}$$

so the full rank-two weak bundle survives inside the protected left-handed sector.

Hence (508) is equivalent to

$$\boxed{\mathcal{E}_W \subset Z_{\text{top}}, \quad \mathcal{E}_W \cap Z_R = 0.} \quad (509)$$

This is exactly the physical $SU(2)_L$ chirality-selection criterion.

CCXLIV. 4. REDUCTION TO A SINGLE COMPLEX OVERLAP UNDER MULTIPLICITY-ONE

Assume now the multiplicity-one irrep hypothesis:

1. $\mathcal{E}_W$ is a multiplicity-one 2-dimensional isotypic component of the low-energy bundle;
2. the protected left-handed target sector inside Z_{top} carrying the weak doublet is also a multiplicity-one copy of the same 2-dimensional irrep.

Choose an orthonormal basis u_1, u_2 of that protected left-handed target doublet, and define

$$\boxed{U_{mA} := \langle u_m, \Psi_A \rangle.} \quad (510)$$

Then $U : \mathcal{E}_W \rightarrow Z_{\text{top}}^{(2)}$ is an intertwiner between two copies of the same irreducible 2-dimensional module. By Schur's lemma,

$$\boxed{U = s \mathbf{1}_2} \quad (511)$$

for some complex number s .

Since

$$G^{(L)} = U^\dagger U,$$

one gets

$$\boxed{\det G^{(L)} = |\det U|^2 = |s|^4.} \quad (512)$$

Therefore the full-rank left-handed weak-bundle test reduces to the single overlap condition

$$\boxed{s \neq 0, \quad s = \langle u_1, \Psi_1 \rangle \text{ in a symmetry-adapted basis.}} \quad (513)$$

:contentReference[oaicite:8]index=8

CCXLIV. 5. PHYSICAL LEFT-HANDED FAMILY INTERPRETATION THEOREM

We can now formulate the K1D theorem in the sharpest honest form.

[Practical K1D theorem] Assume:

1. K1A–K1C hold at practical theorem level, so that $\mathcal{H}_L = \text{Ran}(P_L)$ is the isolated common three-dimensional family space;
2. the mirror-lifting block $\mu : Z_{\text{mir}}^L \rightarrow Z_R$ is invertible;
3. a candidate weak rank-two bundle $\mathcal{E}_W$ exists;
4. the weak-bundle projector test satisfies

$$G^{(R)} = 0, \quad \det G^{(L)} \neq 0.$$

Then the surviving common family space is physically left-handed:

$$\boxed{\mathcal{H}_L \text{ is the physical left-handed family space.}} \quad (514)$$

More precisely, the weak doublet bundle is contained entirely in the protected left-handed sector and has no right-handed massless partner.

By invertibility of μ , all vectorlike mirror pairs are lifted and only the index-protected chiral sector $Z_{\text{top}} \subset Z_L$ remains massless. Thus there is no surviving massless right-handed mirror obstruction.

Now apply the projector test. The condition $G^{(R)} = 0$ implies the candidate weak bundle has no right-handed component. The condition $\det G^{(L)} \neq 0$ implies the projected weak bundle has full rank inside the protected left-handed sector. Hence the weak bundle is entirely left-handed in the surviving massless spectrum.

Therefore the isolated common family space obtained in K1A–K1C is not merely a low-energy three-dimensional space: it is the physical left-handed family space.

CCXLVI. 6. STRONGEST PRACTICAL REDUCTION OF K1D

Under the multiplicity-one irrep hypothesis, K1D reduces even further:

$$\boxed{\mu \text{ invertible} \quad + \quad G^{(R)} = 0 \quad + \quad s \neq 0 \implies K1D.} \quad (515)$$

So the genuinely remaining microscopic content of K1D is finite:

$$\boxed{\text{compute the mirror pairing block } \mu, \quad \text{compute } G^{(R)}, \quad \text{compute one complex overlap } s.} \quad (516)$$

CCXLVII. 7. HONEST STATUS

This closes K1D at practical theorem level, but conditionally.

The reason is exactly the same as before:

1. mirror lifting is exact if μ is invertible, but full microscopic TECT still has to produce such an invertible μ ;
2. the weak-bundle projector test is exact, but the actual defect PDE still has to provide the concrete overlaps.

So the strongest honest conclusion is:

$$\boxed{K1D \text{ is no longer conceptually open.}} \quad (517)$$

It is reduced to the finite microscopic evaluation of

$$\mu, \quad G^{(R)}, \quad s.$$

CCXLVIII. FINAL SUMMARY

K1D asks whether the common isolated family space $\mathcal{H}_L = \text{Ran}(P_L)$ is the physical left-handed family space rather than just an arbitrary low-energy subspace.

This is reduced to two exact finite tests.

First, mirror lifting:

$$Z_L = Z_{\text{top}} \oplus Z_{\text{mir}}^L, \quad Z_R = Z_{\text{mir}}^R.$$

If the mirror pairing block

$$\mu : Z_{\text{mir}}^L \rightarrow Z_R$$

is invertible, then all vectorlike mirror pairs are lifted and only the protected chiral sector

$$Z_{\text{top}} \subset Z_L$$

remains massless.

Second, weak-bundle chirality selection: for a candidate weak frame Ψ_1, Ψ_2 , define

$$G_{AB}^{(L)} = \langle \Psi_A, P_L \Psi_B \rangle, \quad G_{AB}^{(R)} = \langle \Psi_A, P_R \Psi_B \rangle.$$

Then

$$\boxed{G^{(R)} = 0, \quad \det G^{(L)} \neq 0}$$

is equivalent to

$$\boxed{\mathcal{E}_W \subset Z_{\text{top}}, \quad \mathcal{E}_W \cap Z_R = 0.}$$

Therefore:

$$\boxed{\mu \text{ invertible} \quad + \quad G^{(R)} = 0 \quad + \quad \det G^{(L)} \neq 0 \implies \mathcal{H}_L \text{ is the physical left-handed family space.}}$$

Under the multiplicity-one irrep hypothesis, the overlap map is

$$U = s \mathbf{1}_2,$$

so

$$\det G^{(L)} = |s|^4,$$

and K1D reduces further to the finite test

$$\boxed{\mu \text{ invertible}, \quad G^{(R)} = 0, \quad s \neq 0.}$$

Thus K1D is closed at practical theorem level. What remains is only the finite microscopic evaluation of the mirror block and the weak-bundle overlap data.

CCXLIX. FULL SUMMARY OF $K1$: COMMON LEFT-HANDED FAMILY SUBSPACE

CCL. 1. POSITION OF $K1$ INSIDE THE C4 PROGRAM

Inside the C4 matter/family sector, the internal tree is

$$\boxed{K1 : \text{common left-handed family subspace}, \quad K2 : \text{canonical kinetic normalization},} \quad (518)$$

$$\boxed{K3 : \text{nondegenerate splittings}, \quad K4 : \text{hierarchy seed}, \quad K5 : \text{irreducible CP seed}.} \quad (519)$$

The logical order is

$$\boxed{K1 \rightarrow K2 \rightarrow (K3, K4) \rightarrow K5.} \quad (520)$$

Thus $K1$ is the first indispensable CKM precondition: before one can normalize, split, or mix families, one must first construct a common left-handed family space.

CCLI. 2. INTERNAL DECOMPOSITION OF $K1$

The common-family-space condition decomposes as

$$\boxed{K1A \rightarrow K1B \rightarrow K1C \rightarrow K1D,} \quad (521)$$

where

$$\boxed{K1A : \text{common carrier sector exists}, \quad K1B : \text{a three-dimensional family subspace is isolated},} \quad (522)$$

$$\boxed{K1C : \text{sector perturbations preserve the family space effectively}, \quad K1D : \text{physical left-handed interpretation}.} \quad (523)$$

So $K1$ is not one theorem but a chain:

single-defect protected mode $\rightarrow$ three-defect rank-3 low sector $\rightarrow$ sectorally stable family space $\rightarrow$ physical left-handed identifica

CCLII. 3. K1A: ONE DEFECT $\leftrightarrow$ ONE PROTECTED LIGHT MODE

A. 3.1. Conceptual target

K1A asks whether one isolated defect supports exactly one protected light mode in the relevant low sector. At the abstract level, this is already reduced to a Callias-type theorem:

$$\boxed{\text{Callias/Fredholm structure} + \text{asymptotic gap} + \text{topological nontriviality} + \text{Ind}(\mathbb{O}_{\text{def}}^{(0)}) = 1 \implies \text{one exponentially localized protected mode}} \quad (524)$$

B. 3.2. Working-branch reduction

On the explicit working branch, the one-defect problem is already reduced to the finite acceptance data

$$(g_c, g_E, g_O, \mu_\infty, \kappa_*).$$

In the practically most relevant lowest-channel case $|\kappa_*| = 1$, the decisive inequalities reduce to

$$\boxed{\mu_\infty > 0, \quad \xi_E g_E > \xi_c g_c,} \quad (525)$$

together with competitor exclusion and remote-gap control.

The coefficient dictionary is

$$\boxed{\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta^3}, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O,} \quad (526)$$

and

$$\boxed{g_c = \frac{\lambda_\parallel}{C_c}, \quad g_E = \frac{\alpha}{C_E}, \quad g_O = \frac{\beta}{C_O}.} \quad (527)$$

Longitudinally, the finite scalar reduction is

$$\boxed{\lambda_c^{(JJ)} \sim \frac{\alpha_X^2}{M_X^2} \mathcal{N}_{JJ} m_\parallel u_\parallel |G_*| A_{G_*},} \quad (528)$$

while transversely

$$\boxed{\Lambda_E = \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_E, \quad \Lambda_O = \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_O.} \quad (529)$$

So the one-defect problem is no longer conceptual: it is an explicit finite scalar audit on the intended Class II branch.

C. 3.3. Honest status of K1A

K1A is closed at practical theorem level on the explicit canonical working branch. It is not yet a full unconditional microscopic TECT theorem, because the actual TECT defect operator still has to verify the abstract assumptions and the final finite branch data still have to be inserted.

CCLIII. 4. K1B: SPECTRAL ISOLATION OF THE THREE-FAMILY SUBSPACE

A. 4.1. Conceptual target

Assume one defect carries one localized protected light mode $\psi_\star$. For three sufficiently separated defects at positions X_a , define translated approximate modes

$$\psi_a(x) = \psi_\star(x - X_a), \quad a = 1, 2, 3.$$

Because $\psi_\star$ is exponentially localized,

$$\langle \psi_a, \psi_b \rangle = \delta_{ab} + O(e^{-\mu R_{ab}}).$$

So the Gram matrix is

$$G = I + O(e^{-\mu R_{\min}}),$$

hence invertible for sufficiently large separation.

Therefore the approximate low subspace

$$\mathcal{S}_{\text{app}} = \text{span}\{\psi_1, \psi_2, \psi_3\}$$

is already three-dimensional.

B. 4.2. Exact light projector

Let $\mathbb{O}_{3\text{def}}^{(0)}$ be the microscopic reference operator on the three-defect background. If the heavy spectrum stays separated by a gap larger than the tunneling scale, then there exists a contour Γ enclosing exactly the three low eigenvalues. Define

$$P_L = \frac{1}{2\pi i} \oint_{\Gamma} (z - \mathbb{O}_{3\text{def}}^{(0)})^{-1} dz. \quad (530)$$

Then

$$\mathcal{H}_L := \text{Ran}(P_L), \quad \text{rank } P_L = 3. \quad (531)$$

C. 4.3. Honest status of K1B

K1B is closed at practical theorem level, conditional on K1A and on sufficiently large defect separation / heavy-gap control.

CCLIV. 5. K1C: EFFECTIVE PRESERVATION UNDER SECTOR PERTURBATIONS

A. 5.1. Conceptual target

K1C asks whether the isolated common family space $\mathcal{H}_L$ survives sector perturbations effectively. This does not mean strict invariance

$$\delta \mathbb{O}_f \mathcal{H}_L \subset \mathcal{H}_L.$$

The correct meaning is that the exact low-energy dynamics near the family cluster is reducible to a closed finite operator on $\mathcal{H}_L$.

B. 5.2. Exact effective family operator

Write

$$\mathbb{O}_f = \mathbb{O}_{3\text{def}}^{(0)} + \delta\mathbb{O}_f, \quad Q_H = 1 - P_L.$$

Then the exact Feshbach/Schur-complement family operator is

$$\boxed{\mathcal{Y}_f(E) = L_f + K_f(E - H_f)^{-1}K_f^\dagger}, \quad (532)$$

where

$$L_f = P_L \mathbb{O}_f P_L, \quad K_f = P_L \mathbb{O}_f Q_H, \quad H_f = Q_H \mathbb{O}_f Q_H.$$

To second order around the reference energy E_* ,

$$\boxed{\mathcal{Y}_f = P_L \delta\mathbb{O}_f P_L - P_L \delta\mathbb{O}_f Q_H (Q_H \mathbb{O}_{3\text{def}}^{(0)} Q_H - E_*)^{-1} Q_H \delta\mathbb{O}_f P_L + O(\delta\mathbb{O}_f^3)}. \quad (533)$$

Hence sector perturbations preserve the family space effectively whenever the heavy gap survives and the low-heavy mixing is perturbative.

C. 5.3. Honest status of K1C

K1C is closed at practical theorem level, conditional on K1B and on heavy-gap/mixing control.

CCLV. 6. K1D: PHYSICAL LEFT-HANDED INTERPRETATION

A. 6.1. Conceptual target

Even after K1A–K1C, $\mathcal{H}_L$ is still only a common low-energy family subspace. K1D asks whether it is the *physical left-handed* family space.

B. 6.2. Mirror lifting

Decompose the zero-mode sector as

$$Z_L = Z_{\text{top}} \oplus Z_{\text{mir}}^L, \quad Z_R = Z_{\text{mir}}^R.$$

If the mirror block

$$\mu : Z_{\text{mir}}^L \rightarrow Z_R$$

has full rank, then all vectorlike mirror pairs are lifted and only

$$Z_{\text{top}} \subset Z_L$$

remains massless.

Thus the right-handed massless obstruction is reduced to zero after mirror lifting.

C. 6.3. Weak-bundle projector test

Let $\mathcal{E}_W$ be the candidate weak rank-two bundle with local frame Ψ_1, Ψ_2 , and let P_L, P_R project respectively onto the protected left-handed and right-handed sectors. Define

$$\boxed{G_{AB}^{(L)} = \langle \Psi_A, P_L \Psi_B \rangle, \quad G_{AB}^{(R)} = \langle \Psi_A, P_R \Psi_B \rangle}. \quad (534)$$

Then the exact finite chirality-selection test is

$$\boxed{G^{(R)} = 0, \quad \det G^{(L)} \neq 0.} \quad (535)$$

Equivalently,

$$\mathcal{E}_W \subset Z_{\text{top}}, \quad \mathcal{E}_W \cap Z_R = 0.$$

D. 6.4. Multiplicity-one reduction

Under the multiplicity-one irrep hypothesis, the overlap map becomes

$$U = s \mathbf{1}_2,$$

so

$$\det G^{(L)} = |s|^4.$$

Thus K1D reduces further to the finite data

$$\mu, \quad G^{(R)}, \quad s.$$

E. 6.5. Honest status of K1D

K1D is closed at practical theorem level, conditional on mirror lifting and on the microscopic computation of the projector overlaps.

CCLVI. 7. INTEGRATED STATUS OF K1

The strongest honest integrated statement is:

$$\boxed{K1A, K1B, K1C, K1D \text{ are all closed at practical theorem level, conditionally.}} \quad (536)$$

More explicitly:

1. K1A reduces one defect to one protected light mode by a finite acceptance audit;
2. K1B lifts this to an isolated rank-three light family subspace $\mathcal{H}_L = \text{Ran}(P_L)$;
3. K1C shows that all relevant sector perturbations reduce to finite Hermitian operators on the same $\mathcal{H}_L$;
4. K1D identifies $\mathcal{H}_L$ as the physical left-handed family space via mirror lifting and weak-bundle projector tests.

CCLVII. 8. WHAT IS STILL NOT UNCONDITIONAL

K1 is not yet a final unconditional microscopic TECT theorem because several steps still require explicit finite microscopic insertion:

- actual verification of the Callias-type assumptions for the TECT defect operator,
- actual evaluation of the one-defect acceptance data $(g_c, g_E, g_O, \mu_\infty, \kappa_*)$,
- actual heavy-gap/mixing control for the three-defect reference operator,
- actual evaluation of the mirror block μ ,
- actual evaluation of the weak-bundle projector overlaps $G^{(L)}, G^{(R)}$, or equivalently the overlap scalar s .

So the final honest conclusion is:

K1 is structurally complete and practically closed, but still awaits finite microscopic insertion to become a full first-principles TECT theorem.

CCLVIII. FINAL SUMMARY

The *K1* program is the construction of the common physical left-handed family space. It decomposes as

$$\boxed{K1A \rightarrow K1B \rightarrow K1C \rightarrow K1D.}$$

a. K1A. One defect carries exactly one protected light mode. On the explicit working branch, this has already been reduced to the finite one-defect acceptance audit

$$(g_c, g_E, g_O, \mu_\infty, \kappa_*),$$

and in the lowest-channel case to the sharp inequalities

$$\mu_\infty > 0, \quad \xi_E g_E > \xi_c g_c,$$

together with competitor exclusion and remote-gap control.

b. K1B. Three sufficiently separated defects then generate an isolated rank-three low spectral cluster. The exact family space is

$$\boxed{\mathcal{H}_L = \text{Ran}(P_L), \quad P_L = \frac{1}{2\pi i} \oint_{\Gamma} (z - \mathbb{O}_{3\text{def}}^{(0)})^{-1} dz, \quad \text{rank } P_L = 3.}$$

c. K1C. Sector perturbations preserve this family space effectively, in the sense that the exact low-energy dynamics is captured by the finite Hermitian Schur-complement operator

$$\boxed{\mathcal{Y}_f = P_L \delta \mathbb{O}_f P_L - P_L \delta \mathbb{O}_f Q_H (Q_H \mathbb{O}_{3\text{def}}^{(0)} Q_H - E_*)^{-1} Q_H \delta \mathbb{O}_f P_L + O(\delta \mathbb{O}_f^3).}$$

d. K1D. The common family space is physical left-handed if mirror lifting removes the vectorlike sector and the weak-bundle projector test satisfies

$$\boxed{G^{(R)} = 0, \quad \det G^{(L)} \neq 0.}$$

Equivalently,

$$\mathcal{E}_W \subset Z_{\text{top}}, \quad \mathcal{E}_W \cap Z_R = 0.$$

Under the multiplicity-one irrep hypothesis this reduces further to the finite data

$$\mu, \quad G^{(R)}, \quad s.$$

Therefore the strongest honest global status is

$$\boxed{K1A, K1B, K1C, K1D \text{ are all closed at practical theorem level, conditionally.}}$$

K1 is thus structurally complete, but still not a full unconditional microscopic TECT theorem because the remaining content is finite microscopic insertion: the actual defect-operator assumptions, the explicit one-defect audit data, the heavy-gap control, the mirror block μ , and the weak-bundle overlaps.

CCLIX. PROJECT: TECT — MASTER CONTENTS FOR CONTINUATION (K1 CLOSED AT PRACTICAL THEOREM LEVEL, MOVING TO K2)

CCLX. CURRENT STAGE

$$\boxed{\text{Current stage: } K2 \text{ (canonical kinetic normalization).}} \tag{537}$$

The execution order is fixed and must not be violated:

$$\boxed{K1 \rightarrow K2 \rightarrow K3 \rightarrow K4.} \tag{538}$$

No detour to *K5*, CKM fitting, phenomenological matrix fitting, or unrelated auxiliary theorems is allowed before the derivation/proof chain for *K1–K4* is completed.

CCLXI. ROADMAP REMINDER

The internal C4 / flavor roadmap is

$$\boxed{K1 : \text{ common left-handed family subspace,} \quad K2 : \text{ canonical kinetic normalization,}} \quad (539)$$

$$\boxed{K3 : \text{ nondegenerate splittings,} \quad K4 : \text{ hierarchy seed.}} \quad (540)$$

The exact meanings are:

- $K1$: build the common physical left-handed family space;
- $K2$: canonically normalize the kinetic form on that family space;
- $K3$: prove genuine nondegenerate sectoral splittings;
- $K4$: prove the structural off-diagonal hierarchy seed.

CCLXII. GLOBAL STATUS AFTER THE PREVIOUS CHAIN

The strongest honest integrated statement reached so far is:

$$\boxed{K1A, K1B, K1C, K1D \text{ are all closed at practical theorem level, conditionally.}} \quad (541)$$

This is *not* yet a fully unconditional microscopic TECT theorem. What is closed is the logical structure and its finite-reduction form. What remains open is explicit finite microscopic insertion in the full intended branch.

CCLXIII. K1 FULL SUMMARY

A. K1: Common Left-Handed Family Subspace

$K1$ asks for the construction and identification of the common physical left-handed family space. It decomposes as

$$\boxed{K1A \rightarrow K1B \rightarrow K1C \rightarrow K1D.} \quad (542)$$

B. K1A: One defect $\leftrightarrow$ one protected light mode

The single-defect problem was reduced to the one-defect acceptance data

$$(g_c, g_E, g_O, \mu_\infty, \kappa_*).$$

At the radial/operator level, the single-defect defect operator was reduced to the explicit radial first-order system with real dynamical coefficients

$$a(r), \quad b(r), \quad c(r), \quad \mu(r),$$

and the graded Callias core

$$\boxed{Q_{\kappa, \pm}^{\text{core}} = \pm a(r) \frac{d}{dr} + W_\kappa^{\text{core}}(r), \quad W_\kappa^{\text{core}}(r) = \mu(r) + \frac{\kappa b(r)}{r}.} \quad (543)$$

The g_O -term contributes only a purely imaginary phase shift and does not affect L^2 -normalizability. The protected-channel theorem was reduced to:

$$\boxed{\mu_\infty > 0, \quad a_0 > 0, \quad b_0 > 0, \quad \kappa_* = -1, \quad \kappa_{\text{comp}} \geq 1 \implies \text{Ind}(\mathbb{H}_C) = 1} \quad (544)$$

under the lowest-channel practical assumptions and competing-channel exclusion.

In the explicit canonical working branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the practical criterion reduced to the finite scalar checks

$$\boxed{-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho},} \quad (545)$$

with

$$\boxed{R_c = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2.} \quad (546)$$

A more primitive longitudinal witness was also extracted:

$$\boxed{\lambda_\parallel = -\frac{3\sqrt{3}\pi}{512} \Gamma_{JJ} m_\parallel u_\parallel(T^1) \theta_0^3 \kappa^3,} \quad (547)$$

and

$$\boxed{\alpha = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa, \quad \beta = 0.} \quad (548)$$

So the strongest practical *K1A* statement is:

$$\boxed{\Gamma_{JJ} \neq 0, \beta_X \neq 0, m_\parallel \neq 0, u_\parallel(T^1) \neq 0, \theta_0 \kappa \neq 0, \mu_\infty > 0, -1 < R_c < 1, \Delta_{\text{rem}} > 0} \quad (549)$$

$\implies$

$$\boxed{\text{one defect} \leftrightarrow \text{one protected light mode.}} \quad (550)$$

C. K1B: Rank-3 family subspace isolation

Assuming *K1A*, let $\psi_\star$ denote the exponentially localized single-defect protected mode, and define

$$\psi_a(x) = \psi_\star(x - X_a), \quad a = 1, 2, 3,$$

for three sufficiently separated defects.

Then

$$\langle \psi_a, \psi_b \rangle = \delta_{ab} + O(e^{-\mu R_{ab}}),$$

so the Gram matrix is invertible and the approximate low subspace

$$\mathcal{S}_{\text{app}} = \text{span}\{\psi_1, \psi_2, \psi_3\}$$

is three-dimensional.

If the heavy spectrum stays separated by a surviving gap Δ_H , then the exact Riesz projector

$$\boxed{P_L = \frac{1}{2\pi i} \oint_\Gamma (z - \mathbb{O}_{3\text{def}}^{(0)})^{-1} dz} \quad (551)$$

is well-defined and satisfies

$$\boxed{\text{rank } P_L = 3.} \quad (552)$$

Hence

$$\boxed{\mathcal{H}_L := \text{Ran}(P_L), \quad \dim \mathcal{H}_L = 3.} \quad (553)$$

This is the isolated common family subspace.

D. K1C: Effective preservation under sector perturbations

For each sector f , define

$$\mathbb{O}_f = \mathbb{O}_{3\text{def}}^{(0)} + \delta\mathbb{O}_f, \quad Q_H = 1 - P_L.$$

Then the exact Feshbach/Schur-complement family operator on $\mathcal{H}_L$ is

$$\boxed{\mathcal{Y}_f(E) = L_f + K_f(E - H_f)^{-1} K_f^\dagger}, \quad (554)$$

where

$$L_f = P_L \mathbb{O}_f P_L, \quad K_f = P_L \mathbb{O}_f Q_H, \quad H_f = Q_H \mathbb{O}_f Q_H.$$

To second order,

$$\boxed{\mathcal{Y}_f = P_L \delta\mathbb{O}_f P_L - P_L \delta\mathbb{O}_f Q_H (Q_H \mathbb{O}_{3\text{def}}^{(0)} Q_H - E_*)^{-1} Q_H \delta\mathbb{O}_f P_L + O(\delta\mathbb{O}_f^3)}. \quad (555)$$

Thus effective preservation does not mean literal invariance of $\mathcal{H}_L$. It means that the full low-energy sector remains exactly reducible to a finite Hermitian operator on the same $\mathcal{H}_L$, provided the heavy gap survives and the low-heavy mixing is perturbative.

E. K1D: Physical left-handed interpretation

Even after K1A–K1C, $\mathcal{H}_L$ is initially only a common low-energy family space. To interpret it as the *physical left-handed* family space, two additional tests are required.

a. Mirror lifting. Decompose the massless candidate sector as

$$Z_L = Z_{\text{top}} \oplus Z_{\text{mir}}^L, \quad Z_R = Z_{\text{mir}}^R.$$

If the mirror block

$$\mu : Z_{\text{mir}}^L \rightarrow Z_R$$

is invertible, all vectorlike mirror pairs are lifted, leaving only the protected chiral sector $Z_{\text{top}} \subset Z_L$.

b. Weak-bundle projector test. For a candidate weak frame Ψ_1, Ψ_2 , define

$$\boxed{G_{AB}^{(L)} = \langle \Psi_A, P_L \Psi_B \rangle, \quad G_{AB}^{(R)} = \langle \Psi_A, P_R \Psi_B \rangle}. \quad (556)$$

Then the exact chirality-selection test is

$$\boxed{G^{(R)} = 0, \quad \det G^{(L)} \neq 0}. \quad (557)$$

Equivalently,

$$\mathcal{E}_W \subset Z_{\text{top}}, \quad \mathcal{E}_W \cap Z_R = 0.$$

Under the multiplicity-one irrep hypothesis, the overlap map reduces to

$$U = s \mathbf{1}_2, \quad \det G^{(L)} = |s|^4,$$

so K1D reduces further to the finite data

$$\mu, \quad G^{(R)}, \quad s.$$

CCLXIV. INTEGRATED K1 CONCLUSION

The strongest honest integrated conclusion is:

$$\boxed{K1A, K1B, K1C, K1D \text{ are all closed at practical theorem level, conditionally.}} \quad (558)$$

In words:

1. *K1A*: one defect carries one protected light mode;
2. *K1B*: three defects generate an isolated rank-three common family subspace;
3. *K1C*: sector perturbations reduce to finite Hermitian operators on that same family subspace;
4. *K1D*: after mirror lifting and weak-bundle chirality selection, this common family subspace is physically left-handed.

CCLXV. WHAT IS STILL NOT UNCONDITIONAL

K1 is not yet a final unconditional microscopic TECT theorem.

The remaining open content is *finite microscopic insertion*, namely:

- actual verification of the single-defect Callias-type hypotheses in the full TECT defect operator;
- actual intended-branch evaluation of the one-defect finite audit data $(g_c, g_E, g_O, \mu_\infty, \kappa_*)$;
- actual heavy-gap / tunneling comparison for the three-defect reference operator;
- actual microscopic evaluation of the mirror block μ ;
- actual microscopic evaluation of the weak-bundle overlaps $G^{(L)}, G^{(R)}$, or equivalently the overlap scalar s .

So the final honest K1 verdict is:

$$\boxed{K1 \text{ is structurally complete and practically closed, but still awaits finite microscopic insertion to become a full first-principles TECT theorem.}} \quad (559)$$

CCLXVI. STRICT EXECUTION RULE FOR THE NEXT CONVERSATION

From the next conversation onward, the stage marker must be shown first.

The next step is fixed:

$$\boxed{\text{Current stage: } K2A} \quad (560)$$

and the next calculation must begin with the kinetic Gram matrix on $\mathcal{H}_L$:

$$\boxed{Z_{AB}^{(f)} := \langle \phi_A, \mathcal{K}_f \phi_B \rangle,} \quad (561)$$

followed by the proof of positive-definiteness and canonical normalization

$$Z \rightarrow \mathbf{1}.$$

No detour to *K3*, *K4*, *K5*, CKM fitting, or phenomenological matrix ansatz is allowed before *K2* is closed.

CCLXVII. FINAL SUMMARY

K1 is the construction of the common physical left-handed family space, and it decomposes as

$$\boxed{K1A \rightarrow K1B \rightarrow K1C \rightarrow K1D.}$$

a. K1A. One defect carries exactly one protected light mode. On the explicit working branch this has been reduced to the finite one-defect acceptance audit $(g_c, g_E, g_O, \mu_\infty, \kappa_*)$, and in the lowest-channel case to the sharp inequalities

$$\mu_\infty > 0, \quad \xi_E g_E > \xi_c g_c,$$

plus competitor exclusion and remote-gap control.

b. K1B. Three sufficiently separated defects generate an isolated rank-three low spectral cluster with

$$\mathcal{H}_L = \text{Ran}(P_L), \quad \text{rank } P_L = 3.$$

c. K1C. Sector perturbations preserve the family space effectively, in the sense that the exact low-energy dynamics is captured by the finite Hermitian Schur-complement operator

$$\mathcal{Y}_f = P_L \delta \mathbb{O}_f P_L - P_L \delta \mathbb{O}_f Q_H (Q_H \mathbb{O}_{3\text{def}}^{(0)} Q_H - E_*)^{-1} Q_H \delta \mathbb{O}_f P_L + O(\delta \mathbb{O}_f^3).$$

d. K1D. The common family space is physical left-handed if mirror lifting removes the vectorlike sector and the weak-bundle projector test satisfies

$$G^{(R)} = 0, \quad \det G^{(L)} \neq 0.$$

Under the multiplicity-one irrep hypothesis this reduces further to the finite microscopic data

$$\mu, \quad G^{(R)}, \quad s.$$

Therefore the strongest honest integrated statement is

$$K1A, K1B, K1C, K1D \text{ are all closed at practical theorem level, conditionally.}$$

What remains before K1 becomes a full first-principles TECT theorem is finite microscopic insertion: the actual defect-operator assumptions, the explicit one-defect audit data, the heavy-gap control, the mirror block μ , and the weak-bundle overlaps.